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Sheaf cohomology

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Introduction

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PART I

Category theory

Introduction

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Chapter 1

Categories, morphisms and duality

In this chapter we begin by defining the main objects of category theory, that is: categories, and give a few examples. We also study how we can generalize known special types of morphisms and objects into the more general setting of categories.

1.1 Categories

Definition 1.1.1 (Category). A **category** C consists of:

- a class $\mathrm{Ob}(\mathsf{C})$ of **objects**.
- for each pair of objects $X, Y \in \mathrm{Ob}(\mathsf{C})$ a collection of sets $\mathrm{Hom}_{\mathsf{C}}(X, Y)$ of **morphisms from** X **to** Y.
- for each object $X \in \mathrm{Ob}(\mathsf{C})$ a morphisms $\mathbb{1}_X \in \mathrm{Hom}_{\mathsf{C}}(X, X)$ called the **identity** morphism.
- for each pair of morphisms $f \in \mathrm{Hom}_{\mathsf{C}}(X, Y)$ and $g \in \mathrm{Hom}_{\mathsf{C}}(Y, Z)$ a morphism $g \circ f \in \mathrm{Hom}_{\mathsf{C}}(X, Z)$ called their **composite**.

satisfying the following axioms:

- Composition is associative: for each quadruple $X, Y, Z, W \in \mathrm{Ob}(\mathsf{C})$ of objects, if $f \in \mathrm{Hom}_{\mathsf{C}}(Y, Z)$, $g \in \mathrm{Hom}_{\mathsf{C}}(X, Y)$ and $h \in \mathrm{Hom}_{\mathsf{C}}(W, X)$ then

$$(f \circ g) \circ h = f \circ (g \circ h)$$

- Composition satiesfies the unit laws: for each pair of objects $X, Y \in \mathrm{Ob}(\mathsf{C})$ if $f \in \mathrm{Hom}_{\mathsf{C}}(X, Y)$ then

$$\mathbb{1}_Y \circ f = f = f \circ \mathbb{1}_X$$

Classical set theoretical problems, such as Russell's paradox that asserts that there cannot be set of all sets, force us to define the collection of objects in a category as a proper class and not a set.

In some cases the collection of objects is small enough to be a set. We call this types of categories small.

Definition 1.1.2 (Small category). A category C is **small** if its collection of objects $\mathrm{Ob}(\mathsf{C})$ is a set.

Remark 1.1.1 (Size issues). The above definition of a category is actually one of a *locally small* category.

In the general setting, the collection of morphisms between two objects need not to form a set because it is *too big*. There are several ways to fix this: working with proper classes, with Grothendieck universes, ...
We will however be sketchy about this kind of foundational issues. Instead, will refer as "big" categories those whose collection of morphism does not form a set and not bother with the precise definition of what this actually means. By doing this we refrain ourselves to study this kind of categories, but as it turns at this will not be needed.
For the interested reader, [Rie, Section 3.7] provides an account of why this issues may be important in certain situations.

Notation. To indicate that X is an object of a category C we write

$$X \in \mathsf{C} \text{ instead of } X \in \mathrm{Ob}(\mathsf{C})$$

To indicate that $f \in \mathrm{Hom}_{\mathsf{C}}(X, Y)$ is a morphism from X to Y we write

$$f : X \to Y \text{ or } X \xrightarrow{f} Y$$

We will sometimes abbreviate the composition $f \circ g$ as fg.

Before going further, we give some examples of categories that the reader may be used to.

Example 1.1.1.

- The category Set has the collection of all sets as objects and mappings as morphisms.
- The category Grp has groups as objects and group homomorphisms as morphisms.
- The category Ab has abelian groups as objects and group homomorphisms as morphisms. It is a full subcategory of Grp.
- The category Ring has rings as objects and ring homomorphisms as morphisms.[a]
- The category Rng has (not necessarilly unital) rings as objects and ring homomorphisms as morphism.
- The category Top has topological spaces as objects and continuous functions as morphisms.
- For a fixed ring R, the category $\mathsf{Mod}(R)$ has (left) R-modules as objects and R-linear maps as morphisms.
- For a fixed field k, the category Vect_k has k-vector spaces as objects and k-linear maps as morphisms.

[a] We adopt the convetion that all rings have an unit and all ring homomorphisms preserve it.

Example 1.1.2 (Category of open sets)**.** Let X be a topological space. There is a category of open sets $\mathsf{Open}(X)$ consisting of

- As objects, the open subsets of X.
- As morphisms, inclusion of open sets: $U \to V$ if and only if $U \subseteq V$

Example 1.1.3 (Discrete category)**.** Given a category C we can define the **discrete category** C_d taking the same collection of objects as C but only the identity morphisms between them.

Often times when studying a certain type of objects it is useful (or even necessary) to restrict our attention to an smaller but more well-behaved subclass. For example, when studying groups one may want to only consider abelian groups or when studying rings to stay in the unital case.

This notion is formalized using subcategories.

Definition 1.1.3 (Subcategory). Given a category C, a **subcategory** D consists of a subcollection of objects and a subcollection of morphisms in C such that

- If $f : X \to Y$ is a morphism in D then X, Y are objects in D.
- If $f : X \to Y$ and $g : Y \to Z$ are morphisms in D, the composition $gf : X \to Z$ is in D.
- If X is an object in D then the identity map $\mathbb{1}_X$ is in D.

We say that D is a **full subcategory** if for all objects X, Y in D every morphism $f : X \to Y$ in C is also in D.

Example 1.1.4.

- Ab is a full subcategory of Grp.
- The category of finite sets FinSet is a full subcategory of Set.
- Ring is a subcategory of Rng that is not full.
- If k is a field, Vect_k is a full subcategory of $\mathsf{Mod}(k)$.

A similar situation happens often when one wants to combine multiple objects into a single, bigger one. A naive way to do this is to consider their product, for example by considering the cartesian product of two sets or the product topology of two topological spaces. Even though we will introduce later what the categorical product of two objects in a category is, we define it now for categories.

Definition 1.1.4 (Product category). Let C, D be two categories. The **product category** $\mathsf{C} \times \mathsf{D}$ is the category consisting of

- As objects: ordered pairs (C, D), where C is an object in C and D is an object in D, i.e:
$$\mathrm{Ob}(\mathsf{C} \times \mathsf{D}) := \mathrm{Ob}(\mathsf{C}) \times \mathrm{Ob}(\mathsf{D})$$
- As morphisms: ordered pairs $(C \xrightarrow{f} C', D \xrightarrow{g} D')$, where $C \xrightarrow{f} C'$ is a morphism in C and $D \xrightarrow{g} D'$ is a morphism in D, i.e: for all objects $(C, D), (C', D')$ in $\mathsf{C} \times \mathsf{D}$
$$\mathrm{Hom}_{\mathsf{C}\times\mathsf{D}}((C, D), (C', D')) := \mathrm{Hom}_{\mathsf{C}}(C, C') \times \mathrm{Hom}_{\mathsf{D}}(D, D')$$
- Composition is defined *componentwise*: if $(C \xrightarrow{f} C', D \xrightarrow{g} D')$ and $(C' \xrightarrow{f'} C'', D' \xrightarrow{g'} D'')$ are two morphisms in $\mathsf{C} \times \mathsf{D}$ then
$$(f', g') \circ (f, g) := (f' \circ f, g' \circ g) : C \times D \to C'' \times D''$$

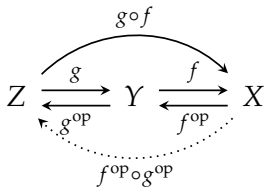
1.1.1 Duality

In the definition of a morphism in a category one has to make a choice of a source and a target. A natural question is to consider what happens if we reverse this arrows. As it turns out, this produces another category that is dual to the original

one.

Definition 1.1.5 (Opposite category)**.** Let C be a category. The **opposite category** C^{op} consists of:

- The same objects as C: $\mathrm{Ob}(\mathsf{C}^{\mathrm{op}}) = \mathrm{Ob}(\mathsf{C})$.
- For each pair of objects $X, Y \in \mathsf{C}^{\mathrm{op}}$: $\mathrm{Hom}_{\mathsf{C}^{\mathrm{op}}}(X, Y) = \mathrm{Hom}_{\mathsf{C}}(Y, X)$.
- If $f^{\mathrm{op}} \colon X \to Y$, $g^{\mathrm{op}} \colon Y \to Z$ are morphisms in C^{op} their composition is defined by
$$f^{\mathrm{op}} \circ g^{\mathrm{op}} = (g \circ f)^{\mathrm{op}}$$
where $f \colon Y \to X$ and $g \colon Z \to Y$ are the associated morphisms in C. Visually



Example 1.1.5. Consider a preordered set $(P, \leq)$ viewed as a small category where a unique morphism $x \to y$ exists if and only if $x \leq y$.
The opposite category P^{op} corresponds to the same set P equipped with the reversed ordering $\geq$. That is, a morphism $x \to y$ exists in P^{op} if and only if $y \leq x$ in P.

Remark 1.1.2. Let $f^{\mathrm{op}} \colon Y \to X$ and $g^{\mathrm{op}} \colon Y \to X$ be two morphisms in the opposite category C^{op}. Then $f^{\mathrm{op}} = g^{\mathrm{op}}$ in C^{op} if and only if $f = g$ in C.

A categorical statement P^* is defined as the **dual** of a statement P in a category C if P^* holds in the opposite category C^{op} if and only if P holds in C.

Note that, by definition, for any category C we have that $(\mathsf{C}^{\mathrm{op}})^{\mathrm{op}} = \mathsf{C}$.

Therefore if a statement is true for all categories C, its dual is true for all categories C^{op}. Since C^{op} is an arbitrary category the dual statement is also true for all categories.

In practice, this means that proving an statement directly proves its dual for free.

1.2 Types of morphisms

As we said in the introduction, category theory shifts the focus from studying the objects itselves to the relations between them.

When starting to learn about set theory, some of the first definitions are those of an injection and a surjection. Then when studying algebraic objects such as groups and rings one learns about monomorphisms and epimorphisms, that are no more and no less than injections and surjections between the underlying sets of the objects that respect the desired algebraic operation.

To translate it to the general categorical framework we cannot speak of objects and thus resort to an equivalent definition.

Definition 1.2.1 (Monomorphism and epimorphism)**.** Let $f \colon X \to Y$ be a morphism in a category C. We say that f is a

- **Monomorphism** (or is **monic**) if for every pair of parallel morphisms

$g, h: Z \to X$ in C, the equality $f \circ g = f \circ h$ implies $g = h$, i.e: f is left-cancellative.
- **Epimorphism** (or is **epic**) if for every pair of parallel morphisms $g, h: Y \to Z$ in C, the equality $g \circ f = h \circ f$ implies $g = h$, i.e: f is right-cancellative.

Example 1.2.1.

- In Set, an monomorphism (resp. epimorphism) is an injective (resp. surjective) function.
- In Grp, an monomorphism (resp. epimorphism) is an injective (resp. surjective) group homomorphism.
- In Top, an monomorphism (resp. epimorphism) is an injective (resp. surjective) continuous map.

One *caveat* of this definition is that the usual definition of a ring epimorphism (i.e: a surjective ring homomorphism) is too restrictive.

Example 1.2.2. In Ring, there are epimorphisms that are not surjective ring homomorphisms.
For example, consider the inclussion $\iota: \mathbb{Z} \hookrightarrow \mathbb{Q}$. For any ring homomorphism $f: \mathbb{Q} \to R$, where R is any ring, since for all $\frac{a}{b} \in \mathbb{Q}$ we have $a, b \in \mathbb{Z}$

$$f\left(\frac{a}{b}\right) = f(a)f(b^{-1}) = f(a)f(b)^{-1}$$

Therefore f is completely determined by its values in $\mathbb{Z}$ and, since $f \circ \iota = f|_{\mathbb{Z}}$ we get that ι is right-cancellative.

As an example of the duality phenomena we described earlier, we show that monomorphisms and monomorphisms are dual concepts.

Lemma 1.2.1 (Monomorphisms and epimorphisms are dual)**.** Let $f: X \to Y$ be a morphism in a category C. Consider the associated morphism $f^{\mathrm{op}}: Y \to X$ in the opposite category C^{op}.
Then f is a monomorphism (resp. epimorphism) in C if and only if f^{op} is an epimorphism (resp. epimorphism) in C^{op}.

Proof. Consider of pair of morphisms $g, h: Z \rightrightarrows X$ in C and their opposites $g^{\mathrm{op}}, h^{\mathrm{op}}: X \rightrightarrows Z$ in C^{op}.

Note that, by definition of composition in C^{op} we have

$$g^{\mathrm{op}} \circ f^{\mathrm{op}} = (f \circ g)^{\mathrm{op}}$$

Therefore we have that $f \circ g = f \circ h$ implies $g = h$ if and only if $(f \circ g)^{\mathrm{op}} = (f \circ h)^{\mathrm{op}}$ implies $g = h$ if and only if $g^{\mathrm{op}} \circ f^{\mathrm{op}} = h^{\mathrm{op}} \circ f^{\mathrm{op}}$ implies $g^{\mathrm{op}} = h^{\mathrm{op}}$. □

Remark 1.2.1. In what follows we will not spell out the proofs of two objects being dual. The proof strategy is basically the same as in the above lemma in all cases.

Another recurrent notion accross mathematics is that of an isomorphism. In short, we say that two objects are isomorphic even though they *look* different they *behave* funtionally the same.

We can extend this notion to general categories.

Definition 1.2.2 (Isomorphism). A morphism $f : X \to Y$ in a category C is an **isomorphism** if there exists another morphism $g : Y \to X$ in C such that

$$g \circ f = \mathbb{1}_X \quad \text{and} \quad f \circ g = \mathbb{1}_Y$$

The morphism g is called the **inverse** of f and is denoted by f^{-1}.

Example 1.2.3.

- In Set, an isomorphism is a bijection.
- In Grp, an isomorphism is a bijective group homomorphism.
- In Top, an isomorphism is a homeomorphism.
- In Ring, an isomorphism is a bijective ring homomorphism.

The usual properties of the inverses and isomorphisms work as one would expect in the general setting.

Proposition 1.2.1. Let $f : X \to Y$ and $g : Y \to Z$ be isomorphisms in a category C. Then:

1. The inverse f^{-1} of f is unique.
2. The identity $\mathbb{1}_X$ is an isomorphism for every $X \in \mathsf{C}$.
3. $g \circ f$ is an isomorphism with $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.
4. If $g \circ f$ is an isomorphism and f is an isomorphism, then g is an isomorphism. Analogously if $g \circ f$ and g are isomorphisms, then f is an isomorphism.

Proof.

1. Suppose that $h_1, h_2 : Y \to X$ are inverses of f. We have
$$h_1 = h_1 \circ \mathbb{1}_Y = h_1 \circ (f \circ h_2) = (h_1 \circ f) \circ h_2 = \mathbb{1}_X \circ h_2 = h_2$$
2. For all $X \in \mathsf{C}$ by definition
$$\mathbb{1}_X \circ \mathbb{1}_X = \mathbb{1}_X$$
i.e: $\mathbb{1}_X$ is its own left and right inverse.
3. We only need to check that $f^{-1} \circ g^{-1}$ is both a left and right inverse of $g \circ f$. On the one hand
$$(f^{-1} \circ g^{-1}) \circ (g \circ f) = f^{-1} \circ (g^{-1} \circ g) \circ f = f^{-1} \circ f = \mathbb{1}_X$$
On the other
$$(g \circ f) \circ (f^{-1} \circ g^{-1}) = g \circ (f \circ f^{-1}) \circ g^{-1} = g \circ g^{-1} = \mathbb{1}_Z$$
4. Let $h := (g \circ f)^{-1} : Z \to X$. We are going to check that $f \circ h : Z \to Y$ is the inverse of g. On the one hand
$$g \circ (f \circ h) = (g \circ f) \circ h = \mathbb{1}_Z$$
On the other, since $h \circ (g \circ f) = \mathbb{1}_X$ precomposing with f and postcomposing with f^{-1} we have
$$\begin{aligned} f \circ (h \circ g \circ f) \circ f^{-1} = f \circ \mathbb{1}_X \circ f^{-1} = \mathbb{1}_Y &\implies (f \circ h) \circ g \circ (f \circ f^{-1}) = \mathbb{1}_Y \\ &\implies (f \circ h) \circ g = \mathbb{1}_Y \end{aligned}$$
□

As it happens in Set, the notions of monomorphisms, monomorphisms and isomorphisms are closely related.

Proposition 1.2.2. Let $f: X \to Y$ be an isomorphism. Then f is both a monomorphism and epimorphism.

Proof. Suppose $f: X \to Y$ is an isomorphism in a category C. Then there exists a morphism $f^{-1}: Y \to X$ in C such that

$$f^{-1} \circ f = \mathbb{1}_X \quad f \circ f^{-1} = \mathbb{1}_Y$$

Consider a pair of morphisms $g, h: Z \rightrightarrows X$ such that $f \circ g = f \circ h$. Precomposing with f^{-1} we get

$$f^{-1} \circ (f \circ g) = f^{-1} \circ (f \circ h) \implies (f^{-1} \circ f) \circ g = (f^{-1} \circ f) \circ h \implies g = h$$

Therefore f is a monomorphism.

Analogously, if we consider pair of morphism $g', h': Y \rightrightarrows Z$ such that $g' \circ f = h' \circ f$ this time by postcomposing with f^{-1} we get $g' = h'$, i.e: f is a epimorphism. □

Example 1.2.4. In general, a morphism $f: X \to Y$ that is both a monomorphism and epimorphism is not an isomorphism.
Indeed, in the category Top of topological spaces consider a set with more than one point X endowed with the discrete X_d and trivial X_t topology.
Consider the identity map $\mathbb{1}_X: X_d \to X_t$, which is continuous. Since $\mathbb{1}_X$ is injective we have that it is a monomorphism and, since it is surjective, it is an epimorphism. However since there does not exists a continuous bijection $X_t \to X_d$, $\mathbb{1}_X$ cannot be an isomorphism.

As we have seen before, one may create an smaller subcategory by restricting their attention to a smaller collection of objects. In this cases, isomorphisms in the smaller category still are in the bigger one.

Proposition 1.2.3. Let D be a subcategory of a category C. If $f: X \to Y$ is an isomorphism in D then f is also an isomorphism in C.

Proof. Since $f: X \to Y$ is an isomorphism in D there exists an inverse $f^{-1}: Y \to X$ in D. Therefore since $\mathrm{Hom}_{\mathsf{D}}(Y, X) \subseteq \mathrm{Hom}_{\mathsf{C}}(Y, X)$, f^{-1} is an inverse of f in C. □

If the subcategory is full we also have the reciprocal.

Proposition 1.2.4. Let D be a full subcategory of a category C. If $f: X \to Y$ is an isomorphism in C and $X, Y \in \mathsf{D}$ then f is also an isomorphism in D.

Proof. Since $f: X \to Y$ is an isomorphism in C there exists an inverse $f^{-1}: Y \to X$ in D. Since D is a full subcategory of D and $X, Y \in \mathsf{D}$ we have that $f^{-1} \in \mathrm{Hom}_{\mathsf{D}}(Y, X)$. Therefore f admits an inverse in D. □

1.3 Special objects

Just like in a category there are special types of morphisms, we also have special objects. For example, every group has a homomorphism from the trivial group and every unital ring has a homomorphism into the ring of integers. This are examples of what are called initial and terminal objects.

Definition 1.3.1 (Initial and terminal objects)**.** An object $I \in \mathsf{C}$ is **initial** if for every object $X \in \mathsf{C}$ there is a unique morphism $I \to X$.
Dually, an object $T \in \mathsf{C}$ is **terminal** if for every object $X \in \mathsf{C}$ there is a unique morphism $X \to T$.

Example 1.3.1.

- In Set, the empty set $\varnothing$ is initial and the singleton $\{*\}$ is terminal.
- In Grp, trivial groups are both initial and terminal.
- In both Ring and Rng, the zero ring is initial. However, the terminal is different: in Ring it is $\mathbb{Z}$ and in Rng is the zero ring.
- In Top, the empty topological space is initial and the singleton $\{*\}$ is terminal.
- There are no initial nor terminal objects in Field, since there are no homomorphisms between fields of different characteristic.

In the above examples we have spoken of *the* initial and *the* terminal instead of *an* initial or *a* terminal. This is justified as this types of objects are categorically indestinguishable, i.e: isomorphic.

Proposition 1.3.1 (Uniqueness of initial and terminal objects)**.** Let X, Y be two initial (resp. terminal) objects in a category C. Then there exists a unique isomorphism $X \xrightarrow{\sim} Y$ in C.

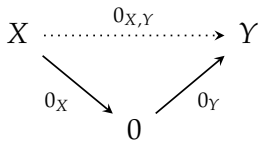
Proof. Since X is initial in C there exists a unique morphism $f : X \to Y$. Analogously there exist a unique morphism $g : Y \to X$.

Note that $g \circ f : X \to X$. Therefore since X is initial $g \circ f = \mathbb{1}_X$. Analogously since $f \circ g : Y \to Y$ we have $f \circ g = \mathbb{1}_Y$.

Therefore f and g are inverses of each other, i.e: $X \cong Y$. Moreover they are unique by definition. □

In some particular cases initial and terminal objects coincide, which means that there is an object with an unique morphism from and to every other object in the category. In this case we say that this object is zero.

Definition 1.3.2 (Zero object)**.** An object $0 \in \mathsf{C}$ is a **zero** object if it is both initial and terminal.
Note that for any two objects $X, Y \in \mathsf{C}$ there exist unique maps $0_X : X \to 0$ and $0_Y : 0 \to Y$. We call this composite map the **zero map** between X and Y and denote it $X \xrightarrow{0_{X,Y}} Y$:



By abuse of notation, we will denote all of these maps by 0.

Example 1.3.2.

- Set and Top have no zero objects since $\varnothing \neq \{*\}$.
- In Grp and Ab, the zero group is the zero object.
- Ring has no zero objects. However in Rng the zero ring is the zero object.

As before, we can speak of *the* zero object (up to isomorphism).

Corollary 1.3.1 (Uniqueness of zero objects)**.** Let $0, \widetilde{0}$ be two zero objects in a category C. Then there exists a unique isomorphism $0 \xrightarrow{\sim} \widetilde{0}$ in C.

Proof. Since 0 and $\widetilde{0}$ are in particular initial we can apply Proposition 1.3.1. □

Chapter 2

Functors and natural transformations

In this chapter we study functors, which encode how we move data across different categories. We also introduce the concept of natural transformations, which in turn encode now moving data across functors, and equivalence of categories.

2.1 Functors

As we have seen, category theory tries to study objects by means of their transformations. Since we can consider categories as their own mathematical object a natural question arises: how do we model transformations between categories?

The answer to this question are functors.

Definition 2.1.1 (Functor). Let C, D be two categories. A (covariant) **functor** $F : \mathsf{C} \to \mathsf{D}$ consists of:

- for each object $X \in \mathsf{C}$, an object $F(X) \in \mathsf{D}$.
- for each morphism $f : X \to Y$ in C, a morphism $Ff : F(X) \to F(Y)$ in D,

such that:

- $F(\mathbb{1}_X) = \mathbb{1}_{F(X)}$ for every $X \in \mathsf{C}$.
- $F(g \circ f) = Fg \circ Ff$ for every composable pair of morphisms f, g in C.

A functor is **contravariant** if it reverses the order of composition: $F(g \circ f) = Ff \circ Fg$. This is equivalent to a covariant functor $F : \mathsf{C}^{\mathrm{op}} \to \mathsf{D}$.
Diagramatically, a covariant functor is represent by

$$\begin{array}{ccc} F\colon \mathsf{C} & \longrightarrow & \mathsf{D} \\ X & & F(X) \\ {\scriptstyle f}\downarrow & \longmapsto & \downarrow{\scriptstyle Ff} \\ Y & & F(Y) \end{array}$$

And a contravariant functor by

$$\begin{array}{ccc} F\colon \mathsf{C} & \longrightarrow & \mathsf{D} \\ X & & F(X) \\ {\scriptstyle f}\downarrow & \longmapsto & \uparrow{\scriptstyle Ff} \\ Y & & F(Y) \end{array}$$

Remark 2.1.1. We will disfavour the covariant/contravariant distinction. Therefore all functors will be covariant (unless otherwise specified) and contravariant functors will be covariant functors from the opposite category.

Example 2.1.1.

- There is a (covariant) endofunctor $P\colon \mathsf{Set} \to \mathsf{Set}$ that sends a set A to its power set $PA = \{A' \subseteq A\}$ and a map $f\colon A \to B$ to the direct image $f_*\colon PA \to PB$, $f_*(A') \coloneqq f(A')$.
  We also get a contravariant version $P\colon \mathsf{Set}^{\mathrm{op}} \to \mathsf{Set}$ that sends a map $f\colon A \to B$ to the inverse image $f^{-1}\colon PB \to PA$, $f^{-1}(B') \coloneqq f^{-1}(B)$.
- There is a functor $F\colon \mathsf{Set} \to \mathsf{Grp}$ that sends a set X to the free group on X.
- There is a functor $-^*\colon \mathsf{Vect}_{\mathbb{K}}^{\mathrm{op}} \to \mathsf{Vect}_{\mathbb{K}}$ that sends a vector space V to its dual V^*.
- There is a functor $\mathrm{Spec}\colon \mathsf{CRing}^{\mathrm{op}} \to \mathsf{Top}$ that sends a commutative ring R to its set $\mathrm{Spec}(R)$ of prime ideals endowed with the Zariski topology.

Example 2.1.2 (Identity functor). Let C be a category. There exists an identity functor

$$\begin{array}{rccc} \mathbb{1}\colon & \mathsf{C} & \longrightarrow & \mathsf{C} \\ & X & & X \\ & f\downarrow & \longmapsto & \downarrow f \\ & Y & & Y \end{array}$$

Example 2.1.3 (Constant functor). Let C be a category. For every object $C \in \mathsf{C}$ there exists a constant functor

$$\begin{array}{rccc} \mathrm{ct}_C\colon & \mathsf{C} & \longrightarrow & \mathsf{C} \\ & X & & C \\ & f\downarrow & \longmapsto & \downarrow \mathbb{1}_C \\ & Y & & C \end{array}$$

Example 2.1.4 (Bifunctor). Let C, C' and D be three categories. Recall that we can construct the product category $\mathsf{C} \times \mathsf{C}'$.
A **bifunctor** is a functor of the form $F\colon \mathsf{C} \times \mathsf{C}' \to \mathsf{D}$. This means that for any $X \in \mathsf{C}$ and $X' \in \mathsf{C}$, $F(X, -)\colon \mathsf{C}' \to \mathsf{D}$ and $F(-, X')\colon \mathsf{C} \to \mathsf{D}$ are functors.
Furthermore for any morphisms $f\colon X \to Y$ in C and $g\colon X' \to Y'$ in C' we have a commutative diagram

$$\begin{array}{ccc} F(X,X') & \xrightarrow{F(X,g)} & F(X,Y') \\ {\scriptstyle F(f,X')}\downarrow & \searrow^{F(f,g)} & \downarrow{\scriptstyle F(f,Y')} \\ F(Y,X') & \xrightarrow[F(Y,g)]{} & F(Y,Y') \end{array}$$

Example 2.1.5. Let C be a category. For every object $C \in \mathsf{C}$ there exists a *covariant* Hom functor

$$\begin{array}{rccc} \mathrm{Hom}(C,-)\colon & \mathsf{C} & \longrightarrow & \mathsf{Set} \\ & X & & \mathrm{Hom}(C,X) \\ & f\downarrow & \longmapsto & \downarrow f\circ \\ & Y & & \mathrm{Hom}(C,Y) \end{array}$$

and a *contravariant* Hom functor

$$\begin{array}{ccc} \mathrm{Hom}(-,C)\colon \mathsf{C} & \longrightarrow & \mathsf{Set} \\ X & & \mathrm{Hom}(C,X) \\ f\downarrow & \longmapsto & \uparrow\circ f \\ Y & & \mathrm{Hom}(C,Y) \end{array}$$

We can combine this two into a bifunctor

$$\begin{array}{ccc} \mathrm{Hom}(-,-)\colon \mathsf{C}^{\mathrm{op}}\times\mathsf{C} & \longrightarrow & \mathsf{Set} \\ (X,Y) & & \mathrm{Hom}(X,Y) \\ (f,g)\downarrow & \longmapsto & \downarrow g\circ -\circ f \\ (Z,W) & & \mathrm{Hom}(Z,W) \end{array}$$

Example 2.1.6. Let D be a subcategory of a category C. There exist an inclussion functor

$$\begin{array}{ccc} \iota\colon \mathsf{D} & \hookrightarrow & \mathsf{C} \\ X & & X \\ f\downarrow & \longmapsto & \downarrow f \\ Y & & Y \end{array}$$

Example 2.1.7 (Preimage functor)**.** Let X, Y be two topological spaces. Consider their categories of open sets $\mathsf{Open}(X), \mathsf{Open}(Y)$.
For every continuous map $f : X \to Y$ we have a preimage functor

$$\begin{array}{ccc} f^{-1}\colon \mathsf{Open}(Y) & \longrightarrow & \mathsf{Open}(X) \\ U & & f^{-1}(U) \\ \downarrow & \longmapsto & \downarrow \\ V & & f^{-1}(V) \end{array}$$

As one would expect, functors and isomorphisms behave well together.

Proposition 2.1.1 (Functors preserve isomorphisms)**.** Let C, D be two categories and $F\colon \mathsf{C} \to \mathsf{D}$ a functor between them. If f is an isomorphism in C then Ff is an isomorphism in D.

Proof. Let $f : X \to Y$ be a isomorphism in C. Then there exists an inverse $f^{-1}\colon Y \to X$ in C such that

$$f^{-1} \circ f = \mathbb{1}_X \quad f \circ f^{-1} = \mathbb{1}_Y$$

Applying F to this equalities we have

$$\begin{aligned} F(f^{-1} \circ f) = F\mathbb{1}_X &\implies Ff^{-1} \circ Ff = \mathbb{1}_{F(X)} \\ F(f \circ f^{-1}) = F\mathbb{1}_Y &\implies Ff \circ Ff^{-1} = \mathbb{1}_{F(Y)} \end{aligned}$$

i.e: Ff has an inverse Ff^{-1} in D. □

Just as with objects and morphisms, there are some particular types of functors that are more useful than others.

Definition 2.1.2 (Functor types). A functor $F\colon \mathsf{C} \to \mathsf{D}$ is:

- **faithful** if for every pair of objects $X, Y \in \mathsf{C}$, the map $\mathrm{Hom}_{\mathsf{C}}(X,Y) \to \mathrm{Hom}_{\mathsf{D}}(F(X),F(Y))$ is injective.
- **full** if for every pair $X, Y \in \mathsf{C}$, the map $\mathrm{Hom}_{\mathsf{C}}(X,Y) \to \mathrm{Hom}_{\mathsf{D}}(F(X),F(Y))$ is surjective.
- **fully faithful** if it is both full and faithful.
- **essentially surjective** if for every object $Y \in \mathsf{D}$ there exists an object $X \in \mathsf{C}$ such that $F(X) \cong Y$.

Example 2.1.8.

- The functor $U\colon \mathsf{Grp} \to \mathsf{Set}$ that sends a group to its underlying set is faithful.
- A subcategory D of a category C is full if and only if the inclussion functor $\iota : \mathsf{D} \to \mathsf{C}$ is full.
- Given a field $\mathbb{K}$, let $\mathsf{C}_{\mathbb{K}}$ be the category whose objects are $\mathbb{K}^n$ for some $n \in \mathbb{N}$. The inclussion $\mathsf{C}_{\mathbb{K}} \to \mathsf{Vect}_{\mathbb{K}}$ is essentially surjective, since all finite dimensional $\mathbb{K}$-vector spaces are isomorphic to $\mathbb{K}^n$ for some n.

A concept that can be abstracted using these definitions is that of a category consisting of "sets with extra structure".

Definition 2.1.3 (Concrete category). A category C is **concrete** if there exists a faithful functor $U\colon \mathsf{C} \to \mathsf{Set}$. Such functor is often times called a **forgetful functor**.

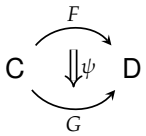
2.2 Natural transformations

We can go further and try to model now how to move data between two functors.

Definition 2.2.1 (Natural transformation). Let $F, G\colon \mathsf{C} \to \mathsf{D}$ be two functors. A **natural transformation** $\psi\colon F \implies G$ consists of a family of morphisms $\psi_X : F(X) \to G(X)$ in D, one for each object $X \in \mathsf{C}$, called the **components** of ψ, such that for every morphism $f : X \to Y$ in C, the following diagram commutes:

$$\begin{array}{ccc} F(X) & \xrightarrow{\psi_X} & G(X) \\ {\scriptstyle Ff}\downarrow & & \downarrow{\scriptstyle Gf} \\ F(Y) & \xrightarrow[\psi_Y]{} & G(Y) \end{array}$$

Diagramatically, a natural transformations is represented by



If each component ψ_X is an isomorphism in D, we say that ψ is a **natural isomorphism**, and we write $F \cong G$.

Example 2.2.1. In $\mathsf{Vect}_{\mathbb{K}}$, the map $\mathrm{ev}\colon V \to V^{**}$ that sends a vector $v \in V$ to the linear evaluation function $\mathrm{ev}_v\colon V^* \to \mathbb{K}$ defines the components of a natural transformation from the identity functor $\mathbb{1}\colon \mathsf{Vect}_{\mathbb{K}} \to \mathsf{Vect}_{\mathbb{K}}$ to the double dual functor $-^{**}\colon \mathsf{Vect}_{\mathbb{K}} \to \mathsf{Vect}_{\mathbb{K}}$.
If we restrict ourselves to the finite-dimensional case then this is a natural isomorphism.

By combining functors and natural transformation we obtain a new example of a

category.

Definition 2.2.2 (Functor category)**.** Let C and D be categories. The **functor category** $\mathsf{Fct}(\mathsf{C}, \mathsf{D})$ has as objecs functors $F\colon \mathsf{C} \to \mathsf{D}$ and as morphisms natural transformations $\eta\colon F \implies G$. Composition and identities are defined componentwise.

Using the notion of natural isomorphisms one can define the notion of an **isomorphism** of categories. Two categories C and D are isomorphic (denoted $\mathsf{C} \cong \mathsf{D}$) if there exists a natural isomorphism between them.

This notion, however, is too restrictive to be useful. The correct notion to identify categories is that of an equivalence.

Definition 2.2.3 (Equivalent categories)**.** Two categories C, D are **equivalent** if there exist functors $F\colon \mathsf{C} \to \mathsf{D}$ and $G\colon \mathsf{D} \to \mathsf{C}$ such that $G \circ F \cong \mathbb{1}_{\mathsf{C}}$ and $F \circ G \cong \mathbb{1}_{\mathsf{D}}$. In this case, F (and G) is called an **equivalence of categories**. We denote this by $\mathsf{C} \simeq \mathsf{D}$.

In general proving that two equivalence of categories by directly constructing the functors F and G is not easy. Instead one can use the following theorem, which we state without proof.

Theorem 2.2.1 ([Rie, Theorem 1.5.9])**.** A functor $F\colon \mathsf{C} \to \mathsf{D}$ defining an equivalence of categories is fully faithful and essentially surjective. Assuming the axiom of choice, any functor with these properties defines an equivalence of categories.

Chapter 3

Limits and colimits

In this chapter we study how different constructions across mathematics can be abstracted into particular cases of what are called limits and colimits. Once these concepts are defined, we spend the rest of the chapter giving familiar constructions that can be made from them. We end the chapter by proving an existence criterion for limits and colimits both in general and for the concrete case of functor categories.

3.1 Diagrams, cones and cocones

A common tool used in mathematics to relate objects and morphisms are diagrams such as commutative squares.

This notions can be easily abstracted using functors.

Definition 3.1.1 (Diagram)**.** Let J be a small category.
A **diagram** X in a category C with shape J is a functor $X\colon \mathsf{J} \to \mathsf{C}$. For every $j \in \mathsf{J}$ we write $X_j := X(j)$.

Example 3.1.1 (Constant diagram)**.** A diagram $X\colon \mathsf{J} \to \mathsf{C}$ is **constant** if it is a constant functor const_c.
In this case the data of the diagram corresponds to the data given by a single object $c \in \mathsf{C}$.

Example 3.1.2. Consider the category

$$\mathsf{J} = \begin{array}{ccc} \bullet & \longrightarrow & \bullet \\ \downarrow & \searrow & \downarrow \\ \bullet & \longrightarrow & \bullet \end{array} = \begin{array}{ccc} 1 & \longrightarrow & 2 \\ \downarrow & \searrow & \downarrow \\ 3 & \longrightarrow & 4 \end{array}$$

A diagram with shape J is a commutative square

$$\begin{array}{ccc} X_1 & \longrightarrow & X_2 \\ \downarrow & & \downarrow \\ X_3 & \longrightarrow & X_4 \end{array}$$

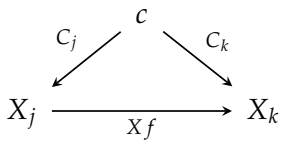
3.1.1 Cones and cocones

To study diagrams from the category theory point of view we must understand their transormations. In this case since diagrams are special types of functors these

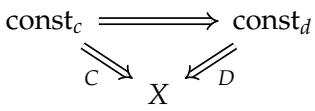
correspond to natural transformations.

Out of all of the possible natural transformations we will only be interested in two particular and dual cases: cones and cocones.

Definition 3.1.2 (Cone). A **cone** C over a diagram $X\colon \mathsf{J} \to \mathsf{C}$ with *apex* c is a natural transformation $C\colon \mathrm{const}_c \Rightarrow X$ from the constant diagram to X. The components $\{C_j\colon c \to X_j\}_{j\in\mathsf{J}}$ are called the *legs* of the cone.
Less abstractly, a cone C over $X\colon \mathsf{J} \to \mathsf{C}$ with *apex* c is a collection of morphisms $C_j\colon c \to X_j$ for every object $j \in \mathsf{J}$ such that for every morphism $f : j \to k$ in J the following diagram in C commutes

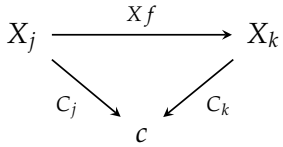


If $C\colon \mathrm{const}_c \Rightarrow X$ and $D\colon \mathrm{const}_d \Rightarrow X$ are two cones over X a morphism of cones is a natural transformation $\mathrm{const}_c \Rightarrow \mathrm{const}_d$ making the following diagram commute

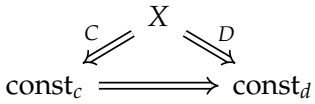


Therefore for any diagram $X : \mathsf{J} \to \mathsf{C}$ we have a category of cones $\Delta \downarrow X$.

Definition 3.1.3 (Cocone). A **cocone** C over a diagram $X\colon \mathsf{J} \to \mathsf{C}$ with *nadir* c is a natural transformation $C\colon X \Rightarrow \mathrm{const}_c$ from the constant diagram to X. The components $\{C_j\colon X_j \to c\}_{j\in\mathsf{J}}$ are called the *legs* of the cone.
Less abstractly, a cocone C over $X\colon \mathsf{J} \to \mathsf{C}$ with *nadir* c is a collection of morphisms $C_j\colon X_j \to c$ for every object $j \in \mathsf{J}$ such that for every morphism $f\colon j \to k$ in J the following diagram in C commutes

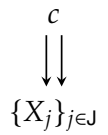


If $C\colon X \Rightarrow \mathrm{const}_c$ and $D\colon X \Rightarrow \mathrm{const}_d$ are two cocones over X a morphism of cocones is a natural transformation $\mathrm{const}_c \Rightarrow \mathrm{const}_d$ making the following diagram commute

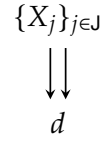


Therefore for any diagram $X\colon \mathsf{J} \to \mathsf{C}$ we have a category of cocones $X \downarrow \Delta$.

Notation. For a diagram $X\colon \mathsf{J} \to \mathsf{C}$ we will denote a cone $C\colon \mathrm{const}_c \Rightarrow X$ by



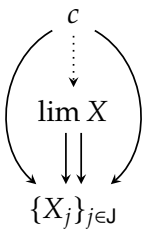
and a cocone $D\colon X \Rightarrow \mathrm{const}_d$ by



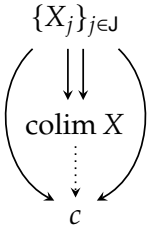
3.2 Limits and colimits

Limits and colimits unify diverse geometric and algebraic constructions by identifying them as the "best possible" approximations of a diagram, formalizing their universal properties. By viewing them as terminal or initial objects in a category of cones or cocones, we reduce the collective behavior of an entire system of objects and arrows down to a single, uniquely determined universal object.

Definition 3.2.1 (Limit). A **limit** of a diagram $X\colon \mathsf{J} \to \mathsf{C}$ is a terminal object in the category of cones $\Delta \downarrow X$.
Less abstractly, a limit is a cone $\lim X\colon \operatorname{const}_{\lim X} \Rightarrow X$ such that for any other cone $C\colon \operatorname{const}_c \Rightarrow X$ there exists a unique morphism $c \to \lim X$ making the following diagram commute



Definition 3.2.2 (Colimit). A **colimit** of a diagram $X\colon \mathsf{J} \to \mathsf{C}$, denoted $\operatorname{colim} X$, is an initial object in the category of cocones $X \downarrow \Delta$.
Less abstractly, a colimit is a cocone $\operatorname{colim} X\colon X \Rightarrow \operatorname{const}_{\operatorname{colim} X}$ such that for any other cocone $C\colon X \Rightarrow \operatorname{const}_c$ there exists a unique morphism $\operatorname{colim} X \to c$ making the following diagram commute



Remark 3.2.1. Limits and colimits do not need to exist in general for all diagrams and categories.
Categories where all limits and colimits exist are called complete and cocomplete categories respectively. Set is an example of both a complete and cocomplete category, and so are all of the categories we are going to study.
For more information we refer to [Rie, Section 3.6].

As a direct consequence of their definition as initial or initial by Proposition 1.3.1 we have that these objects are unique up to categorical isomorphism.

Proposition 3.2.1 (Uniqueness of limits and colimits). If a diagram $X\colon \mathsf{J} \to \mathsf{C}$ has a limit/colimit, it is unique up to unique isomorphism.

Before going into the examples, we fix some terminology around the interaction of functors with limits and colimits.

Definition 3.2.3 (Functors and limits). Let $F\colon \mathsf{C} \to \mathsf{D}$ be a functor and $X\colon \mathsf{J} \to \mathsf{C}$ a diagram. We say that

- F **preserves** limits if whenever (L, φ) is the limit of X in C, $(F(L), F(\varphi))$ is the limit of $F \circ X$ in D.
- F **reflects** limits if whenever (L, φ) is a cone over X such that $(F(L), F(\varphi))$ is the limit of $F \circ X$ in D then (L, φ) is the limit of X in C.

Replacing cones with cocones and limits with colimits we get dual statements.

3.2.1 Examples of limits and colimits

We now define some special cases of limits and colimits that are generally useful. However, it is important to take into account that this constructions do not exist in general for all categories.

Products and coproducts

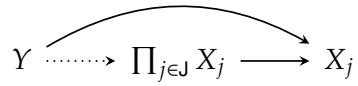
Let J be a small and discrete category.

Since the only morphisms in J are the identity morphisms, a diagram $X\colon \mathsf{J} \to \mathsf{C}$ consists in a collection of objects $\{X_j\}_{j\in\mathsf{J}}$.

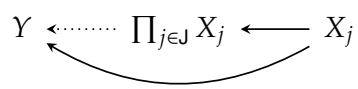
Therefore a cone with apex c over X is an object $c \in \mathsf{C}$ together with morphisms $c \to X_j$ for each $j \in \mathsf{J}$ and a cocone with nadir $c \in \mathsf{C}$ over X is an object c together with morphisms $X_j \to c$ for each $j \in \mathsf{J}$.

Note that any collection of objects $\{X_j\}_{j\in\mathsf{J}}$ in C may be considered as a diagram $X\colon \mathsf{J} \to \mathsf{C}$.

Definition 3.2.4 (Product). Let $\{X_j\}_{j\in\mathsf{J}}$ be a collection of objects in a category C.
We define the **product** of $\{X_j\}_{j\in\mathsf{J}}$, denoted $\prod_{j\in\mathsf{J}} X_j$, as the limit of the diagram $X\colon \mathsf{J} \to \mathsf{C}$.
Less abstractly, a **product** is an object $\prod_{j\in\mathsf{J}} X_j \in \mathsf{C}$ together with maps $\pi_k\colon \prod_{j\in\mathsf{J}} X_j \to X_k$ for every $j \in \mathsf{J}$ such that, for any map $Y \to X_j$ there exists a unique morphism $Y \to \prod_{j\in\mathsf{J}} X_j$ making the following diagram commute



Definition 3.2.5 (Coproduct). Let $\{X_j\}_{j\in\mathsf{J}}$ be a collection of objects in a category C.
We define the **coproduct** of $\{X_j\}_{j\in\mathsf{J}}$, denoted $\coprod_{j\in\mathsf{J}} X_j$, as the colimit of the diagram $X\colon \mathsf{J} \to \mathsf{C}$.
Less abstractly, a **coproduct** is an object $\coprod_{j\in\mathsf{J}} X_j \in \mathsf{C}$ together with maps $\iota_k\colon X_k \to \coprod_{j\in\mathsf{J}} X_j$ for every $j \in \mathsf{J}$ such that, for any map $X_j \to Y$ there exists a unique morphism $\coprod_{j\in\mathsf{J}} X_j \to Y$ making the following diagram commute



Remark 3.2.2. Consider the case where $\mathsf{J} = \varnothing$ is the empty category, i.e: $\mathrm{Ob}(\mathsf{J}) = \varnothing$. Suppose that products exist in C.

Let's see how are ∅-shaped products (empty products).
Note that the family of objecs $\{X_j\}_{j\in\varnothing} \subseteq \mathsf{C}$ is empty, and so there are no projection maps $\pi_k : \prod_{j\in\varnothing} X_j \to X_k$ nor morphisms $\{f_j : Y \to X_j\}_{j\in\varnothing}$ for all $Y \in \mathsf{C}$. Therefore the product condition is vacuosly true, i.e: there exists a unique map $Y \to \prod_{j\in\varnothing} X_j$ for all $Y \in \mathsf{C}$. This means that $\prod_{j\in\varnothing} X_j$ is a terminal object in C.
Dually, if C has coproducts then the empty coproduct is initial in C.

Equalisers and coequalisers

Consider the *parallel pair* category

$$\mathsf{J} = \quad \bullet \rightrightarrows \bullet$$

A diagram $X\colon \mathsf{J} \to \mathsf{C}$ corresponds to a pair of morphisms $f, g\colon A \rightrightarrows B$ in C.

Therefore a cone $C\colon \mathrm{const}_c \Rightarrow X$ is a morphism $c\colon C \to A$ such that $fc = gc$:

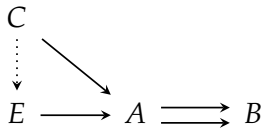
$$C \xrightarrow{c} A \underset{g}{\overset{f}{\rightrightarrows}} B$$

Analogously a cocone $D\colon X \Rightarrow \mathrm{const}_d$ is a morphism $d\colon B \to D$ such that $df = dg$:

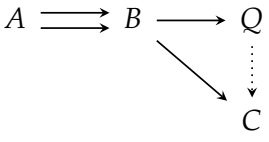
$$A \underset{g}{\overset{f}{\rightrightarrows}} B \xrightarrow{d} D$$

Note that any pair of morphisms $f, g\colon A \rightrightarrows B$ can be considered as a diagram $X\colon \mathsf{J} \to \mathsf{C}$.

Definition 3.2.6 (Equaliser). Let $f, g\colon A \rightrightarrows B$ be two morphisms in C, considered as a diagram $X\colon \mathsf{J} \to \mathsf{C}$. The **equaliser** of f and g is the limit of this diagram.
Less abstractly, it is an object $E \in \mathsf{C}$ together with a morphism $\mathrm{eq}\colon E \to A$ such that for any morphism $c\colon C \to A$ with $fc = gc$ there exists a unique map $C \to E$ the following diagram commute



Definition 3.2.7 (Coequaliser). Let $f, g\colon A \rightrightarrows B$ be two morphisms in C, considered as a diagram $X\colon \mathsf{J} \to \mathsf{C}$. The **coequaliser** of f and g is the limit of this diagram.
Less abstractly, it is an object $Q \in \mathsf{C}$ together with a morphism $\mathrm{coeq}\colon B \to Q$ such that for any morphism $d\colon B \to C$ with $df = dg$ there exists a unique map $Q \to C$ making the following diagram commute



Proposition 3.2.2. Equaliser arrows are monomorphisms. Dually, coequaliser arrows are epimorphisms.

Proof. Let $\mathrm{eq}\colon E \to X$ be a equaliser. Consider two morphisms $f, g\colon C \rightrightarrows E$ in C such that $\mathrm{eq} \circ f = \mathrm{eq} \circ g$. By the uniqueness of the map $\mathrm{eq}\colon E \to X$ we have $f = g$. □

Example 3.2.1 (Equalisers and coequalisers in Set)**.** In Set the equaliser of two maps $f, g\colon A \rightrightarrows B$ is $E = \{a \in A \mid f(a) = g(a)\}$ together with the inclussion $E \hookrightarrow A$. The coequaliser of f and g is $Q = {}^{B}\!/_{\sim}$ where $\sim$ is the smallest equivalence relation such that for all $a \in A \quad f(a) \sim g(a)$ together with the natural projection

$$\begin{aligned} \mathrm{coeq}\colon B &\twoheadrightarrow Q \\ b &\mapsto [b] \end{aligned}$$

A special case of equalisers and coequalisers are kernels and cokernels.

Definition 3.2.8 (Kernel)**.** Let C be a category with a zero object $0 \in \mathsf{C}$. The **kernel** $\ker f$ of a morphism $f\colon A \to B$ is the equaliser

$$\ker f \rightarrowtail A \underset{0}{\overset{f}{\rightrightarrows}} B$$

Less abstractly, it is an object $\ker f$ together with a morphism $i\colon \ker f \to X$ such that for every $C \in \mathsf{C}$ and every morphism $g\colon C \to X$ such that $f \circ g = 0$ there is a unique morphism $\phi\colon C \to \ker f$ such that $g = p \circ \phi$

Definition 3.2.9 (Cokernel)**.** Let C be a category with a zero object $0 \in \mathsf{C}$. The **cokernel** $\operatorname{coker} f$ of a morphism $f\colon A \to B$ is the coequaliser

$$A \underset{0}{\overset{f}{\rightrightarrows}} B \twoheadrightarrow \operatorname{coker} f$$

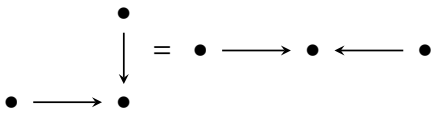
Less abstractly, it is an object $\operatorname{coker} f$ together with a morphism $p\colon Y \to \operatorname{coker} f$ such that for every $C \in \mathsf{C}$ and every morphism $g\colon Y \to C$ such that $g \circ f = 0$ there is a unique morphism $\phi\colon \operatorname{coker} f \to C$ such that $g = \phi \circ p$.
We denote

$${}^{A}\!/_{B} := \operatorname{coker}(A \to B)$$

Remark 3.2.3. By Proposition 3.2.2, kernels are monomorphisms and cokernels are epimorphisms.

Pullbacks and pushouts

Consider the category

$\mathsf{J} =$ 

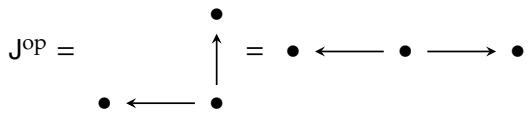
A diagram $X\colon \mathsf{J} \to \mathsf{C}$ consists of a pair of morphisms $A \xrightarrow{f} Z \xleftarrow{g} B$.

Therefore a cone $\mathrm{const}_C \Rightarrow X$ consists of an object $C \in \mathsf{C}$ together with morphisms $C \to X$ and $C \to Y$ making the following diagram commute.

$$\begin{array}{ccc} C & \longrightarrow & A \\ \downarrow & & \downarrow{\scriptstyle f} \\ B & \underset{g}{\longrightarrow} & Z \end{array}$$

Note that any pair of morphisms $A \xrightarrow{f} Z \xleftarrow{g} B$ can be considered as a diagram $X\colon \mathsf{J} \to \mathsf{C}$.

Consider now the opposite category



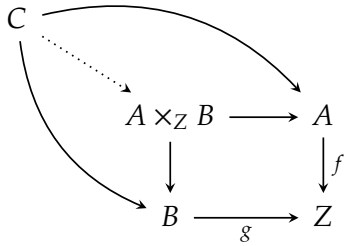
A diagram $Y\colon \mathsf{J}^{\mathrm{op}} \to \mathsf{C}$ consists of a pair of morphisms $A \xleftarrow{f} Z \xrightarrow{g} B$.

Therefore a cocone $X \Rightarrow \mathrm{const}_D$ consists of an object $D \in \mathsf{D}$ together with morphisms $A \to D$ and $B \to D$ making the following diagram commute

$$\begin{array}{ccc} Z & \xrightarrow{f} & A \\ {\scriptstyle g}\downarrow & & \downarrow \\ B & \longrightarrow & D \end{array}$$

Note that any pair of morphisms $A \xleftarrow{f} Z \xrightarrow{g} B$ can be considered as a diagram $X\colon \mathsf{J}^{\mathrm{op}} \to \mathsf{C}$.

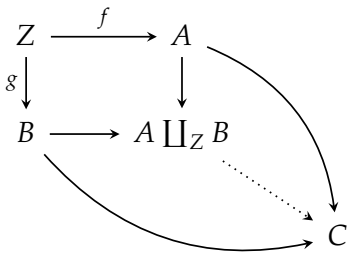
Definition 3.2.10 (Pullback). Let $A \xrightarrow{f} Z \xleftarrow{g} B$ be a pair of morphisms in C, considered as a diagram $X\colon \mathsf{J} \to \mathsf{C}$.
The **pullback** (or **fibre product**) of f and g is the limit of this diagram, denoted by $A \times_Z B$.
Less abstractly, it is an object $A \times_Z B \in \mathsf{C}$ together with a pair of morphisms $\pi_A\colon A \times_Z B \to A, \pi_B : A \times_Z B \to B$ such that for any pair of morphisms $C \to A, C \to B$ making the square commute, there exists a unique morphism $C \to A \times_Z B$ such that



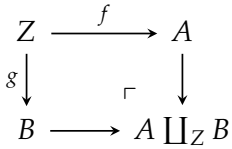
Pullback diagrams are often denoted by

$$\begin{array}{ccc} A \times_Z B & \longrightarrow & A \\ \downarrow & \ulcorner & \downarrow{\scriptstyle f} \\ B & \xrightarrow[g]{} & Z \end{array}$$

Definition 3.2.11 (Pushout). Let $A \xleftarrow{f} Z \xrightarrow{g} B$ be a pair of morphisms in C, considered as a diagram $X\colon \mathsf{J}^{\mathrm{op}} \to \mathsf{C}$.
The **pushout** of f and g is the colimit of this diagram, denoted by $X \coprod_Z Y$.
Less abstractly, it is an object $A \coprod_Z B \in \mathsf{C}$ together with a pair of morphisms $\iota_A\colon A \to A \coprod_Z B, \iota_B : B \to A \coprod_Z B$ such that for any pair of morphisms $A \to C, B \to C$ making the square commute, there exists a unique morphism $A \coprod_Z B \to C$ such that



Pushout diagrams are often denoted by



Example 3.2.2 (Pullbacks and pushouts in Set)**.** In Set the pullback of two morphisms $A \xrightarrow{f} Z \xleftarrow{g} B$ is the set

$$A \times_Z B = \{(a, b) \in A \times B \mid f(a) = g(b)\}$$

together with the projection maps

$$\begin{aligned} \pi_A : A \times_Z B &\to A \\ (a, b) &\mapsto a \end{aligned} \qquad \begin{aligned} \pi_B : A \times_Z B &\to B \\ (a, b) &\mapsto b \end{aligned}$$

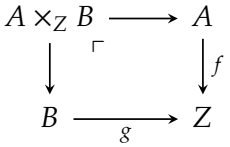
The pushout of two morphisms $A \xleftarrow{f} Z \xrightarrow{g} B$ is the set

$$A \coprod_Z B = {}^{A \amalg B}\!/_{\sim}$$

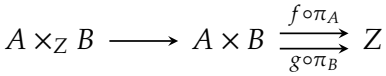
where $\sim$ is the equivalence relation generated by $f(a) \sim g(a)$ for all $a \in A$ together the inclussion maps

$$\begin{aligned} \iota_A : A &\to A \coprod_Z B \\ a &\mapsto [a, 0] \end{aligned} \qquad \begin{aligned} \iota_B : B &\to A \coprod_Z B \\ b &\mapsto [b, 1] \end{aligned}$$

Remark 3.2.4. Notice how pullbacks in Set can be seen as a equaliser in Set from the product maps $f \circ \pi_A, g \circ \pi_B : A \times B \to Z$. Considering pushouts in Set we get a dual observation.
Indeed, it is not difficult to check that if C has products then the pullback



is equivalent to the equaliser



Dually if C has coproducts the pushout

$$\begin{array}{ccc} Z & \xrightarrow{f} & A \\ {\scriptstyle g}\downarrow & \ulcorner & \downarrow \\ B & \longrightarrow & A \coprod_Z B \end{array}$$

is equivalent to the coequaliser

$$Z \underset{\iota_B \circ g}{\overset{\iota_A \circ f}{\rightrightarrows}} A \coprod B \longrightarrow A \coprod_Z B$$

For a proof see [Rie, Lemmas 3.4.16, 3.4.17].

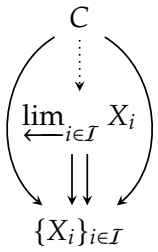
Inverse and direct limits

Definition 3.2.12 (Directed set)**.** A **directed set** is a preordered set $(\mathcal{I}, \leq)$ such that for every $i, j \in \mathcal{I}$, there exists $k \in \mathcal{I}$ with $i \leq k$ and $j \leq k$. We can view this as a category following Example 1.1.5.

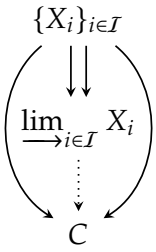
Example 3.2.3. Let $(X, \mathcal{T})$ be a topological space. Then $(\mathcal{T}, \subseteq)$ forms a directed set. This means that the category of open sets $\mathsf{Open}(X)$ can be seen as a directed set.

Definition 3.2.13 (Direct and inverse systems)**.** A **direct system** in C over $\mathcal{I}$ is a diagram $X \colon \mathcal{I} \to \mathsf{C}$. Less abstractly, it consists of a family of objects $\{X_i\}_{i \in \mathcal{I}}$ and morphisms $f_{ij} \colon X_i \to X_j$ for all $i \leq j$, such that $f_{ii} = \mathrm{id}_{X_i}$ and $f_{jk} \circ f_{ij} = f_{ik}$ for all $i \leq j \leq k$. Dually, an **inverse system** in C over $\mathcal{I}$ is a diagram $X \colon \mathcal{I}^{\mathrm{op}} \to \mathsf{C}$. Less abstractly, it consists of a family of objects $\{X_i\}_{i \in \mathcal{I}}$ and morphisms $p_{ij} \colon X_j \to X_i$ for all $i \leq j$, such that $p_{ii} = \mathrm{id}_{X_i}$ and $p_{ij} \circ p_{jk} = p_{ik}$ for all $i \leq j \leq k$.

Definition 3.2.14 (Inverse limit)**.** The **inverse limit** (or **projective limit**) of an inverse system $X \colon \mathcal{I} \to \mathsf{C}$ is the limit of this diagram, denoted as $\varprojlim_{i \in \mathcal{I}} X_i$.
Less abstractly, it is an object $\varprojlim_{i \in \mathcal{I}} X_i \in \mathsf{C}$ together with morphisms $\pi_i \colon \varprojlim_{i \in \mathcal{I}} X_i \to X_i$ such that $p_{ij} \circ \pi_j = \pi_i$ for all $i \leq j$. For any other object C and morphisms $c_i \colon C \to X_i$ compatible with the system $(p_{ij} \circ c_j = c_i)$, there exists a unique morphism $C \to \varprojlim_{i \in \mathcal{I}} X_i$ making the following diagram commute:



Definition 3.2.15 (Direct limit)**.** The **direct limit** (or **inductive limit**) of a direct system $X \colon \mathcal{I} \to \mathsf{C}$ is the colimit of this diagram, denoted as $\varinjlim_{i \in \mathcal{I}} X_i$.
Less abstractly, it is an object $\varinjlim_{i \in \mathcal{I}} X_i \in \mathsf{C}$ together with morphisms $\iota_i \colon X_i \to \varinjlim_{i \in \mathcal{I}} X_i$ such that $\iota_j \circ f_{ij} = \iota_i$ for all $i \leq j$. For any other object C and morphisms $c_i \colon X_i \to C$ compatible with the system $(c_j \circ f_{ij} = c_i)$, there exists a unique morphism $\varinjlim_{i \in \mathcal{I}} X_i \to C$ making the following diagram commute:



Example 3.2.4 (Inverse limits in Set). In Set, if $(\{X_i\}_{i\in\mathcal{I}}, \{p_{ij}\}_{i\leq j})$ is an inverse system indexed by a directed set $(\mathcal{I}, \leq)$, its inverse limit is

$$\varprojlim_{i\in\mathcal{I}} X_i = \left\{ (x_i)_{i\in\mathcal{I}} \in \prod_{i\in\mathcal{I}} X_i \mid \forall i \leq j \quad x_i = p_{ij}(x_j) \right\}$$

together with the natural projection maps $\pi_k((x_i)_{i\in\mathcal{I}}) = x_k$.

Example 3.2.5 (Direct limits in Set). In Set, if $(\{X_i\}_{i\in\mathcal{I}}, \{f_{ij}\}_{i\leq j})$ is a direct system indexed by a directed set $(\mathcal{I}, \leq)$, its direct limit is

$$\varinjlim_{i\in\mathcal{I}} X_i = \coprod_{i\in\mathcal{I}} X_i \Big/ \sim$$

where $\sim$ is the equivalence relation defined by $x_i \sim x_j$ if and only if there exists $k \in \mathcal{I}$ with $k \geq i, j$ such that

$$f_{ik}(x_i) = f_{jk}(x_j)$$

together with the natural insertion maps $\iota_k(x_k) = [x_k]$.

3.3 Existence of limits

We finish this chapter by giving a condition for the existence of finite limits and colimits

Theorem 3.3.1. A category has all finite limits if and only if it has a terminal object, binary products and equalisers.
Dually it has all finite colimits if and only if it has a initial object, binary coproducts and coequalisers.

Proof.

$\Longrightarrow$ Direct since all of these are particular cases of limits.

$\Longleftarrow$ Consider a diagram $X : \mathsf{J} \Rightarrow \mathsf{C}$.
Since binary products exist in C we can construct inductively

$$P := \prod_{j\in\mathsf{J}} X_j \in \mathsf{C}$$

For every morphism $u : j \to k$ in J we can consider two maps from P to X_k: the projection $\pi_k : P \to X_k$ and the composition $Xu \circ \pi_j : P \to X_j \to X_k$. Therefore we can consider a pair of parallel morphisms

$$\prod_{j\in\mathsf{J}} X_j \underset{t}{\overset{s}{\rightrightarrows}} \prod_{u:j\to k\in\mathsf{J}} X_k$$

where s is defined in each component as π_k and t is $Xu \circ \pi_j$ in each component. The equaliser of s and t (which exists by hypothesis) is then the limit of X. Indeed, let $\mathrm{eq}\colon L \to \prod_{j\in \mathsf{J}} X_j$ be this equaliser. This comes with a set of morphisms $\{\phi_j\colon L \to X_j\}_{j\in\mathsf{J}}$ such that

$$\phi_j = \pi_j \circ \mathrm{eq}$$

By definition $s \circ \mathrm{eq} = t \circ \mathrm{eq}$, so for every $u\colon j \to k$ in J we have

$$\pi_k \circ \mathrm{eq} = (Xu \circ \pi_j) \circ \mathrm{eq}$$

Therefore

$$\phi_k = Xu \circ \phi_j$$

and so L is a cone over X.
To see that it is a limit cone consider another cone $(W, \{\psi_j\colon W \to X_j\})$ such that for any $u\colon j \to k$ we have $\psi_k = Xu \circ \psi_j$. By definition of the product there exists a unique morphism $h\colon W \to \prod_{j\in\mathsf{J}} W_j$ such that $\pi_j \circ h = \psi_j$.
Note that $s \circ h = t \circ h$. Indeed the components of $s \circ h$ are $\pi_k \circ h = \psi_k$ and the components of $t \circ h$ are $Xu \circ \pi_j \circ h = Xu \circ \psi_j = \psi_k$.
Therefore by the definition of equaliser there exists a unique morphism $\widetilde{h}\colon W \to L$ such that $\mathrm{eq} \circ \widetilde{h} = h$. Finally since for any $j \in \mathsf{J}$

$$\phi_j \circ \widetilde{h} = \pi_j \circ \mathrm{eq} \circ \widetilde{h} = \pi_j \circ h = \psi_j$$

we have that W factors uniquely through L and so L is the limit of X. □

In the particular case of functor categories we have a stronger result: these categories has limits (resp. colimits) if the target category has them.

Proposition 3.3.1. Let C and D be two categories and $\mathsf{Fct}(\mathsf{C},\mathsf{D})$ be its functor category. If D has limits (resp. colimits) of shape J then $\mathsf{Fct}(\mathsf{C},\mathsf{D})$ has limits (resp. colimits) of shape J. Furthermore, they are computed objectwise.

Proof. Consider a J-shaped diagram $X\colon \mathsf{J} \Rightarrow [\mathsf{C},\mathsf{D}]$. We wish to construct a limit object $L \in \mathsf{Fct}(\mathsf{C},\mathsf{D})$ together with a limit cone $\pi_j\colon L \to X_j$ for all $j \in \mathsf{J}$.

For any object $c \in \mathsf{C}$ the *evaluation* functor

$$\begin{array}{ccc} \mathrm{ev}_c\colon [\mathsf{C},\mathsf{D}] & \longrightarrow & \mathsf{D} \\ F & & F(c) \\ \eta \Downarrow & \longmapsto & \downarrow \eta_c \\ G & & G(c) \end{array}$$

allows us to define a diagram $X^c := \mathrm{ev}_c \circ X : J \Rightarrow \mathsf{D}$. By hypothesis there exists

$$\lim_{j\in\mathsf{J}} X^c(j) =: L(c) \qquad \pi^c_j\colon L(c) \to X^c(j)$$

To make L into a functor we need to define its action on morphisms. Consider a morphism $f\colon c \to d$ in C. Since X_j is a functor for all $j \in \mathsf{J}$ we have an induced morphism $X_j f\colon X_j(c) \to X_j(d)$. Let's consider the family of morphism

$$\left\{X_j f \circ \pi^c_j\colon L(c) \to X_j(d)\right\}_{j\in\mathsf{J}}$$

We want to show that this forms a cone over $L(d)$ with apex $L(c)$

Let $u : j \to k$ be a morphism in J. Since $Xu : X_j \Rightarrow X_k$ is a natural transformation we have a commutative diagram

$$\begin{array}{ccc} X_j(c) & \xrightarrow{(Xu)_c} & X_k(x) \\ {\scriptstyle X_j f}\downarrow & & \downarrow{\scriptstyle X_k f} \\ X_j(d) & \xrightarrow[(Xu)_d]{} & X_k(d) \end{array}$$

Since π^c is a limit over X^c we also now that $(Xu)_c \circ \pi^c_j = \pi^c_k$. Therefore we have

$$(Xu)_d \circ (X_j f \circ \pi^c_j) = ((Xu)_d \circ X_j f) \circ \pi^c_j = X_k f \circ (Xu)_c \circ \pi^c_j = X_k f \circ \pi^c_k$$

i.e: it is a valid cone over $L(d)$. By the universal property of $L(d) = \lim_{j \in \mathsf{J}} X^d(j)$ there exists a unique morphism $Lf : L(c) \to L(d)$ such that for all $j \in \mathsf{J}$ the following diagram commutes

$$\begin{array}{ccc} L(c) & \xrightarrow{Lf} & L(d) \\ {\scriptstyle \pi^c_j}\downarrow & & \downarrow{\scriptstyle \pi^d_j} \\ X^c(j) & \xrightarrow[X_j f]{} & X^d(j) \end{array}$$

The functiorality of L follows directly from the uniqueness of this induced map into $L(d)$. Moreover the above commutative square is the naturallity condition for π_j with respect to f. Therefore, for each $j \in \mathsf{J}$ the collection $\{\pi^c_j\}_{c \in \mathsf{C}}$ is a natural transformation $\pi_j : L \Rightarrow X_j$. It is an easy check to see that this collection forms a cone over X in the functor category $\mathsf{Fct}(\mathsf{C}, \mathsf{D})$.

To finish the proof, we need to check the universal property. Consider another functor $M \in \mathsf{Fct}(\mathsf{C}, \mathsf{D})$ equipped with a cone $\tau_j : M \Rightarrow X_j$.

For each $c \in \mathsf{C}$ the evaluation $\tau^c_j := \mathrm{ev}_c \circ M : M(c) \to X^c(j)$ is a cone over X^c. This induces a unique morphism $\phi_c : M(c) \to L(c)$ such that $\pi^c_j \circ \phi_c = \tau^c_j$.

To show that $\phi \equiv \{\phi_c\}_{c \in \mathsf{C}}$ is a natural transformation let $f : c \to d$ be a morphism in C. We want to see that the following diagram commutes

$$\begin{array}{ccc} M(c) & \xrightarrow{\phi_c} & L(c) \\ {\scriptstyle Mf}\downarrow & & \downarrow{\scriptstyle Lf} \\ M(d) & \xrightarrow[\phi_d]{} & L(d) \end{array}$$

Postcompossing with π^d_j we get, on the one hand

$$\pi^d_j \circ (Lf \circ \phi_c) = (\pi^d_j \circ Lf) \circ \phi_c = X_j f \circ \pi^c_j \circ \phi_c = X_j f \circ \tau^c_j$$

And on the other

$$\pi^d_j \circ (\phi_d \circ Mf) = (\pi^d_j \circ \phi_d) \circ Mf = \tau^d_j \circ Mf$$

Since τ_j is a natural transformation we have

$$X_j f \circ \tau_j^c = \tau_j^d \circ M f$$

Therefore since the projections agree for all $j \in \mathsf{J}$ by uniqueness of the maps π_j we get the commutativity of the diagram. Its uniqueness as a morphism of cones follows directly from the objectwise uniqueness of each ϕ_c. □

Chapter 4

More on functors

In this chapter we introduce one of the most fundamental results in category theory: the Yoneda lemma. We also define what an adjoint pair of functions is and state its relation with limits and colimits.

4.1 The Yoneda lemma

A common theme in these part of the thesis has been stressing that we want to study objects not by themselves but through its relation with their environment. In particular we can apply these way of thought to categories themselves using functors, and to functors using natural transformations.

Note that every category C has a functor of the form $\mathsf{C} \longrightarrow \mathsf{Set}$: the (covariant) Hom functor that works by fixing an object $X \in category C$ and sending $Y \in \mathsf{C}$ to its set of morphisms $\mathrm{Hom}(X, Y) \in \mathsf{Set}$.

Therefore we could try to study how a functor $F\colon \mathsf{C} \longrightarrow \mathsf{Set}$ works on an object $X \in \mathsf{C}$ by studying its relation to $\mathrm{Hom}(X, -)$. This was done by Yoneda and Grothendieck independently and it's what is known as the Yoneda lemma.

Instead of giving its proof, we will give an intuitive explanaition the philosophy behind it. But first, here is the statement.

Theorem 4.1.1 (Yoneda - [Rie, Theorem 2.2.4]). Let C be a category, $X \in \mathsf{C}$ an object, and $F\colon \mathsf{C} \longrightarrow \mathsf{Set}$ a functor. There is a bijection:

$$\Phi : \mathrm{Hom}(\mathrm{Hom}(X, -), F) \xrightarrow{\sim} F(X)$$

given by $\alpha \mapsto \alpha_X(\mathbb{1}_X)$. This bijection is natural in both X and F.

Let's exemplify how this works using the following situation: suppose you are playing a game with a friend that consists on guessing a card he is thinking from a poker deck. Since randomly guessing is boring, you can ask any yes or no question about the card to guide your response. Therefore you could start asking "is the card red?" or "is the number bigger than 5?" until you are only left with one possible option. However, what if you were prohibitted from asking direct questions about the card and instead only could ask about its relation with other cards? In this case there is also a winning strategy, for example instead of asking if the card is red you

may ask "is the card color the same as the king of hearts?". Therefore a card is completely determined by its relation to the other cards.

This is the philosophy behind Yonneda's lemma: objects are completely determined by its environment.

4.2 Adjunctions

Throughout mathematics several constructions appear that are "approximate inverses" of each other. Probably the most recurrent is that of "free" objects constructed from a base space: for example the free group generated by a set is the group consisting of all of the words formed with the elements of the set and their formal inverses. Another example is that of giving a topology to a set by considering the discrete topology. These are both examples of a general phenomenon called adjunction.

Definition 4.2.1 (Adjoint functors). Let $L\colon \mathsf{C} \to \mathsf{D}$ and $R\colon \mathsf{D} \to \mathsf{C}$ be a pair of functors. We say that (L, R) is a pair of **adjoint functors** or that L is left-adjoint to R and R is right-adjoint to L if there exists a natural isomorphism of bifunctors

$$\mathrm{Hom}_{\mathsf{D}}(L(-), -) \cong \mathrm{Hom}_{\mathsf{C}}(-, R(-)) \tag{4.2.1}$$

We represent this diagramatically using

$$\mathsf{C} \underset{R}{\overset{L}{\rightleftarrows}} \mathsf{D} \quad (\perp)$$

Remark 4.2.1 (Unit and counit of an adjunction). Let (L, R) be a pair of adjoint functors. The natural isomorphism (4.2.1) gives the natural isomorphisms

$$\mathrm{Hom}_{\mathsf{D}}(L(-), L(-)) \cong \mathrm{Hom}_{\mathsf{C}}(-, (G \circ L)(-))$$
$$\mathrm{Hom}_{\mathsf{D}}((L \circ R)(-), -) \cong \mathrm{Hom}_{\mathsf{C}}(R(-), R(-))$$

In particular we get natural transformations

$$\eta\colon \mathbb{1}_{\mathsf{C}} \Rightarrow R \circ L \qquad \varepsilon\colon L \circ R \Rightarrow \mathbb{1}_{\mathsf{D}}$$

that satisfy the *triangular identities*

$$\begin{array}{ccc} L & \xRightarrow{L\eta} & L \circ R \circ L \\ & \searrow^{\mathbb{1}_F} & \Big\downarrow \varepsilon F \\ & & L \end{array} \qquad \begin{array}{ccc} R & \xRightarrow{\eta R} & R \circ L \circ R \\ & \searrow^{\mathbb{1}_R} & \Big\downarrow F\varepsilon \\ & & R \end{array}$$

Conversely, if L, R are two pair of functors and ε, η are natural transformations as above then (L, R) form an adjoint pair. For a proof of this see [Rie, Proposition 4.2.6].

Remark 4.2.2. Left and right adjoints are dual concepts.

Just as we can speak of *the* initial and terminal object in a category we can speak of *the* adjoint of a functor, this type up to natural isomorphism.

Proposition 4.2.1 (Adjoints are unique). The right adjoint (resp. left adjoint) to a functor, provided it exists, is unique up to natural isomorphism

Proof. Let $L\colon \mathsf{C} \to \mathsf{D}$ be a functor.

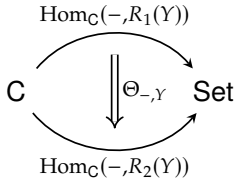
Let $R_1, R_2\colon \mathsf{D} \to \mathsf{C}$ be two functors which are right adjoint to L. This means that for all $X \in \mathsf{C}$ and $Y \in \mathsf{D}$ we have bijections

$$\phi_{X,Y}\colon \operatorname{Hom}_{\mathsf{D}}(L(X), Y) \xrightarrow{\sim} \operatorname{Hom}_{\mathsf{C}}(X, R_1(Y))$$
$$\psi_{X,Y}\colon \operatorname{Hom}_{\mathsf{D}}(L(X), Y) \xrightarrow{\sim} \operatorname{Hom}_{\mathsf{C}}(X, R_2(Y))$$

Using this we can define a bijection

$$\Theta_{X,Y} = \psi_{X,Y} \circ \phi_{X,Y}^{-1}\colon \operatorname{Hom}_{\mathsf{C}}(X, R_1(Y)) \xrightarrow{\sim} \operatorname{Hom}_{\mathsf{C}}(X, R_2(Y))$$

Since ψ and ϕ are natural in X so is Θ. Therefore for a fixed $Y \in \mathsf{D}$ we have a natural isomorphism



By the Yoneda lemma $\Theta_{-,Y}$ induces a unique isomorphism in C

$$\alpha_Y\colon R_1(Y) \xrightarrow{\sim} R_2(Y)$$

which ensemble into a natural isomorphism

$$\alpha\colon R_1 \Rightarrow R_2$$ □

We end the chapter by stating relation between adjoint functors and (co)limits that will prove really useful later on.

Theorem 4.2.1 ([Rie, Theorem 4.5.2]). Right adjoints preserve all limits that exist in their codomain.
Dually left adjoints preserve all colimits that exist in their domain.

PART II

Homological algebra

Introduction

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Chapter 5

Additive categories

In the previous chapter, we established the abstract language of categories, emphasizing that the structure of a mathematical ecosystem is governed by its morphisms rather than its internal elements. We now turn our attention to categories where the collections of morphisms possess an algebraic structure of their own. In many familiar environments, such as the category of abelian groups or modules over a ring, parallel morphisms can be added together and subtracted. Formalizing this observation leads us to the concept of an additive category.

The defining trait of an additive category is that its hom-sets are not merely sets; they are abelian groups, and this algebraic framework plays nicely with morphism composition

Definition 5.0.1 (Additive category)**.** A category C is **additive** if it satisfies

Ad.1) For every pair of objects $X, Y \in \mathsf{C}$ the set $\mathrm{Hom}_{\mathsf{C}}(X, Y)$ is endowed with an abelian group structure and the composition maps are bilinear with respect to the group operation.
Ad.2) There exists a zero object $0 \in \mathsf{C}$.
Ad.3) C admits finite coproducts.
Ad.4) C admits finite products.

Example 5.0.1.

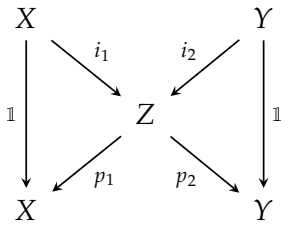
- The category of abelian groups Ab is additive. More generally, for any ring R the category of R-modules $\mathsf{Mod}(R)$ is additive.
- The category of groups Grp is not additive, since given two group homomorphisms f, g its sum is not necessarilly a homomorphism.

Notation. If $X, Y \in \mathsf{C}$ we denote by $X \oplus Y$ its coproduct and call it their **direct sum**.

In an arbitrary category, products and coproducts are completely independent structures. However, the presence of an underlying abelian group structure on hom-sets forces these two notions to seamlessly collapse into one another.

Lemma 5.0.1. Let C be a category satisfying Ad.1 and Ad.2.
Suppose that C verifies

Ad.5) For any pair of objects $X, Y \in \mathsf{C}$ there exists $Z \in \mathsf{C}$ and morphisms



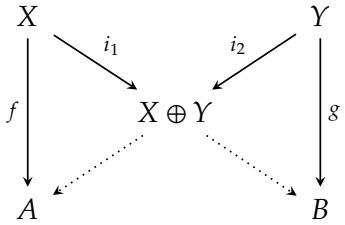
such that

$$p_1 \circ i_1 = \mathbb{1}_X \quad p_1 \circ i_2 = 0$$
$$p_2 \circ i_2 = \mathbb{1}_Y \quad p_2 \circ i_1 = 0$$
$$i_1 \circ p_1 + i_2 \circ p_2 = \mathbb{1}_Z$$

Then conditions Ad.4, Ad.3 and Ad.5 are equivalent and the objects $X \times Y$, $X \oplus Y$ and Z are naturally isomorphic.

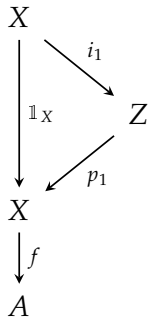
Proof. We will show that conditions Ad.3 and Ad.5 are equivalent and that $X \oplus Y \cong Z$. A dual argument shows that Ad.4 and Ad.5 are equivalent and that $X \times Y \cong Z$.

Ad.3 $\implies$ Ad.5 Let $X \oplus Y$ be the coproduct of X and Y, that comes with morphisms $i_1 \colon X \to X \oplus Y$ and $i_2 \colon Y \to X \oplus Y$.
By definition, for all morphisms $f \colon X \to A$, $g \colon Y \to B$ there exist unique morphisms $X \oplus Y \to A$, $X \oplus Y \to B$ such that



In particular if $f = \mathbb{1}_X$ and $g = \mathbb{1}_Y$

Ad.5 $\implies$ Ad.3 We are going to see that Z verifies the universal property of the coproduct.
Let $f : X \to A$ be a morphism. We want to find a morphism $\widetilde{f} \colon Z \to A$ such that $f = \widetilde{f} \circ i_1$. We have the following



Therefore if $\widetilde{f} \coloneqq f \circ p_1$ since $i_1 \circ p_1 = \mathbb{1}_X$ we have $\widetilde{f} \circ i_1 = f$.
Analogously we see that any morphism $g : Y \to B$ can can be factored through Z. Therefore since Z verifies the universal property of the coproduct, $Z \cong X \oplus Y$. □

Remark 5.0.1. Given objects $A_1, \ldots, A_n$ and $B_1, \ldots, B_m$ in an additive category C,

we can represent morphisms $f : A_1 \oplus \cdots \oplus A_n \to B_1 \oplus \cdots \oplus B_m$ as $n \times m$ matrices

$$\begin{pmatrix} f_{11} & \cdots & f_{1n} \\ \vdots & \ddots & \vdots \\ f_{m1} & \cdots & f_{mn} \end{pmatrix} \text{ where } f_{k\ell} \coloneqq p_k \circ f \circ i_\ell : A_\ell \to B_k$$

Using that $\sum_k i_k \circ p_k = 1$, it follows that addition and composition of matrices obey the usual rules of matrix addition and multiplication respectively.

Without elements to test, checking whether a morphism is monic or epic might seem daunting. Fortunately, the group structure of our hom-sets gives us a clean, element-free translation of injectivity and surjectivity based entirely on the zero morphism.

Lemma 5.0.2. A morphism $f : X \to Y$ in an additive category is a monomorphism if and only if for every morphism $g : Z \to X$

$$f \circ g = 0 \implies g = 0$$

Dually f is an epimorphism if and only if for every morphism $g : Y \to Z$

$$g \circ f = 0 \implies g = 0$$

Proof. We will check the monomorphism caracterization.

$\implies$ Let $f : X \rightarrowtail Y$ be a monomorphism. Consider a morphism $g : Z \to X$ such that $f \circ g = 0$. Since $f \circ 0 = 0$ applying left-cancellativity we have that $g = 0$.

$\impliedby$ Let $g_1, g_2 : Z \rightrightarrows X$ be two morphisms such that $f \circ g_1 = f \circ g_2$.
Since $\mathrm{Hom}(Z, Y)$ is an abelian group we have

$$0 = f \circ g_1 - f \circ g_2 = f \circ (g_1 - g_2)$$

Therefore by hypothesis $g_1 - g_2 = 0 \iff g_1 = g_2$, proving that f is a monomorphism. □

Being additive also works well when considering functor categories.

Lemma 5.0.3. Let I be a small category. If C is additive then the functor category $\mathsf{Fct}(\mathsf{I}, \mathsf{C})$ is additive.

Proof.

Ad.1 For any two natural transformations $\phi, \psi : \mathsf{I} \Rightarrow \mathsf{C}$ we can define their addition componentwise: $(\phi + \psi)_i = \phi_i + \psi_i$ for all $i \in \mathsf{I}$. Since $\phi_i, \psi_i \in \mathrm{Hom}_{\mathsf{C}}(-,-)$ this is well defined.

Ad.2 Consider the zero functor

$$\begin{array}{ccc} 0 : \mathsf{I} & \longrightarrow & \mathsf{C} \\ X & & 0 \\ f \downarrow & \longmapsto & \downarrow 0 \\ Y & & 0 \end{array}$$

This is clearly the zero object in $\mathsf{Fct}(\mathsf{I}, \mathsf{C},)$.

Ad.3, Ad.4 Direct from Proposition 3.3.1. □

Now it's time to introduce functors into play. In general a functor does not preserve the abelian group structure of the hom-sets. Therefore the correct way to move between additive categories without loosing information is by using additive functors.

Definition 5.0.2 (Additive functor). Let C, D be additive categories. A functor $F\colon \mathsf{C} \to \mathsf{D}$ is **additive** if for all $X, Y \in \mathsf{C}$

$$\mathrm{Hom}_{\mathsf{C}}(X, Y) \to \mathrm{Hom}_{\mathsf{D}}(F(X), F(Y))$$

is a morphism of abelian groups.

Example 5.0.2. For a fixed R-module M the functor $\mathrm{Hom}_R(M, -)\colon \mathsf{Mod}(R) \to \mathsf{Ab}$ is additive.

Instead of checking that the morphism $\mathrm{Hom}_{\mathsf{C}}(X, Y) \to \mathrm{Hom}_{\mathsf{D}}(F(X), F(Y))$ is a homomorphism for all objects X and Y we can use the following characterization.

Proposition 5.0.1. Let C, D be additive categories. A functor $F\colon \mathsf{C} \to \mathsf{D}$ is additive if and only if it commutes with finite direct sums, i.e: if $F(0) \cong 0$ and for all $X, Y \in \mathsf{C}$

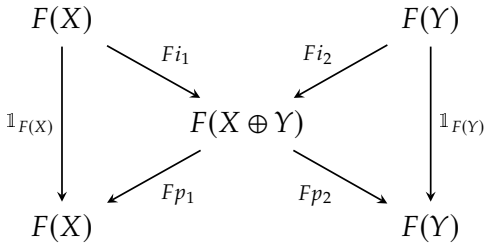
$$F(X \oplus Y) = F(X) \oplus F(Y)$$

Proof.

$\Longrightarrow$ Since F is a functor $F(\mathbb{1}_0) = \mathbb{1}_{F(0)}$. Furthermore since F is additive if $0_{X,Y}$ is the zero morphism between $X, Y \in \mathsf{C}$ then $F(0_{X,Y}) = 0_{F(X),F(Y)}$ because group homomorphisms preserve the identity. In particular taking $X = Y = 0$ since $0_{0,0} = \mathbb{1}_0$ we have

$$\mathbb{1}_{F(0)} = F(\mathbb{1}_0) = F(0_{0,0}) = 0_{F(0),F(0)}$$

which means that $F(0)$ is the zero object in D.
Let $X, Y \in \mathsf{C}$ be two objects and $X \oplus Y$ be its direct sum. Applying F to the diagram in Lemma 5.0.1 defining $Z \cong X \oplus Y$ we get



Note that

$$Fp_1 \circ Fi_1 = F(p_1 \circ i_1) = F(\mathbb{1}_X) = \mathbb{1}_{F(X)} \qquad Fp_1 \circ Fi_2 = F(p_1 \circ i_2) = F(0) = 0$$
$$Fp_2 \circ Fi_2 = F(p_2 \circ i_2) = F(\mathbb{1}_Y) = \mathbb{1}_{F(Y)} \qquad Fp_2 \circ Fi_1 = F(p_2 \circ i_1) = F(0) = 0$$

Furthermore since F is additive

$$Fp \circ Fi + Fj \circ Fq = F(p \circ i + j \circ q) = F\mathbb{1}_{X\oplus Y} = \mathbb{1}_{F(X\oplus Y)}$$

Therefore by Lemma 5.0.1 $F(X \oplus Y) \cong F(X) \oplus F(Y)$.

$\Longleftarrow$ Applying Lemma 5.0.1 to $X \oplus X$ and $Y \oplus Y$ we get maps $\Delta_X : X \to X \oplus X$ and $\nabla_Y : Y \oplus Y \to Y$ such that

$$p_1 \circ \Delta_X = \mathbb{1}_X = p_2 \circ \Delta_X$$
$$\nabla_Y \circ i_1 = \mathbb{1}_Y = \nabla_Y \circ i_2$$

Then for any pair of morphisms $f, g : X \to Y$ in D we have

$$f + g = \nabla_Y \circ (f \oplus g) \circ \Delta_X$$

where $f \oplus g$ is the unique morphism such that

$$p_k \circ (f \oplus g) \circ i_j = \begin{cases} f & j = k = 1 \\ g & j = k = 2 \\ 0 & \text{otherwise} \end{cases}$$

Applying F to this expression we get

$$F(f + g) = F(\nabla_Y \circ (f \oplus g) \circ \Delta_X) = F\nabla_Y \circ F(f \oplus g) \circ F\Delta_X$$

Since by hypothesis F preserves finite direct sums

$$F\nabla_Y \circ F(f \oplus g) \circ F\Delta_X = \nabla_{F(Y)} \circ (Ff \oplus Fg) \circ \Delta_{F(X)} = Ff + Fg$$

Therefore $F(f + g) = Ff + Fg$, which means that F is additive. □

Being additive is preserved by adjoints.

Proposition 5.0.2. Let $L \dashv R \colon \mathsf{C} \to \mathsf{D}$ be a pair of adjoint functors between additive categories. Then L is additive if and only if R is additive.

Proof. Since left adjoints preserve colimits and right adjoints preserve limits we have

$$L(X \oplus Y) = L(X) \oplus L(Y)$$
$$R(X \oplus Y) = R(X \times Y) = R(X) \times R(Y) = R(X) \oplus R(Y)$$

Note that if R is additive then $L(0) = 0$ since

$$\mathrm{Hom}_{\mathsf{C}}(L(0), 0) = \mathrm{Hom}_{\mathsf{D}}(0, R(0)) = \mathrm{Hom}_{\mathsf{D}}(0, 0)$$

and viceversa.

Therefore by Proposition 5.0.1 we have the result. □

Being additive, however, is not necessarilly preserved when taking full subcategories. For example, the full subcategory of Ab consisting of all but the trivial group is clearly not additive. This motivates the following definition.

Definition 5.0.3 (Full additive subcategory). Let C be an additive category. A full subcategory D of C is a **full additive subcategory** if the inclussion functor $\iota : D \hookrightarrow \mathsf{C}$ is additive.

Remark 5.0.2. By Proposition 5.0.1 a full subcategory D of C is a full additive subcategory if and only if $0 \in \mathsf{D}$ and D is closed under finite direct sums.

5.1 Complexes

From now until the end of the chapter, let C be an additive category.

It's time to define now the building blocks of homological algebra: complexes. The idea behind is to not study individual objects in an additive categories but "sequences" of them.

First we make explicit the notion of a "sequence" of objects.

Definition 5.1.1 (Differential object). A **differential object** $(X^\bullet, \mathrm{d}_X^\bullet)$ in C is a sequence of objects $X^k \in \mathsf{C}$ and morphisms d_X^k for every $k \in \mathbb{Z}$:

$$\cdots \longrightarrow X^{k-1} \xrightarrow{\mathrm{d}_X^{k-1}} X^k \xrightarrow{\mathrm{d}_X^k} X^{k+1} \longrightarrow \cdots$$

A morphism of differential objects $f^\bullet \colon (X^\bullet, \mathrm{d}_X^\bullet) \to (Y^\bullet, \mathrm{d}_Y^\bullet)$ is a collection of morphisms $\{f^k \colon X^k \to Y^k\}_{k\in\mathbb{Z}}$ such that, for all $k \in \mathbb{Z}$, the following diagram commutes

$$\begin{array}{ccccccc} \cdots & \longrightarrow & X^k & \xrightarrow{\mathrm{d}_X^k} & X^{k+1} & \longrightarrow & \cdots \\ & & \downarrow{\scriptstyle f^k} & & \downarrow{\scriptstyle f^{k+1}} & & \\ \cdots & \longrightarrow & Y^k & \xrightarrow[\mathrm{d}_Y^k]{} & Y^{k+1} & \longrightarrow & \cdots \end{array}$$

Therefore we get a category $\mathsf{Diff}(\mathsf{C})$ of differential objects in C. By abuse of notation we will often denote differential objects $(X^\bullet, \mathrm{d}_X^\bullet)$ as $X^\bullet$.

Remark 5.1.1. We can view the category of differential objects $\mathsf{Diff}(\mathsf{C})$ as the functor category $\mathsf{Fct}(\mathbb{Z}, \mathsf{C})$. Therefore by Lemma 5.0.3 $\mathsf{Diff}(\mathsf{C})$ is an additive category.

Remark 5.1.2. For all $n \in \mathbb{Z}$ there exists a fully faithful functor

$$\iota_n \colon \mathsf{C} \longrightarrow \mathsf{Diff}(\mathsf{C})$$

which sends an object $X \in \mathsf{C}$ to the differential object

$$\cdots \longrightarrow 0 \longrightarrow X \longrightarrow 0 \longrightarrow \cdots$$

and a morphism $f \colon X \to Y$ in C to

$$\begin{array}{ccccccc} \cdots & \longrightarrow & 0 & \longrightarrow & X & \longrightarrow & 0 & \longrightarrow & \cdots \\ & & \downarrow & & \downarrow{\scriptstyle f} & & \downarrow & & \\ \cdots & \longrightarrow & 0 & \longrightarrow & Y & \longrightarrow & 0 & \longrightarrow & \cdots \end{array}$$

Therefore we can identify an object $X \in \mathsf{C}$ as a differential object using ι_0. When the context is clear, we will directly write $X \in \mathsf{Diff}(\mathsf{C})$ instead of $\iota_0(X) \in \mathsf{Diff}(\mathsf{C})$.

We will focus in a particular case of differential objects: those where $\mathrm{d}^{k+1} \circ \mathrm{d}^k = 0$. This choice will be better justified when we introduce exact sequences in the next chapter.

Definition 5.1.2 (Complex). A **complex** in C is a differential object $(X^\bullet, \mathrm{d}_X^\bullet)$ such that for all $k \in \mathbb{Z}$

$$\mathrm{d}_X^{k+1} \circ \mathrm{d}_X^k = 0$$

We denote by $\mathsf{Comp}(\mathsf{C})$ the full additive subcategory of $\mathsf{Diff}(\mathsf{C})$ consisting of complexes in C.

Before continuing, we fix some terminology about complexes.

Definition 5.1.3. A complex $X^\bullet$ is

1. **Bounded above** if $X^n = 0$ of $n \gg 0$.
2. **Bounded below** if $X^n = 0$ of $n \ll 0$.
3. **Bounded** if it is bounded above and below.

We denote by $\mathsf{Comp}^*(\mathsf{C})$ $(* = +, -, \mathrm{b})$ the full additive subcategory of $\mathsf{Comp}(\mathsf{C})$ consisting of bounded above (resp. bounded below, bounded) complexes. To

emphasize that we are working with unbounded complexes we may write $\mathsf{Comp}^{\mathrm{ub}}(\mathsf{C}) := \mathsf{Comp}(\mathsf{C})$.

Remark 5.1.3. Note that for any $n \in \mathbb{N}$, the inclussion functor ι_n can be viewed as a functor

$$\iota_n : \mathsf{C} \longrightarrow \mathsf{Comp}^*(\mathsf{C})$$

Therefore we can identify $X \in \mathsf{C}$ with $\iota_0(X) \in \mathsf{Comp}^*(\mathsf{C})$.

5.2 Homotopy of complexes

As it turns out, isomorphisms between complexes are unnecessarilly strong for our needs and we can do with a less restrictive relation between them.

Definition 5.2.1 (Homotopy).

1. A morphism of complexes $f^\bullet : X^\bullet \to Y^\bullet$ is **homotopic to zero** if for all $p \in \mathbb{Z}$ there exists a morphism $s^p : X^p \to Y^{p-1}$ in C such that

$$f^p = s^{p+1} \circ \mathrm{d}_X^p + \mathrm{d}_Y^{p-1} \circ s^p$$

   Visually

$$\begin{array}{ccccccccc} \cdots & \longrightarrow & X^{p-1} & \longrightarrow & X^p & \longrightarrow & X^{p+1} & \longrightarrow & \cdots \\ & & \downarrow{\scriptstyle f^{p-1}} & \swarrow{\scriptstyle s^p} & \downarrow{\scriptstyle f^p} & \swarrow{\scriptstyle s^{p+1}} & \downarrow{\scriptstyle f^{p+1}} & & \\ \cdots & \longrightarrow & Y^{p-1} & \longrightarrow & Y^p & \longrightarrow & Y^{p+1} & \longrightarrow & \cdots \end{array}$$

2. Two morphisms $f^\bullet, g^\bullet : X^\bullet \to Y^\bullet$ are **homotopic** if $f^\bullet - g^\bullet$ is homotopic to zero. We denote this by $f^\bullet \sim g^\bullet$.
3. A complex $X^\bullet \in \mathsf{Comp}(\mathsf{C})$ is **homotopic to zero** if $\mathbb{1}_{X^\bullet} \sim 0$.
4. A morphism of complexes $f^\bullet : X^\bullet \to Y^\bullet$ is a **homotopy equivalence** if there exists $g^\bullet : Y^\bullet \to X^\bullet$ such that $g^\bullet \circ f^\bullet \sim \mathbb{1}_{X^\bullet}$ and $f^\bullet \circ g^\bullet \sim \mathbb{1}_{Y^\bullet}$. In this case we say that $X^\bullet$ and $Y^\bullet$ are **homotopy equivalent** and write $X^\bullet \sim Y^\bullet$.

This relation is well behaved with respect to both composition and the action of an additive functor.

Lemma 5.2.1. Let $f^\bullet : X^\bullet \to Y^\bullet$ and $g^\bullet : Y^\bullet \to Z^\bullet$ be two morphisms of complexes in $\mathsf{Comp}(\mathsf{C})$. If either $f^\bullet$ or $g^\bullet$ is homotopic to 0 then $g^\bullet \circ f^\bullet$ is homotopic to 0.

Proof. Assume that $f^\bullet$ is homotopic to 0. The proof consists on chasing the following diagram:

$$\begin{array}{ccccc} X^{p-1} & \longrightarrow & X^p & \longrightarrow & X^{p+1} \\ \downarrow{\scriptstyle f^{p-1}} & \swarrow{\scriptstyle s^p} & \downarrow{\scriptstyle f^p} & \swarrow{\scriptstyle s^{p+1}} & \downarrow{\scriptstyle f^{p+1}} \\ Y^{p-1} & \longrightarrow & Y^p & \longrightarrow & X^{p+1} \\ \downarrow{\scriptstyle g^{p-1}} & & \downarrow{\scriptstyle g^p} & & \downarrow{\scriptstyle g^{p+1}} \\ Z^{p-1} & \longrightarrow & Z^p & \longrightarrow & Z^{p+1} \end{array}$$

We have that for all $p \in \mathbb{Z}$ there exists $s^p : X^p \to Y^{p-1}$ such that

$$f^p = s^{p+1} \circ \mathrm{d}_X^p + \mathrm{d}_Y^{p-1} \circ s^p$$

Therefore

$$\begin{aligned} g^p \circ f^p &= g^p \circ \left(s^{p+1} \circ \mathrm{d}_X^p + \mathrm{d}_Y^{p-1} \circ s^p \right) \\ &= \left(g^p \circ s^{p+1} \right) \circ \mathrm{d}_X^p + \left(g^p \circ \mathrm{d}_Y^{p-1} \right) \circ s^p \\ &= \left(g^p \circ s^{p+1} \right) \circ \mathrm{d}_X^p + \mathrm{d}_Z^{p-1} \circ \left(g^{p-1} \circ s^p \right) \end{aligned}$$ □

Lemma 5.2.2. Let $F \colon \mathsf{C} \to \mathsf{D}$ be an additive functor. If $f^\bullet \colon X^\bullet \to Y^\bullet$ is homotopic to zero then $F f^\bullet$ is also homotopic to zero.

Proof. Since $f^\bullet$ is homotopic to zero for all $p \in \mathbb{Z}$ there exists a morphism $s^p \colon X^p \to Y^{p-1}$ such that

$$f^p = s^{p+1} \circ \mathrm{d}_X^p + \mathrm{d}_Y^{p-1} \circ s^p$$

Therefore

$$F f^p = F(s^{p+1} \circ \mathrm{d}_X^p + \mathrm{d}_Y^{p-1} \circ s^p) = F s^{p+1} \circ \mathrm{d}_{F(X)}^p + \mathrm{d}_{F(Y)}^{p-1} \circ F s^p$$ □

Chapter 6

Abelian categories

We continue now our goal of generalizing the category of modules to a more general category. To do this we consider now additive categories where the first isomorphism theorem works.

Before doing so, we fix some terminology and notation around the necessary building blocks of this theorem.

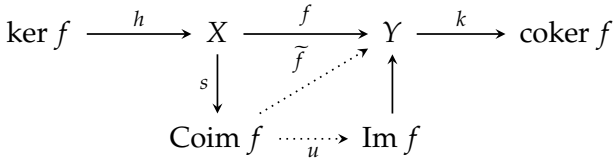
Definition 6.0.1. Let $f: X \to Y$ be a morphism in an additive category C that admits kernels and cokernels. Consider the following

$$\ker f \xrightarrow{h} X \xrightarrow{f} Y \xrightarrow{k} \operatorname{coker} f$$

We denote

$$\operatorname{Im} f := \ker k \qquad \operatorname{Coim} f := \operatorname{coker} h$$

I.e: we have the following diagram



Note that, since $f \circ h = 0$, f factors uniquely through $\operatorname{Coim} f$ producing the morphism $\widetilde{f}$. Furthermore since $k \circ f = k \circ \widetilde{f} = 0$ we have that $\widetilde{f}$ factors uniquely through $\operatorname{Im} f$. Therefore we have a canonical morphism

$$u: \operatorname{Coim} f \to \operatorname{Im} f$$

Now we are ready to define our categories of interest.

Definition 6.0.2 (Abelian category). An additive category C is **abelian** if it verifies

Ab.1) Any morphism in C admits a kernel and a cokernel.

Ab.2) For any morphism f in C the canonical morphism $\operatorname{Coim} f \to \operatorname{Im} f$ is an isomorphism.

As a direct consecuence of Theorem 3.3.1 we obtain that limits and colimits exist in these categories

Lemma 6.0.1. An abelian category admits all finite limits and colimits.

In the category of modules, an easy way to check whether homomorphisms are monomorphisms (resp. epimorphisms) is to check their kernels (resp. cokernels).

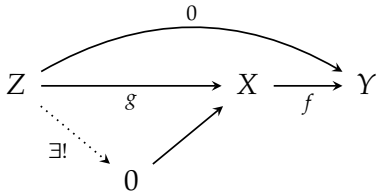
This still works in abelian categories.

Lemma 6.0.2. Let $f: X \to Y$ be a morphism in an abelian category. Then f is a monomorphism if and only if $0 \to X$ is its kernel.
Dually, f is an epimorphism if and only if $Y \to 0$ is its cokernel.

Proof. We will do the kernel one.

$\Longrightarrow$ Suppose f is a monomorphism.
If $g: Z \to X$ is a morphism such that $f \circ g = 0$ then by Lemma 5.0.2 $g = 0$. In particular g factors uniquely through $0 \to X$, i.e: $0 \to X$ is the kernel of f.

$\Longleftarrow$ Suppose now that $0 \to X$ is the kernel of f.
Consider a morphism $g: Z \to X$ such that $f \circ g = 0$. By the kernel properties we have the following diagram



Therefore by uniqueness $g = 0$, making f a monomorphism by Lemma 5.0.2. □

As in the case of additive categories, functor categories play nicely with abelian categories.

Proposition 6.0.1. Let I be a small category and C an abelian category. Then the functor category $\mathsf{Fct}(\mathsf{I}, \mathsf{C})$ is abelian.

Proof. By Lemma 5.0.3 we have that $\mathsf{Fct}(\mathsf{I}, \mathsf{C})$ is additive.

Furthermore since C has kernels and cokernels, which are no more than special cases of limits and colimits, by Proposition 3.3.1 $\mathsf{Fct}(\mathsf{I}, \mathsf{C})$ has kernels and cokernels and they are computed elementwise.

Finally since $\operatorname{Coim} f(i) \xrightarrow{\sim} \operatorname{Im} f(i)$ for all $i \in \mathsf{I}$ we have that $\operatorname{Coim} f \xrightarrow{\sim} \operatorname{Im} f$. □

Furthermore since kernel, cokernels and direct sums are computed term-wise we directly obtain that $\mathsf{Comp}^*(\mathsf{C})$ is abelian.

Corollary 6.0.1. If C is abelian then the categories of complexes $\mathsf{Comp}^*(\mathsf{C})$ $(* = +, -, \mathrm{b})$ are abelian.

6.1 Exactness

In what remains of the chapter we will assume that C is an abelian category.

Definition 6.1.1.

1. A complex $X \xrightarrow{f} Y \xrightarrow{g} Z$ is **exact** if $\operatorname{Im} f \xrightarrow{\sim} \ker g$. Note that this is equivalent to $\operatorname{coker} f \xrightarrow{\sim} \operatorname{Im} g$.
2. More generally, a sequence of morphisms $X^p \xrightarrow{\mathrm{d}^p} \cdots \to X^n$ with $\mathrm{d}^{i+1} \circ \mathrm{d}^i = 0$ for all $i \in \{p, \dots, n-1\}$ is **exact** if $\operatorname{Im} \mathrm{d}^i \xrightarrow{\sim} \ker \mathrm{d}^{i+1}$ for all $i \in \{p, \dots, n-1\}$.

3. A **short exact sequence** is an exact sequence of the form

$$0 \to X \to Y \to Z \to 0$$

Remark 6.1.1. A short sequence $0 \to X \xrightarrow{f} Y \xrightarrow{g} Z \to 0$ is short exact if and only if f is a monomorphism, g is an epimorphism and $\operatorname{Im} f = \ker g$.
Indeed, it is exact at X iff $\ker f = \operatorname{Im} 0 = 0$ and is exact at Z iff $\operatorname{coker} g = \operatorname{Im} 0 = 0$.

Example 6.1.1. Any morphism $f : X \to Y$ may be decomposed in short exact sequences

$$0 \longrightarrow \ker f \rightarrowtail X \twoheadrightarrow \operatorname{Coim} f \longrightarrow 0$$

$$0 \longrightarrow \operatorname{Im} f \rightarrowtail Y \twoheadrightarrow \operatorname{coker} f \longrightarrow 0$$

with $\operatorname{Coim} f \cong \operatorname{Im} f$.
Indeed, by definition we have $\operatorname{Coim} f = \operatorname{coker}(\ker f \to X)$ and $\operatorname{Im} = \ker(Y \to \operatorname{coker} f)$.

A stronger condition than exactness is split exactness. Roughly speaking, a short exact sequence is split exact if the only information is at its ends.

Proposition 6.1.1. Let

$$0 \longrightarrow X \xrightarrow{f} Y \xrightarrow{g} Z \longrightarrow 0$$

be a short exact sequence in an abelian category C. We say that the sequence **splits** if any of the following equivalent conditions are satisfied

1. There exists $h : Z \to Y$ such that $g \circ h = \mathbb{1}_Z$.
2. There exists $k : Y \to X$ such that $k \circ f = \mathbb{1}_X$.
3. There exists $\varphi = (k, h)$ and $\psi = \binom{f}{g}$ such that $Y \xrightarrow{\varphi} X \oplus Z$ and $X \oplus Z \xrightarrow{\psi} Y$ are inverses to each other.
4. The sequence is homotopic to 0.
5. The sequence is isomorphic to $0 \to X \to X \oplus Z \to Z \to 0$.

Proof.

$1 \implies 3$ Since $g = g \circ h \circ g$ we have that $g \circ (\mathbb{1}_Y - h \circ g) = 0$, i.e: $\mathbb{1}_Y - h \circ g$ factors through $\ker g$ and so it factors through X. Therefore there exists $h : Y \to Z$ such that $\mathbb{1}_Y - h \circ g = f \circ k$.

$3 \implies 1$ Since $g \circ f = 0$ we have $g = g \circ h \circ g$, i.e: $(g \circ h - \mathbb{1}_Z) \circ g$. Therefore since g is an epimorphism we have $g \circ h - \mathbb{1}_Z = 0 \iff g \circ h = \mathbb{1}_Z$.

$2 \iff 3$ Analogous to $1 \iff 3$.

$3 \iff 4$ By definition the sequence is homopic to zero if and only if there exists

$$\begin{array}{ccccccccc} 0 & \longrightarrow & X & \xrightarrow{f} & Y & \xrightarrow{g} & Z & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \mathbb{1}_X} & \swarrow{\scriptstyle k} & \downarrow{\scriptstyle \mathbb{1}_Y} & \swarrow{\scriptstyle h} & \downarrow{\scriptstyle \mathbb{1}_Z} & & \\ 0 & \longrightarrow & X & \xrightarrow[f]{} & Y & \xrightarrow[g]{} & Z & \longrightarrow & 0 \end{array}$$

such that $\mathbb{1}_X = k \circ f$, $\mathbb{1}_Y = h \circ g$ and $\mathbb{1}_Z = h \circ g + f \circ k$.

$3 \iff 5$ Trivial. □

When studying exactness between different categories we don't want to loose information on whether a sequence is exact or not. Therefore the correct notion of

functors in this case is that of exact functors.

Definition 6.1.2. Let C, D be two abelian categories and
$$0 \longrightarrow X \longrightarrow Y \longrightarrow Z \longrightarrow 0$$
be a short exact sequence in an C.
An additive functor $F\colon \mathsf{C} \to \mathsf{D}$ is **left-exact** if
$$0 \longrightarrow F(X) \longrightarrow F(Y) \longrightarrow F(Z)$$
is exact in D.
Dually, it is **right-exact** if
$$F(X) \longrightarrow F(Y) \longrightarrow F(Z) \longrightarrow 0$$
is exact in D.
If F is both left and right exact we say that it is **exact**.

We state the following characterization of exact functors, which we will not prove.

Lemma 6.1.1 ([kashiwaraIntroductionCategoriesHomological]). Let $F\colon \mathsf{C} \to \mathsf{D}$ be an additive functor.

1. The following are equivalent
   - (a) F is left exact (resp. right exact).
   - (b) F commutes with kernels: for any morphism $f\colon X \to Y$, $F(\ker f) \xrightarrow{\sim} \ker F f$ (resp. with cokernels).
   - (c) F commutes with finite limits (resp. finite colimits).
   - (d) For any short exact sequence $0 \to X \to Y \to Z$ (resp. $X \to Y \to Z \to 0$) in C the sequence $0 \to F(X) \to F(Y) \to F(Z)$ is exact in D (resp. $F(X) \to F(Y) \to F(Z) \to 0$ is exact).
2. The following are equivalent
   - (a) F is exact.
   - (b) For any exact sequence $X \to Y \to Z$ in C the sequence $F(X) \to F(Y) \to F(Z)$ is exact in D.
   - (c) F commutes with finite limits and finite colimits.

As a direct application of Lemma 6.1.1 we have the following examples.

Example 6.1.2. The Hom bifunctor $\mathrm{Hom}_{\mathsf{C}}\colon \mathsf{C}^{\mathrm{op}} \times \mathsf{C} \to \mathsf{Ab}$ is left exact with respect to each of its arguments.

Example 6.1.3. Recall Theorem 4.2.1.
Right adjoints are left exact. Dually, left adjoints are right exact.

Example 6.1.4. Let I be a small category. If C admits I-indexed limits then the functor $\lim\colon \mathsf{Fct}(\mathsf{I}^{\mathrm{op}}, \mathsf{C}) \to \mathsf{C}$ is left exact. Dually if C admits I-indexed colimits then the functor $\mathrm{colim}\colon \mathsf{Fct}(\mathsf{I}, \mathsf{C}) \to \mathsf{C}$ is right exact.

Example 6.1.5. Let R be a ring and $\mathcal{I}$ an index set. The two functors
$$\prod_{i\in\mathcal{I}}\colon \mathsf{Fct}(\mathcal{I}, \mathsf{Mod}(R)) \to \mathsf{Mod}(R) \qquad \bigoplus_{i\in\mathcal{I}}\colon \mathsf{Fct}(\mathcal{I}, \mathsf{Mod}(R)) \to \mathsf{Mod}(R)$$
are exact.

Example 6.1.6. Let R be a ring and $\mathcal{I}$ a small directed category. The functor $\varinjlim_{i\in\mathcal{I}}\colon \mathsf{Fct}(\mathcal{I}, \mathsf{Mod}(R)) \to \mathsf{Mod}(R)$ is exact.

6.1.1 Diagram lemmas

Since abelian categories may not necessarily consists of sets with elements (for example the category of complexes). Therefore proving facts about this categories requires, in principle, to work over the full abstraction of category theory. However using the Freyd-Mitchell embedding theorem one can *cheat* a little.

Theorem 6.1.1 (Freyd-Mitchell embedding - [Wei94, Section 1.6]). Let C be a small abelian category. Then there exists a ring R (with unit, not necessarilly commutative) and a fully faithful exact functor $F\colon \mathsf{C} \to \mathsf{Mod}(R)$ which embeds C as a full subcategory of $\mathsf{Mod}(R)$.

In short, to prove something about a diagram in an abelian category we may assume that we are working on the category of modules over a ring R and then diagram-chase elements. Furthermore, any fact about kernels, cokernels and so one that hold on $\mathsf{Mod}(R)$ also holds in any abelian category.

Another strategy to prove statements in abelian categories is the *philosophy of generalized elements* [Mac78, Theorem VIII.4.3].

Lemma 6.1.2 (Five lemma). Consider a commutative diagram

$$\begin{array}{ccccccccc} X^0 & \xrightarrow{\alpha^0} & X^1 & \xrightarrow{\alpha^1} & X^2 & \xrightarrow{\alpha^2} & X^3 \\ \downarrow{\scriptstyle f^0} & & \downarrow{\scriptstyle f^1} & & \downarrow{\scriptstyle f^2} & & \downarrow{\scriptstyle f^3} \\ Y^0 & \xrightarrow[\beta^0]{} & Y^1 & \xrightarrow[\beta^1]{} & Y^2 & \xrightarrow[\beta^2]{} & Y^3 \end{array}$$

where the rows are exact.
If f^0 is an epimorphism and f^1, f^3 are monomorphisms then f^2 is a monomorphism. Dually if f^3 is an monomorphism and f^0, f^2 are epimorphisms then f^1 is a epimorphism.

Proof. Let $x_2 \in X^2$ such that $f^2(x_2) = 0$. We want to see that $x_2 = 0$.

We have that $(f^3 \circ \alpha^2)(x_2) = 0$ which, since f^3 is a monomorphism, implies that $\alpha^2(x_2) = 0$. Since the first row is exact there exists $x_1 \in X^1$ such that $\alpha(x_1) = x_2$.

Let $y_1 \coloneqq f^1(x_1)$. Since $(\beta^1 \circ f^1)(x_1) = 0$ and the second row is exact there exists $y_0 \in X^0$ such that $\beta^0(y_0) = f^1(x_1)$.

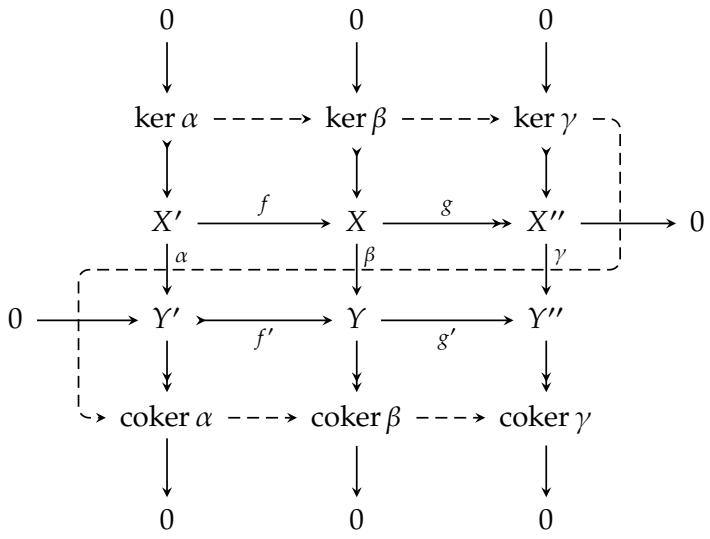
Since f^0 is an epimorphism there exists $x_0 \in X^0$ such that $y_0 = f^0(x_0)$.

Since $(f^1 \circ \alpha^0)(x_0) = f^1(x_1)$ and f^1 is a monomorphism we have $\alpha^0(x_0) = x_1$. Therefore $x_2 = \alpha^1(x_1) = 0$. □

Lemma 6.1.3 (Snake lemma). Consider a commutative diagram

$$\begin{array}{ccccccccc} & & X' & \xrightarrow{f} & X & \xrightarrow{g} & X'' & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \beta} & & \downarrow{\scriptstyle \gamma} & & \\ 0 & \longrightarrow & Y' & \xrightarrow[f']{} & Y & \xrightarrow[g']{} & Y'' & & \end{array}$$

where the rows are exact.
Then there exists a morphism $\delta\colon \ker \gamma \to \operatorname{coker} \alpha$ giving rise to an exact sequence



Proof. We will first check that $\ker \alpha \to \ker \beta \to \ker \gamma$ is exact. Let $x \in \ker \beta$ with $g(x) = 0$. Since the first row is exact there exists $x' \in X'$ with $f(x') = x$. Then $(f' \circ \alpha)(x') = (\beta \circ f)(x') = 0$. Since f' is a monomorphism we get $\alpha(x') = 0$ and thus $x' \in \ker \alpha$.

A dual argument shows that $\operatorname{coker} \alpha \to \operatorname{coker} \beta \to \operatorname{coker} \gamma$ is exact.

It only remains to construct δ. Let $x'' \in \ker \gamma$ and choose $x \in X$ with $g(x) = x''$. Set $y := \beta(y)$. Since $g'(y) = 0$ there exists $y' \in Y'$ such that $f'(y') = y$. Therefore we can define $\delta(x'')$ to be the image of y' in $\operatorname{coker} \alpha$. □

6.2 Complexes and cohomology

Recall that the categories $\mathsf{Comp}^*(\mathsf{C})$ are abelian for $* = +, -, \mathrm{b}, \mathrm{ub}$.

Since for a complex $X^\bullet$ we always have $\operatorname{Im} \mathrm{d}_X^n \subseteq \ker \mathrm{d}_X^{n-1}$ for all $n \in \mathbb{N}$, there is a failure of exactness if and only if $\operatorname{Im} \mathrm{d}_X^n \neq \ker \mathrm{d}_X^{n-1}$. We introduce cohomology as a way of measuring how far away is a complex from being exact.

Definition 6.2.1 (Cohomology). Let

$$X^\bullet := \cdots \longrightarrow X^{n-1} \xrightarrow{\mathrm{d}_X^{n-1}} X^n \xrightarrow{\mathrm{d}_X^n} X^{n+1} \longrightarrow \cdots$$

be a complex in C (i.e: $X^\bullet \in \mathsf{Comp}(\mathsf{C})$). We define

$$\begin{aligned} Z^n(X^\bullet) &:= \ker \mathrm{d}_X^n \\ B^n(X^\bullet) &= \operatorname{Im} \mathrm{d}_X^{n-1} \\ \mathrm{H}^n(X^\bullet) &:= {}^{Z^n(X^\bullet)}\!/_{B^n(X^\bullet)} = \operatorname{coker}(B^n(X^\bullet) \to Z^n(X^\bullet)) \end{aligned}$$

We call $\mathrm{H}^n(X^\bullet)$ the n-th **cohomology** of $X^\bullet$.
Note that this defines a functor $\mathrm{H}^n : \mathsf{Comp}(\mathsf{C}) \to \mathsf{C}$.

Remark 6.2.1. A complex $X^\bullet$ is exact at X^n if and only if $\mathrm{H}^i(X^\bullet) = 0$.

Remark 6.2.2. We have short exact sequences

$$X^{n-1} \xrightarrow{\mathrm{d}_X^{n-1}} \ker \mathrm{d}_X^n \longrightarrow \mathrm{H}^n(X^\bullet) \longrightarrow 0$$

$$0 \longrightarrow \mathrm{H}^n(X^\bullet) \longrightarrow \operatorname{coker} \mathrm{d}_X^{n-1} \xrightarrow{\mathrm{d}_X^n} X^{n+1}$$

We can glue this into an exact sequence

$$0 \longrightarrow \mathrm{H}^n(X^\bullet) \longrightarrow \operatorname{coker} \mathrm{d}_X^{n-1} \xrightarrow{\mathrm{d}_X^n} \ker \mathrm{d}_X^{n+1} \longrightarrow \mathrm{H}^{n+1}(X^\bullet) \longrightarrow 0$$

The first thing we are going to see is that cohomology is a homotopy invariant.

Lemma 6.2.1. Let C be an abelian category and $f^\bullet \colon X^\bullet \to Y^\bullet$ a morphism in $\mathsf{Comp}(\mathsf{C})$ homotopic to zero. Then $\mathrm{H}^n f^\bullet \colon \mathrm{H}^n(X^\bullet) \to \mathrm{H}^n(Y^\bullet)$ is the zero morphism for all $n \in \mathbb{Z}$.
In particular if $f^\bullet, g^\bullet : X^\bullet \to Y^\bullet$ are homotopic $\mathrm{H}^n f^\bullet = \mathrm{H}^n g^\bullet$ for all $n \in \mathbb{Z}$.

Proof. Let

$$f^n = s^{n+1} \circ \mathrm{d}_X^n + \mathrm{d}_Y^{n-1} \circ s^n$$

Then $\mathrm{d}_X^n = 0$ in $\ker \mathrm{d}_X^n$ and the image of d_Y^{n-1} is zero on $\ker \mathrm{d}_Y^n \big/ \operatorname{Im} \mathrm{d}_Y^{n-1}$.

Therefore $\mathrm{H}^n f^\bullet : Z^n(X^\bullet) \big/ B^n(X^\bullet) \to Z^n(Y^\bullet) \big/ B^n(Y^\bullet)$ is the zero morphism. □

Corollary 6.2.1. Let $X^\bullet \sim Y^\bullet$ be two homotopic complexes. Then $\mathrm{H}^n(X^\bullet) \cong \mathrm{H}^n(Y^\bullet)$.

Proof. Since $X^\bullet \sim Y^\bullet$ there exist maps $f^\bullet : X^\bullet \to Y^\bullet$ and $g^\bullet : Y^\bullet \to X^\bullet$ such that $g^\bullet \circ f^\bullet \sim \mathbb{1}_{X^\bullet}$ and $f^\bullet \circ g^\bullet \sim \mathbb{1}_{Y^\bullet}$.

Therefore by Lemma 6.2.1 for all $n \in \mathbb{Z}$ we have that

$$\mathrm{H}^n(g^\bullet \circ f^\bullet) \cong \mathrm{H}^n \mathbb{1}_{X^\bullet} = \mathbb{1}_{\mathrm{H}^n(X^\bullet)}$$
$$\mathrm{H}^n(f^\bullet \circ g^\bullet) \cong \mathrm{H}^n \mathbb{1}_{Y^\bullet} = \mathbb{1}_{\mathrm{H}^n(Y^\bullet)}$$

Furthermore, since H^n is a functor

$$\mathrm{H}^n(g^\bullet \circ f^\bullet) = \mathrm{H}^n g^\bullet \circ \mathrm{H}^n f^\bullet$$
$$\mathrm{H}^n(f^\bullet \circ g^\bullet) = \mathrm{H}^n f^\bullet \circ \mathrm{H}^n g^\bullet$$

which means that $\mathrm{H}^n f^\bullet$ and $\mathrm{H}^n g^\bullet$ are inverses of each other. □

Since cohomology is a functor we can also study how it acts on morphisms. For some of these morphisms, the resulting cohomology objects will be the same and thus they are "equivalent" in some sense. This leads to the notion of a quasi-isomorphism.

Definition 6.2.2 (Quasi-isomorphism). Let C be an abelian category. A morphism $f^\bullet \colon X^\bullet \to Y^\bullet$ is a **quasi-isomorphism** (abbreviated q.is) if $\mathrm{H}^n f^\bullet$ is an isomorphism for all $n \in \mathbb{Z}$. In this case we say that $X^\bullet$ and $Y^\bullet$ are **quasi-isomorphic**.
In particular $X^\bullet$ is quasi-isomorphic to zero if and only if $X^\bullet$ is exact.

Recall that we can identify an object $X \in \mathsf{C}$ with a complex $\cdots \to 0 \to X \to 0 \to \cdots$. Using quasi-isomorphisms we can associated this bounded complex an associated unbounded complex that has the same cohomology.

Definition 6.2.3 (Resolution). Let C be an abelian category and $X \in \mathsf{C}$. A **right resolution** of X is a quasi-isomorphism $X \to Y^\bullet$ with $Y^\bullet \in \mathsf{Comp}^+(\mathsf{C})$.
Dually, a **left resolution** is a quasi-isomorphism $Z^\bullet \to X$ with $Z^\bullet \in \mathsf{Comp}^-(\mathsf{C})$.

Remark 6.2.3. Let $X \to Y^\bullet$ be a right resolution of X.
This means that we have a morphism of complexes

$$\begin{array}{ccccccccc}
\cdots & \longrightarrow & 0 & \longrightarrow & X & \longrightarrow & 0 & \longrightarrow & \cdots \\
 & & \downarrow & & \downarrow{\scriptstyle \varepsilon} & & \downarrow & & \\
\cdots & \longrightarrow & 0 & \longrightarrow & Y^0 & \longrightarrow & Y^1 & \longrightarrow & \cdots
\end{array}$$

Note that this is exactly the same data as an exact complex

$$0 \longrightarrow X \xrightarrow{\varepsilon} Y^0 \longrightarrow Y^1 \longrightarrow \cdots$$

We end the chapter with a key result: taking cohomology transforms a short exact sequence into a long exact sequence in a functorial way.

Theorem 6.2.1 (Long exact sequence in cohomology). Let

$$0 \longrightarrow X^\bullet \longrightarrow Y^\bullet \longrightarrow Z^\bullet \longrightarrow 0$$

be a short exact sequence in an abelian category $\mathsf{Comp}(\mathsf{C})$.

1. For all $k \in \mathbb{Z}$ the sequence $\mathrm{H}^k(X^\bullet) \to \mathrm{H}^k(Y^\bullet) \to \mathrm{H}^k(Z^\bullet)$ is exact.
2. For all $k \in \mathbb{Z}$ there exists $\delta^k \colon \mathrm{H}^k(Z^\bullet) \to \mathrm{H}^{k+1}(X^\bullet)$ making the long sequence

$$\begin{array}{l}
\cdots \longrightarrow \mathrm{H}^{k-1}(Y^\bullet) \longrightarrow \mathrm{H}^{k-1}(Z^\bullet) \xrightarrow{\delta^{k-1}} \\
\mathrm{H}^k(X^\bullet) \longrightarrow \mathrm{H}^k(Y^\bullet) \longrightarrow \mathrm{H}^k(Z) \xrightarrow{\delta^k} \\
\mathrm{H}^{k+1}(X^\bullet) \longrightarrow \mathrm{H}^{k+1}(Y^\bullet) \longrightarrow \cdots
\end{array}$$

exact. Furthermore δ^k can be constructed to be functorial with respect to short exact sequences of $\mathsf{Comp}(\mathsf{C})$.

Proof. Consider the diagram

$$\begin{array}{ccccccccc}
 & & 0 & & 0 & & 0 & & \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
 & & \mathrm{H}^k(X^\bullet) & & \mathrm{H}^k(Y^\bullet) & & \mathrm{H}^k(Z^\bullet) & & \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
 & & \operatorname{coker} \mathrm{d}_X^{k-1} & \longrightarrow & \operatorname{coker} \mathrm{d}_Y^{k-1} & \longrightarrow & \operatorname{coker} \mathrm{d}_Z^{k-1} & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & \ker \mathrm{d}_X^{k+1} & \longrightarrow & \ker \mathrm{d}_Y^{k+1} & \longrightarrow & \ker \mathrm{d}_Z^{k+1} & & \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
 & & \mathrm{H}^{k+1}(X^\bullet) & & \mathrm{H}^{k+1}(Y^\bullet) & & \mathrm{H}^{k+1}(Z^\bullet) & & \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
 & & 0 & & 0 & & 0 & &
\end{array}$$

where the columns are exact by Remark 6.2.2 and the rows are exact by hypothesis. Therefore the result follows from the Snake lemma. □

Chapter 7

Derived functors

In this chapter C refers to an abelian category.

7.1 Injective and projective objects

We begin by studying objects that have nice properties with respect to exact sequence.

Definition 7.1.1. Let C be a category. An object $I \in \mathsf{C}$ is **injective** if all the diagrams of the form

$$\begin{array}{ccc} X & \xrightarrow{k} & I \\ {\scriptstyle f}\downarrow & & \\ Z & & \end{array}$$

where $X \rightarrowtail Z$ is a monomorphism admit an extension

$$\begin{array}{ccc} X & \xrightarrow{k} & I \\ {\scriptstyle f}\downarrow & \nearrow_{h} & \\ Z & & \end{array}$$

Dually, an object $P \in \mathsf{C}$ is **projective** if all the diagrams of the form

$$\begin{array}{ccc} & & X \\ & & \downarrow{\scriptstyle f} \\ P & \xrightarrow[k]{} & Z \end{array}$$

where $X \twoheadrightarrow Z$ is an epimorphism admit an extension

$$\begin{array}{ccc} & & X \\ & \nearrow_{h} & \downarrow{\scriptstyle f} \\ P & \xrightarrow[k]{} & Z \end{array}$$

Notation. If C is any category we denote by $\mathcal{I}_{\mathsf{C}}$ its full subcategory made of injective objects and by $\mathcal{P}_{\mathsf{C}}$ its full subcategory of projective objects.

A more useful characterization of this objects in terms of exact functors is the following.

Proposition 7.1.1. Let C be an abelian category. An object $I \in \mathsf{C}$ is injective if and only if the functor $\mathrm{Hom}_{\mathsf{C}}(-, I)$ is exact.
Dually, an object $P \in \mathsf{C}$ is projective if and only if the functor $\mathrm{Hom}_{\mathsf{C}}(P, -)$ is exact.

Proof. By Example 6.1.2 we only need to check right exactness.

We will do the injective version this time:

$\Longrightarrow$ Suppose $I \in \mathcal{I}_{\mathsf{C}}$ is injective and consider a short exact sequence

$$0 \longrightarrow X \xrightarrow{f} Y \xrightarrow{g} Z \longrightarrow 0$$

Using Lemma 6.1.1 (recall that Hom is a contravariant functor), we want to see that

$$0 \longrightarrow \mathrm{Hom}(Z, I) \xrightarrow{g^*} \mathrm{Hom}(Y, I) \xrightarrow{f^*} \mathrm{Hom}(X, I)$$

is exact.
Since by Example 6.1.2 this sequence is exact at $\mathrm{Hom}(Z, I)$ and $\mathrm{Hom}(Y, I)$, we only need to check exactness at $\mathrm{Hom}(X, I)$, i.e: we want to see that f^* is surjective.
Consider a morphism $\alpha \colon X \to I$. Since f is a monomorphism and I is injective there exists a morphism $\beta \colon Y \to I$ such that $\beta \circ f = f^*(\beta) = \alpha$, making f^* surjective as wanted.

$\Longleftarrow$ Suppose $\mathrm{Hom}(-, I)$ is exact and consider a monomorphism $f \colon X \rightarrowtail Y$ in C.
We have the following short exact sequence

$$0 \longrightarrow X \xrightarrow{f} Y \longrightarrow \operatorname{coker} f \longrightarrow 0$$

Applying $\mathrm{Hom}(-, I)$ to this sequence we get

$$0 \longrightarrow \mathrm{Hom}(\operatorname{coker} f, I) \longrightarrow \mathrm{Hom}(Y, I) \xrightarrow{f^*} \mathrm{Hom}(X, I) \longrightarrow 0$$

which is exact since $\mathrm{Hom}(-, I)$ is exact. In particular we have that f^* is surjective.
Therefore if $\gamma \colon X \to I$ is any morphism there exists $\beta \in \mathrm{Hom}(X, I)$ such that $\gamma = \beta \circ f$, which is exactly the universal property of injective objects. □

We would like to build resolutions using injective and projective objects. This is however not always possible.

Definition 7.1.2. A category has **enough injectives** if for every object $X \in \mathsf{X}$ there exists an injective object $I \in \mathcal{I}_{\mathsf{C}}$ and a monomorphism $X \rightarrowtail I$.
Dually, C has **enough projectives** if for every object $X \in \mathsf{A}$ there exists a projective object $P \in \mathcal{P}_{\mathsf{C}}$ and an epimorphism $P \twoheadrightarrow X$.

Since as we will see later the category of sheaves has enough injective but not enough projectives, we will study the injective case. However all of the results have dual counterparts for projective objects.

Assuming that we have enough injectives, every object $X \in \mathsf{C}$ admits a right resolution $X \to I^\bullet$ made of injectives.

Theorem 7.1.1. Let C be an abelian category with enough injectives. Then every object $X \in \mathsf{C}$ admits a right resolution $X \to I^\bullet$ where $I^n \in \mathcal{I}_{\mathsf{C}}$ is injective for all

$n \in \mathbb{Z}$.
Dually, if C has enough projectives every object $X \in \mathsf{C}$ admits a left resolution $P^\bullet \to X$ where $P^n \in \mathcal{P}_\mathsf{C}$ is projective for all $n \in \mathbb{Z}$.

Proof. Let $X \in \mathsf{C}$.

We want to construct an exact sequence

$$0 \longrightarrow X \longrightarrow I^0 \longrightarrow \cdots \longrightarrow I^n \longrightarrow \cdots$$

We will do it by induction over n.

For the base case, since C has enough injectives there exists an injective object $I_0 \in \mathcal{I}_\mathsf{C}$ and a monomorphism $X \rightarrowtail I_0$.

Suppose now that we have a exact sequence

$$0 \longrightarrow X \longrightarrow I^0 \longrightarrow \cdots \longrightarrow I^{n-1} \longrightarrow I^n$$

Since C is abelian there exists

$$\operatorname{coker}(I^{n-1} \to I^n) \colon I^n \twoheadrightarrow B^n$$

Note that

$$I^{n-1} \longrightarrow I^n \twoheadrightarrow B^n \longrightarrow 0$$

is exact. Since C has enough injectives there exists $I^{n+1} \in \mathcal{I}_\mathsf{C}$ and a monomorphism $B^n \rightarrowtail I^{n+1}$ Therefore

$$\begin{array}{ccccc} & & B^n & & \\ & & \uparrow & \searrow & \\ I^{n-1} & \longrightarrow & I^n & \longrightarrow & I^{n+1} \end{array}$$

is exact and induction follows. □

Furthermore, injective objects allow us to extend morphisms $f : X \to Y$ to morphisms $f^\bullet \colon I^\bullet \to J^\bullet$ between injective resolutions of X and Y.

Theorem 7.1.2. Let C be a category with enough injectives. Let $X \xrightarrow{\varepsilon} I^\bullet, Y \xrightarrow{\eta} J^\bullet$ be two right-resolutions of $X, Y \in \mathsf{C}$.
If $J^\bullet$ is made of injectives then any morphism $f \colon X \to Y$ can be extended to a morphism $f^\bullet \colon I^\bullet \to J^\bullet$ in the sense that $\eta \circ f^0 = f \circ \varepsilon$. Furthermore this extension is unique up to homotopy.
We call $f^\bullet$ a **lifting** of f.

Proof. We will first prove existence by constructing $f^\bullet$. We will do it by induction.

For the base case $n = 0$ we have:

$$\begin{array}{ccc} X & \xrightarrow{\varepsilon} & I^0 \\ {\scriptstyle f}\downarrow & & \\ Y & \xrightarrow[\eta]{} & J^0 \end{array}$$

Since ε is a monomorphism and J^0 is injective there exists a unique $f^0 \colon I^0 \to J^0$ making the diagram commute:

$$\begin{array}{ccc} X & \xrightarrow{\varepsilon} & I^0 \\ {\scriptstyle f}\downarrow & & \vdots{\scriptstyle f^0} \\ Y & \xrightarrow[\eta]{} & J^0 \end{array}$$

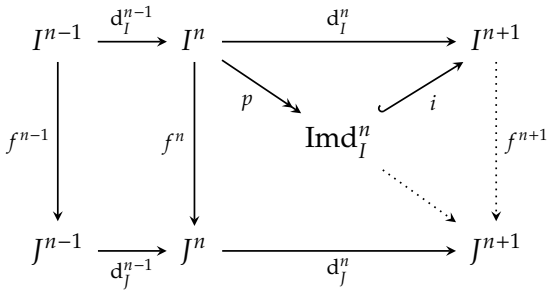
Suppose now that we have constructed $f^i \colon I^i \to J^i$ for $i \in \{0, \dots, n\}$ making the following diagram commute:

$$\begin{array}{ccccccccccc} X & \xrightarrow{\varepsilon} & I^0 & \xrightarrow{\mathrm{d}_I^0} & \cdots & \longrightarrow & I^{n-1} & \xrightarrow{\mathrm{d}_I^{n-1}} & I^n & \xrightarrow{\mathrm{d}_I^n} & I^{n+1} \\ {\scriptstyle f}\downarrow & & {\scriptstyle f^0}\downarrow & & & & {\scriptstyle f^{n-1}}\downarrow & & {\scriptstyle f^n}\downarrow & & \\ Y & \xrightarrow[\eta]{} & J^0 & \xrightarrow[\mathrm{d}_J^0]{} & \cdots & \longrightarrow & J^{n-1} & \xrightarrow[\mathrm{d}_J^{n-1}]{} & J^n & \xrightarrow[\mathrm{d}_J^n]{} & J^{n+1} \end{array}$$

Consider $g \coloneqq \mathrm{d}_J^n \circ f^n \colon I^n \to J^{n+1}$. Since $\mathrm{d}_I^{n-1} \circ f^n = f^{n-1} \circ d_J^{n-1}$ and $\mathrm{d}_J^{n-1} \circ \mathrm{d}_J^n = 0$ we have that g vanishes on $\operatorname{Im} \mathrm{d}_I^{n-1}$. Therefore g induces a morphism

$$I^n \big/ \operatorname{Im} \mathrm{d}_I^{n-1} \to J^{n+1}$$

Since $I^\bullet$ is exact we have $I^n \big/ \operatorname{Im} \mathrm{d}_I^{n-1} \cong \operatorname{Im} \mathrm{d}_I^n \subseteq I^{n+1}$. Therefore since J^{n+1} is injective the map $\operatorname{Im} \mathrm{d}_I^{n-1} \to J^{n+1}$ can be extended to the whole I^{n+1} giving a morphism $f^{n+1} \colon I^{n+1} \to J^{n+1}$. Diagramatically:



This completes the existence part of the proof.

To see uniqueness, suppose that we have two liftings $f^\bullet, g^\bullet \colon I^\bullet \to J^\bullet$ of f. We want to find an homotopy $h^n \colon I^n \to J^{n-1}$ such that for all $n \in \mathbb{N}$

$$\phi^n \coloneqq f^n - g^n = \mathrm{d}_J^{n-1} \circ h^n + h^{n+1} \circ \mathrm{d}_I^n$$

Again, we will do this by induction over n.

For the base case $n = 1$ note that $\phi^{-1} = f - f = 0$ (since both $f^\bullet$ and $g^\bullet$ are liftings of f). Therefore

$$\phi^0 \circ \varepsilon = \eta \circ (f - f) = 0$$

Since $\phi^0 \circ \varepsilon = 0$ we have that ϕ^0 factors through $\operatorname{coker} \varepsilon_1 \cong \operatorname{Im} \mathrm{d}_I^0$. Using that J^0 is injective we can construct $h^1 \colon I^1 \to J^0$ such that $h^1 \circ \mathrm{d}_I^0 = \phi^0$.

Assume now that we have constructed $h^i \colon I^i \to J^{i-1}$ for $i \in \{1, \dots, n\}$. We want to construct $h^{n+1} \colon I^{n+1} \to J^n$ such that

$$\psi \coloneqq \phi^n - \mathrm{d}_J^{n-1} \circ h^n = h^{n+1} \circ \mathrm{d}_I^n$$

Note that

$$\begin{aligned}
\psi \circ \mathrm{d}_I^{n-1} &= \phi^n \circ \mathrm{d}_I^{n-1} - \mathrm{d}_J^{n-1} \circ h^n \circ \mathrm{d}_I^{n-1} \\
&= \phi^n \circ \mathrm{d}_I^{n-1} - \mathrm{d}_J^{n-1} \circ \left(\phi^{n-1} - \mathrm{d}_J^{n-2} \circ h^{n-1}\right) \\
&= \left(\phi^n \circ \mathrm{d}_I^{n-1} - \mathrm{d}_J^{n-1} \circ \phi^{n-1}\right) + \left(\mathrm{d}_J^{n-1} \circ \mathrm{d}_J^{n-2}\right) \circ h^{n-1} \\
&= 0
\end{aligned}$$

Therefore ψ vanishes on $\operatorname{Im} \mathrm{d}_I^{n-1}$ and thus it factors through $I^n / \operatorname{Im} \mathrm{d}_I^{n-1} \cong \operatorname{Im} \mathrm{d}_I^n \subseteq I^{n+1}$. Using that J^n is injective this map extends to $h^{n+1} \colon I^{n+1} \to J^n$. □

In particular we get if two injective complexes resolve the same object $X \in \mathsf{C}$ their cohomology is the same.

Corollary 7.1.1. Let $I^\bullet, J^\bullet$ be two injective resolutions of an object $X \in \mathsf{C}$. Then

$$\mathrm{H}^\bullet(I^\bullet) \cong \mathrm{H}^\bullet(J^\bullet)$$

Proof. Taking $\mathbb{1}_X$ in Theorem 7.1.2 we get that $I^\bullet$ and $J^\bullet$ are homotopic. Therefore the result follows from Corollary 6.2.1. □

Before studying more in detail how injective objects interact with diagrams, we see that they are preserved by some right adjoint additive functors.

Proposition 7.1.2. Let $R \colon \mathsf{D} \to \mathsf{C}$ be a additive functor that is right adjoint to an exact functor $L \colon \mathsf{C} \to \mathsf{D}$. If $I \in \mathsf{D}$ is an injective object then $R(I)$ is also injective.
Dually, if $L \colon \mathsf{C} \to \mathsf{D}$ is an additive functor that is left adjoint to an exact functor $R \colon \mathsf{D} \to \mathsf{C}$ and $P \in \mathsf{C}$ is a projective object then $L(P)$ is also projective.

Proof. By Proposition 7.1.1 we want to show that $\operatorname{Hom}_{\mathsf{C}}(-, R(I))$ is exact.

Given a monomorphism $f \colon X \to Y$ in C we have

$$\begin{array}{ccc}
\operatorname{Hom}_{\mathsf{D}}(L(Y), I) & \xrightarrow{Lf^*} & \operatorname{Hom}_{\mathsf{D}}(L(A), I) \\
\downarrow{\scriptstyle \sim} & & \downarrow{\scriptstyle \sim} \\
\operatorname{Hom}_{\mathsf{C}}(Y, R(I)) & \xrightarrow[f^*]{} & \operatorname{Hom}_{\mathsf{C}}(A, R(I))
\end{array}$$

Since L is exact and I is injective Lf^* is surjective. Therefore f^* is surjective and the result follows. □

7.1.1 More diagram lemmas

Our first observation is that injective objects are closely related to split sequences.

Lemma 7.1.1. Let
$$0 \longrightarrow X \xrightarrow{f} Y \xrightarrow{g} Z \longrightarrow 0$$
be a short exact sequence in C with X injective. Then the sequence splits.

Proof. By definition taking $k = \mathbb{1}_X$ there exists $h : Y \to X$ such that $k \circ f = \mathbb{1}_X$. Then we just need to apply Proposition 6.1.1. □

Remark 7.1.1. If $F \colon \mathsf{C} \to \mathsf{D}$ is an additive functor between abelian categories and the hypothesis of Lemma 7.1.1 are satisfied the sequence
$$0 \longrightarrow F(X) \longrightarrow F(Y) \longrightarrow F(Z) \longrightarrow 0$$
splits and in particular it is exact.

Lemma 7.1.2. Let $X, Z \in \mathsf{C}$. Then $X \oplus Z$ is injective if and only if X and Z are injective.

Proof. This follows from the fact that if F, G are two additive functors between abelian categories then $F \oplus G$ is exact if and only if F and G are exact and $\mathrm{Hom}(-, I \oplus J) \cong \mathrm{Hom}(-, I) \oplus \mathrm{Hom}(-, J)$. □

Proposition 7.1.3. Let
$$0 \longrightarrow X \longrightarrow Y \longrightarrow Z \longrightarrow 0$$
be a short exact sequence in C with X and Z injective. Then Y is injective.

Proof. By Lemma 7.1.1 since X is injective the sequence splits. Therefore the sequence is isomorphic to
$$0 \longrightarrow X \longrightarrow X \oplus Z \longrightarrow Z \longrightarrow 0$$
Finally since X and Z are injective by Lemma 7.1.2 $X \oplus Z \cong Y$ is injective. □

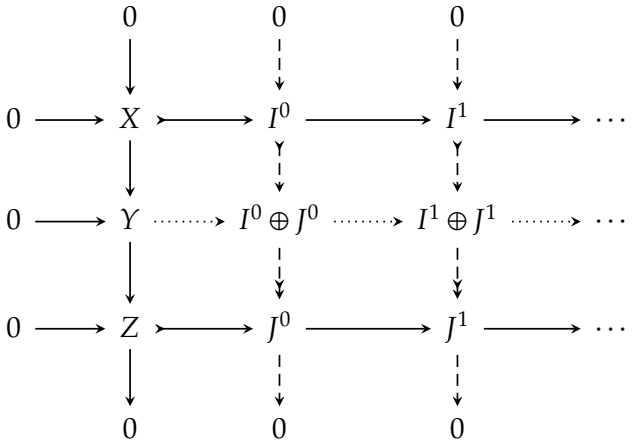
Using this we can prove the Horseshoe lemma, which allows us to extend short exact sequences between objects to short exact sequences between their injective resolutions.

Theorem 7.1.3 (Horseshoe lemma)**.** Consider a commutative diagram
$$\begin{array}{ccccccccc}
 & & 0 & & & & & & \\
 & & \downarrow & & & & & & \\
0 & \longrightarrow & X & \rightarrowtail & I^0 & \longrightarrow & I^1 & \longrightarrow & \cdots \\
 & & \downarrow & & & & & & \\
0 & \longrightarrow & Y & & & & & & \\
 & & \downarrow & & & & & & \\
0 & \longrightarrow & Z & \rightarrowtail & J^0 & \longrightarrow & J^1 & \longrightarrow & \cdots \\
 & & \downarrow & & & & & & \\
 & & 0 & & & & & &
\end{array}$$

where the column is exact and the rows are injective resolutions. Then $I^\bullet \oplus J^\bullet$ is an injective resolution of Y and the left-hand column lifts to an exact sequence of complexes

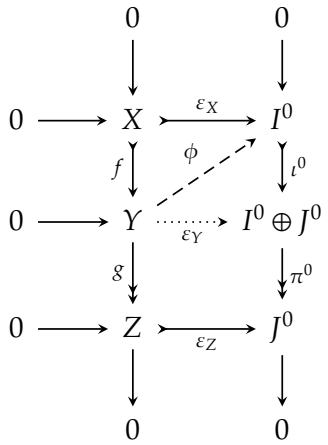
$$0 \longrightarrow I^\bullet \xrightarrow{\iota} I^\bullet \oplus J^\bullet \xrightarrow{\pi} J^\bullet \longrightarrow 0$$

where $\iota^n : I^n \rightarrowtail I^n \oplus J^n$ and $\pi^n : I^n \oplus J^n \twoheadrightarrow J^n$ are the natural inclussion and projection respectively. Diagramatically



Proof. We will do it by induction over n. Note that for all $n \in \mathbb{N}$ by Lemma 7.1.2 $I^n \oplus J^n$ is injective.

For the base case $n = 0$ we have



Since I^0 is injective and $f : X \rightarrowtail B$ is a monomorphism there exists a morphism $\phi : Y \to I^0$ such that $\phi \circ f = \varepsilon_X$. Define

$$\begin{aligned} \varepsilon_Y : Y &\to I^0 \oplus J^0 \\ y &\mapsto (\phi, \varepsilon_Z \circ g)(y) \end{aligned}$$

Let's see that the top and bottom square commute.

For the top one, let $x \in X$. On the one hand $\left(\iota^0 \circ \varepsilon_X\right)(x) = (\varepsilon(x), 0)$. On the other using that $\phi \circ f = \varepsilon_X$ and $g \circ f = 0$ we have

$$\left(\varepsilon_Y \circ f\right)(x) = \left(\phi \circ f, \varepsilon_Z \circ g \circ f\right)(y) = (\varepsilon(x), 0)$$

Now, to see the commutativity of the bottom square note that for all $y \in Y$

$$\left(\pi^0 \circ \varepsilon_Y\right)(y) = \left(\varepsilon_Z \circ g\right)(y)$$

It only remains to see that ε_Y is a monomorphism. Suppose that $\varepsilon_Y(y) = 0$. Then $\phi(y) = 0$ and $\left(\varepsilon_Z \circ g\right)(y) = 0$. Since ε_Z is a monomorphism $g(y) = 0$. Because $\ker g = \operatorname{Im} f$ there exists $x \in X$ such that $f(x) = y$. Then $\left(\phi \circ f\right)(x) = \varepsilon_X(x) = 0$. Using that ε_X is a monomorphism $x = 0$, which means that $y = f(0) = 0$ and thus ε_Y is a monomorphism.

This completes the base case.

We have the following diagram:

$$\begin{array}{ccccccccc}
0 & \longrightarrow & X & \xrightarrow{f} & Y & \xrightarrow{g} & Z & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle \varepsilon_X} & & \downarrow{\scriptstyle \varepsilon_X} & & \downarrow{\scriptstyle \varepsilon_X} & & \\
0 & \longrightarrow & I^0 & \xrightarrow{\iota^0} & I^0 \oplus J^0 & \xrightarrow{\pi^0} & J^0 & \longrightarrow & 0
\end{array}$$

Since the vertical maps are monomorphisms their kernels are 0. If we denote by X^1, Y^1 and Z^1 their cokernels by the Snake lemma we get a short exact sequence:

$$0 \longrightarrow X^1 \longrightarrow Y^1 \longrightarrow Z^1 \longrightarrow 0$$

Furthermore the shifted complexes $I^{\geq 1}$ and $J^{\geq 1}$ are injective resolutions of X^1 and Z^1 respectively. Therefore we are in the exact same situation as in the base case and by induction we can construct $I^n \oplus J^n$ and short exact sequences

$$0 \longrightarrow X^n \longrightarrow Y^n \longrightarrow Z^n \longrightarrow 0$$

for all $n \geq 0$.

To finish the proof we need to define differentials $\mathrm{d}_Y^n : I^n \oplus J^n \to I^{n+1} \oplus J^{n+1}$. We define these as by composing the quotient map $I^n \oplus J^n \to Y^{n+1}$ with the monomorphism $Y^{n+1} \rightarrowtail I^{n+1} \oplus J^{n+1}$ constructed in the inductive step. In matrix form

$$\mathrm{d}_Y^n := \begin{pmatrix} \mathrm{d}_X^n & \delta^n \\ 0 & \mathrm{d}_Z^n \end{pmatrix}$$

where d_X and d_Z are the differentials of the resolutions $I^\bullet$ and $J^\bullet$ and δ^n is the connecting morphism from the Snake lemma. By construction we have $\ker \mathrm{d}_Y^n = \operatorname{Im} \mathrm{d}_Y^{n-1}$, making $I^\bullet \oplus J^\bullet$ exact. □

> **Remark 7.1.2.** In the proof of Theorem 7.1.3 we have made implicit use of the Freyd-Mitchell embedding theorem. A diagramatic proof of (the dual of) this result can be seen at [Wei94, Lemma 2.8.8].

7.2 Derived functors

In this section C and D will be abelian categories, which we will assume that have enough injectives.

We will consider left-exact functors $F\colon \mathsf{C} \to \mathsf{D}$, but there are dual definitions and statements for right-exact functors assuming C and D have enough projectives.

If

$$0 \to X \xrightarrow{f} Y \xrightarrow{g} Z \to 0$$

is a short exact sequence in C we have an associated exact sequence.

$$0 \to F(X) \xrightarrow{Ff} F(Y) \xrightarrow{Fg} F(Z)$$

Our objective is to see how we can extend this sequence to the right. A naive approach would be to do something like this

$$0 \to F(X) \xrightarrow{Ff} F(Y) \xrightarrow{Fg} F(Z) \to \operatorname{Coim} Fg \to 0$$

However we lose to much information by doing so.

The machinery that enables us to extend this sequences are derived functors.

Definition 7.2.1. Let C be an abelian category with enough injectives. Let $F\colon \mathsf{C} \to \mathsf{D}$ be a left-exact functor. We define the i-th **right derived functor** of F

$$\mathrm{R}^i F\colon \mathsf{C} \to \mathsf{D}$$

as, for any object $X \in \mathsf{C}$:

$$\mathrm{R}^i F(X) \coloneqq \mathrm{H}^i(F(I^\bullet))$$

where $I^\bullet$ is an injective resolution of X, and for any morphism $f : X \to Y$

$$\mathrm{R}^i Ff \coloneqq \mathrm{H}^i(Ff^\bullet)$$

where $f^\bullet$ is a lifting of f.

Remark 7.2.1. Since F is left exact the sequence $0 \to F(X) \to F(I^0) \to F(I^1)$ is exact. Therefore we always have

$$\mathrm{R}^0 F(X) \cong F(X)$$

Furthermore, under this identification if $f : X \to Y$ is a morphism in C then

$$\mathrm{R}^0 Ff = Ff$$

Remark 7.2.2. If $X \in \mathsf{C}$ is injective then for all $i > 0$

$$\mathrm{R}^i F(X) = 0$$

Indeed, one can resolve X by $0 \to X \to X \to 0$.

As a sanity check, we see that taking the right derived functor still gives us an additive functor.

Proposition 7.2.1. For all $i \geq 0$ the right derived functors $\mathrm{R}^i F\colon \mathsf{C} \to \mathsf{D}$ are additive.

Proof. Let $f, g\colon X \to Y$ be two morphisms in C. Clearly, if $f^\bullet$ and $g^\bullet$ are liftings of f and g respectively then $f^\bullet + g^\bullet$ is a lifting of $f+g$. Therefore using that F and H^i are additive

$$\mathrm{R}^i F(f+g) = \mathrm{H}^i(F(f^\bullet + g^\bullet)) = \mathrm{H}^i\left(Ff^\bullet + Fg^\bullet\right) = \mathrm{H}^i(Ff^\bullet) + \mathrm{H}^i(Fg^\bullet) = \mathrm{R}^i F f + \mathrm{R}^i F g$$

□

Having seen this and using the properties of injective objects we now prove that this machinery works for our goal.

Theorem 7.2.1. Let

$$0 \longrightarrow X \longrightarrow Y \longrightarrow Z \longrightarrow 0$$

be a short exact sequence in C. There exists a long exact sequence:

$$\begin{array}{l} 0 \longrightarrow F(X) \longrightarrow F(Y) \longrightarrow F(Z) \\ \quad\hookrightarrow \mathrm{R}^1 F(X) \longrightarrow \mathrm{R}^1 F(Y) \longrightarrow \mathrm{R}^1 F(Z) \\ \quad\hookrightarrow \mathrm{R}^2 F(X) \longrightarrow \mathrm{R}^2 F(Y) \longrightarrow \cdots \end{array}$$

Proof. Choose injective resolutions $I^\bullet$ and $J^\bullet$ of X and Z respectively. By the Horseshoe lemma $I^\bullet \oplus J^\bullet$ is an injective resolution of Y and we have a short exact sequence of injective complexes

$$0 \longrightarrow I^\bullet \longrightarrow I^\bullet \oplus J^\bullet \longrightarrow J^\bullet \longrightarrow 0$$

Since I^n is injective by Lemma 7.1.1 the sequences

$$0 \longrightarrow I^n \longrightarrow I^n \oplus J^n \longrightarrow J^n \longrightarrow 0$$

split. Therefore since F is additive we have a short exact sequence

$$0 \longrightarrow F(I^\bullet) \longrightarrow F(I^\bullet \oplus J^\bullet) \longrightarrow F(J^\bullet) \longrightarrow 0$$

Writing its corresponding long exact sequence in cohomology we get

$$\begin{array}{l} 0 \longrightarrow F(X) \longrightarrow F(Y) \longrightarrow F(Z) \\ \quad\hookrightarrow \mathrm{R}^1 F(X) \longrightarrow \mathrm{R}^1 F(Y) \longrightarrow \mathrm{R}^1 F(Z) \\ \quad\hookrightarrow \mathrm{R}^2 F(X) \longrightarrow \mathrm{R}^2 F(Y) \longrightarrow \cdots \end{array}$$

□

If we tried to compute right derived functors directly using the definition we would soon run into problems. Indeed even though injective resolutions exist we don't know a priori how complicated the injective objects of our category are.

As we are going to see, the computation of right derived functors on objects can be done by replacing injective resolutions with right resolutions made of objects with vanishing cohomology.

Definition 7.2.2. An object $Q \in \mathsf{C}$ is **F-acyclic** if for all $i > 0$

$$\mathrm{R}^i F(Q) = 0$$

An **F-acyclic resolution** of an object $X \in \mathsf{C}$ is a right resolution $X \to Q^\bullet$ made of F-acyclic objects.

Theorem 7.2.2. Let $Q^\bullet$ be an F-acylic resolution of $X \in \mathsf{C}$. Then for all $i \geq 0$

$$\mathrm{R}^i F(X) \cong \mathrm{H}^i(F(Q^\bullet))$$

Proof. Let $I^\bullet$ be an injective resolution of X. By Theorem 7.1.2 there exists a morphism $f^\bullet \colon Q^\bullet \to I^\bullet$ lifting $\mathbb{1}_X$:

$$\begin{array}{ccc} X & \longrightarrow & Q^\bullet \\ {\scriptstyle \mathbb{1}_X}\downarrow & & \downarrow{\scriptstyle f^\bullet} \\ X & \longrightarrow & I^\bullet \end{array}$$

This gives a morphism of complexes $Ff \colon F(Q^\bullet) \to F(I^\bullet)$, which induces maps on cohomology

$$\phi_i \colon \mathrm{H}^i(F(Q^\bullet)) \to \mathrm{H}^i(F(I^\bullet)) = \mathrm{R}^i F(X)$$

for all $i \geq 0$. We will prove that ϕ_i are isomorphisms by induction over i.

For the base case $i = 0$ since F is left exact the sequences $0 \to F(X) \to F(Q^0) \to F(Q^1)$ and $0 \to F(X) \to F(I^0) \to F(I^1)$ are exact. Therefore

$$\mathrm{H}^0(F(Q^\bullet)) \cong F(X) \cong \mathrm{H}^0(F(I^\bullet))$$

Suppose now that ϕ_i is an isomorphism for all $i \in \{0, \dots, n\}$. Consider the short exact sequence

$$0 \longrightarrow X \longrightarrow Q^0 \longrightarrow Z^1 \longrightarrow 0$$

where $Z^1 := \mathrm{Im}(Q^0 \to Q^1)$. By Theorem 7.2.1 we have a long exact sequence

$$\begin{array}{ccccccc} 0 & \longrightarrow & F(X) & \longrightarrow & F(Q^0) & \longrightarrow & F(Z^1) \\ & & & & & \xrightarrow{\delta^0} & \\ & & \mathrm{R}^1 F(X) & \longrightarrow & \mathrm{R}^1 F(Q^0) & \longrightarrow & \mathrm{R}^1 F(Z^1) \\ & & & & & \xrightarrow{\delta^1} & \\ & & \mathrm{R}^2 F(X) & \longrightarrow & \mathrm{R}^2 F(Q^0) & \longrightarrow & \cdots \end{array}$$

Since Q^0 is F-acyclic for all $i > 0$ we get an isomorphism

$$\mathrm{H}^{i+1}(F(X)) = \mathrm{R}^{i+1} F(X) \cong \mathrm{R}^i F(Z^1) = \mathrm{H}^i(F(Z^1))$$

On the other hand, the truncated complex

$$0 \longrightarrow Z^1 \longrightarrow Q^1 \longrightarrow Q^2 \longrightarrow \cdots$$

is an F-acyclic resolution of Z^1. Applying the induction hypothesis for all $j \in \{1, \dots, n\}$ we have

$$\mathrm{R}^j F(Z^1) \cong \mathrm{H}^j(F(Q^{\bullet,\geq 1})) = \mathrm{H}^{j+1}(F(Q^\bullet))$$

In particular for $j = n$ combining the two isomorphisms

$$\mathrm{R}^{n+1} F(X) \cong \mathrm{H}^{n+1} F(Q^\bullet)$$

It remains to check the case where $i = 1$. From the exact sequence

$$0 \longrightarrow Z^1 \longrightarrow Q^1 \longrightarrow Z^2 \longrightarrow 0$$

where $Z^2 \coloneqq \mathrm{Im}\left(Q^1 \to Q^2\right)$ using that F is left exact we get

$$F(Z^1) \cong \ker\left(F(Q^1) \to F(Q^2)\right)$$

Therefore using the exact sequence

$$F(Q^0) \longrightarrow F(Z^1) \longrightarrow \mathrm{R}^1 F(X) \longrightarrow 0$$

we have

$$\begin{aligned} \mathrm{R}^1 F(X) &\cong {F(Z^1)}\Big/{\mathrm{Im}\left(F(Q^0) \to F(Q^1)\right)} \\ &\cong {\ker\left(F(Q^1) \to F(Q^2)\right)}\Big/{\mathrm{Im}\left(F(Q^0) \to F(Q^1)\right)} \\ &= \mathrm{H}^1 F(Q^\bullet) \end{aligned}$$

□

Remark 7.2.3. One may be tempted to define right derived functors directly by computing the cohomology using acylic resolution instead of injective ones. However using this definition is not clear (or at least it is not clear to the author) how would one define its action on a morphism since for non-injective objects the existence of a lifting is not guaranteed.

We end this chapter by considering the composition of right derived functors. Since in general $F(I)$ will not be injective even if I is (see Proposition 7.1.2), we need to add more conditions on the categories and functors for the composition to be well defined.

Theorem 7.2.3. Let $F\colon \mathsf{C} \to \mathsf{D}$ and $G\colon \mathsf{D} \to \mathsf{E}$ be two left exact functors of abelian categories. Suppose that C and D have enough injectives.

1. If G is exact then $\mathrm{R}^i(G \circ F) \cong G \circ \mathrm{R}^i F$.
2. If F is exact then there is a natural morphism $\mathrm{R}^i(G \circ F) \to \mathrm{R}^i G \circ F$.
3. Let $X \in \mathsf{C}$ be F-acyclic and suppose that F sends injective objects to G-acyclic ones. Then $\mathrm{R}^i(G \circ F)(X) \cong (\mathrm{R}^i G)(F(X))$.

Proof. For $X \in \mathsf{C}$ let $X \xrightarrow{\text{q.is}} I^\bullet$ be an injective resolution of X. Then

$$\mathrm{R}^i(G \circ F)(X) = \mathrm{H}^i\left((G \circ F)(I^\bullet)\right)$$

1. If G is exact then we have an isomorphism
$$\mathrm{H}^i\left((G \circ F)(I^\bullet)\right) \cong G(\mathrm{H}^i(F(I^\bullet)))$$
2. Consider an injective resolution $F(X) \xrightarrow{\text{q.is}} J^\bullet$ of $F(X)$. By Theorem 7.1.2 there exists a morphism $F(I^\bullet) \to J^\bullet$. Applying G we get a morphism
$$(G \circ F)(I^\bullet) \to G(J^\bullet)$$
Therefore since $\mathrm{H}^i\left((G \circ F)(I^\bullet)\right) = \mathrm{R}^i(G \circ F)(X)$ and $\mathrm{H}^i(G(J^\bullet)) = \mathrm{R}^i G(F(X))$ the result follows.
3. By hypothesis $F(I^\bullet)$ is quasi-isomorphic to $F(X)$. Therefore $(\mathrm{R}^i G)(F(X)) \cong \mathrm{H}^i(G(F(I^\bullet)))$ by Theorem 7.2.2 and $\mathrm{H}^i(G(F(I^\bullet))) \cong \mathrm{R}^i(G \circ F)(X)$. □

PART III

Sheaves

Introduction

The transition from classical geometry and topology to modern algebraic geometry and complex manifold theory in the 20th century was fundamentally driven by a single, profound challenge: how to systematically bridge the gap between local data and global structures. In classical analysis, one frequently encounters properties that are inherently local such as the analyticity of a function, the existence of local solutions to a differential equation, or the triviality of a coordinate patch. However, the most common topological and geometric invariants of a space are global, such as the Euler characteristic or the fundamental group.

Before sheaf theory, mathematicians wrestled with these transitions using ad hoc, often cumbersome tools. A primary example was Henri Poincaré's and later Georges de Rham's work on differential forms, which demonstrated that global topological information about a manifold could be extracted from local analytic data (smooth forms). Similarly, in the late 19th and early 20th centuries, the Italian school of algebraic geometry and mathematicians like Karl Weierstrass and Henri Poincaré encountered deep obstructions when attempting to stitch local meromorphic functions into global ones (the Mittag-Leffler and Cousin problems).

The explicit breakthrough came during World War II. While imprisoned in the prisioner camp Oflag XVII-A in Austria, French mathematician Jean Leray formulated the concept of a *faisceau* (sheaf) around 1944–1945. Leray's insight was born out of a desire to compute the homology of a space by studying the local behavior of functions or coefficients that vary continuously across it.

Shortly thereafter, Henri Cartan and his seminar in Paris refined Leray's definitions, recognizing that sheaves provided the exact language needed to study complex analytic geometry. In 1955, Jean-Pierre Serre published his seminal paper Faisceaux Algébriques Cohérents (FAC) [Ser55], which adapted sheaf theory from the Euclidean topology of complex manifolds to the Zariski topology of abstract algebraic varieties. This pivotal move detached algebraic geometry from its reliance on ambient projective spaces.

Finally, Alexander Grothendieck completely subsumed the theory into category theory in his monumental Tôhoku paper [Gro]. Grothendieck showed that the category of sheaves of abelian groups on a topological space possesses enough injectives, thereby providing the foundations for sheaf cohomology as a derived functor.

This part of the thesis builds the structural machinery of sheaf theory from first principles, establishing the necessary prerequisites for the homological treatments in later chapters.

- Chapter 8 (Presheaves and Sheaves) transitions from the concrete motivation of smooth functions on manifolds to abstract sheaf categories. We define presheaves, sheaves, and stalks, and construct the sheafification functor which forces an arbitrary presheaf to satisfy the local-to-global principle. We conclude by proving that the category of sheaves of abelian groups forms an abelian category, setting the stage for homological algebra.
- Chapter 9 (Operations on Sheaves) transitions from a static space to continuous transformations. We construct adjoint functors associated with a continuous map. We will also study specialized geometric embeddings, such as open and closed immersions and see how many classical sheaf constructions can be viewed through this lens.

The study of sheaves and their geometric operations is grounded in several foundational texts and modern reference manuals, each offering a distinct perspective on the theory.

For a comprehensive introduction to the classical topological and homological aspects of the theory, the classical works of Godement [God98] and Bredon [Bre97] remain essential, while Gallier and Quaintance [GQ22] provide a highly accessible entry point into modern sheaf cohomology. The transition of these concepts into abstract algebraic geometry is developed in Hartshorne's classic textbook [Har77] and the intuitive, comprehensive lecture notes of Vakil [Vak25]. For readers seeking a more rigorous and category-theoretic treatment of the subject there are the volumes by Kashiwara and Schapira [KS90; KS06] and Grothendieck's Tohôku paper [Gro]. For an exposition focused more on concrete applications rather than in the general theory one can check Rosiak [Ros22].

For an extensive account of the historical development of sheaf theory from its wartime origins under Jean Leray to its modern formulation, the reader is strongly encouraged to consult the detailed preamble in Kashiwara and Schapira [KS90].

Chapter 8

Presheaves and sheaves

In previous chapters, we developed the framework of category theory and homological algebra to study mathematical objects and their global transformations. However, many geometric and topological problems are inherently local. To bridge the gap between local properties and global structures, we introduce the concepts of presheaves and sheaves.

Intuitively, a presheaf allows us to attach data (such as modules, groups, or rings) to the open sets of a topological space in a way that respects the restriction from larger open sets to smaller ones. A sheaf goes a step further by requiring that this data satisfies a local-to-global principle: sections that are locally compatible can be uniquely glued together into a global section.

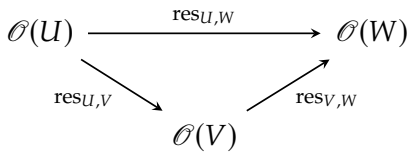
This chapter begins with a motivating example centered around smooth functions, tracking how these concrete structures naturally abstract into general definitions. We will explore the local properties of sheaves through the construction of stalks, establish the sheafification functor as a left adjoint to the forgetful functor, and conclude by examining the rich homological properties of sheaves valued in an abelian category.

8.1 A motivating example

We begin this chapter by giving a concrete example of sheaves following [Vak25] that will help us motivate the abstractions.

Consider smooth (C^∞) functions on the topological space $X = \mathbb{R}^n$ (or more generally on a manifold X). The sheaf of smooth functions on X is the data of all smooth functions on all open subsets of X. Note that for each open subset $U \subseteq X$ we have an $\mathbb{R}$-module of smooth functions $\mathscr{O}(U)$.

Given an smooth function f in an open set U we can restrict it to an open subset $V \subseteq U$. In other words, we have a restriction map $\mathrm{res}_{U,V} \colon \mathscr{O}(U) \to \mathscr{O}(V)$. Furthermore if we have $W \subseteq V \subseteq U$ open subsets restricting f to V and then to W is the same as restrinting f directly to W, which means that the following diagram is commutative



Notice how if we replace smooth functions with, say, continuous, real-analytic or bounded functions this example would work analogously. Therefore all of this "nice" types of functions share some common properties. The abstraction of this data — the collection of $\mathbb{R}$-modules plus the restriction maps — is what we will call a presheaf (of $\mathbb{R}$-modules) over X.

However in this particular our presheaf is more rich. Indeed, lets consider an open cover $\mathcal{U} := \{U_i\}_{i\in\mathcal{I}}$ of an open set $U \subseteq X$.

On the one hand if we have two smooth function $f_1, f_2 \in \mathscr{O}(U)$ such that both agree on every U_i then they must be the same function to begin with. In the language of the restriction maps we have that if $\mathrm{res}_{U_i,U_j} f_1 = \mathrm{res}_{U_i,U_j} f_2$ for all $U_i, U_j \in \mathcal{U}$ then $f_1 = f_2$. This means that we can identify functions on an open set by looking at them on a covering by smaller open set.

On the other suppose now we have a collection of smooth function $f_i \in \mathscr{O}(U_i)$ for all $i \in \mathcal{I}$ such that they agree on all pairwise overlaps. Then we can *glue* them together forming an smooth function f in all of U. Again, with the notation of the restriction maps we have that if $\mathrm{res}_{U_i\cap U_j} f_i = \mathrm{res}_{U_i\cap U_j} f_j$ for all $i, j \in \mathcal{I}$ there exists $f \in \mathscr{O}(U)$ such that $\mathrm{res}_{U_i} f = f_i$.

This two properties are what will differentiate a presheaf from a sheaf. Note that they still hold for continous or real-analytic functions, but they no longer work in the bounded functions case (the second property fails).

8.2 Presheaves and sheaves

We are now ready to start abstracting this data into a more general setting, where we can use the full power of the category theory we have developed previously.

Definition 8.2.1 (Presheaf). Let X be a topological space and let C be a category. The category of C-valued presheaves over X is the functor category

$$\mathsf{PSh}_X(\mathsf{C}) := \mathsf{Fct}(\mathsf{Open}(X)^{\mathrm{op}}, \mathsf{C})$$

Less abstractly, a C-valued presheaf $\mathcal{F}$ is the collection of data:

- For each open set $U \subseteq X$ an object $\mathcal{F}(U) \in \mathsf{C}$ called *sections of* $\mathcal{F}$ *over* U.
- For every inclusion of open sets $V \subseteq U$ a morphism $\mathrm{res}_{U,V} : \mathcal{F}(U) \to \mathcal{F}(V)$ in C called *restriction* such that
  - For every open set $U \subseteq X$, $\mathrm{res}_{U,U} = \mathbb{1}_U$.
  - For every inclusion of open sets $W \subseteq V \subseteq U$, $\mathrm{res}_{U,V} \circ \mathrm{res}_{V,W} = \mathrm{res}_{U,W}$.

A morphism of presheaves $\phi : \mathcal{F} \to \mathcal{G}$ is a collection of morphisms

$$\{\phi(U) : \mathcal{F}(U) \to \mathcal{G}(U)\}_{U\in\mathsf{Open}(X)}$$

in C making the following diagram commute for all $V \subseteq U$

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\phi(U)} & \mathcal{G}(U) \\ {\scriptstyle \mathrm{res}_{U,V}}\downarrow & & \downarrow{\scriptstyle \mathrm{res}_{U,V}} \\ \mathcal{F}(V) & \xrightarrow[\phi(V)]{} & \mathcal{G}(V) \end{array}$$

Example 8.2.1. Smooth, real-analytic and bounded functions on $X = \mathbb{R}^n$ (or more generally in a manifold X) are presheaves.

Since presheaves are particular instances of functors we have already seen several examples back in chapter 2. One example that will be useful later is the following.

Definition 8.2.2 (Constant presheaf)**.** Let X be a topological space. For any object $A \in \mathsf{C}$ the **constant presheaf** with value A, denoted $\underline{A}_X^{\mathrm{pre}}$ is the C-valued presheaf associated to the constant functor

$$\begin{array}{ccc} \underline{A}_X^{\mathrm{pre}} \colon \mathsf{Open}(X)^{\mathrm{op}} & \longrightarrow & \mathsf{C} \\ U & & A \\ {\scriptstyle V\subseteq U}\downarrow & \longmapsto & \downarrow{\scriptstyle \mathbb{1}_A} \\ V & & A \end{array}$$

Note that this defines a functor

$$\underline{-}_X^{\mathrm{pre}} \colon \mathsf{C} \to \mathsf{PSh}_X(\mathsf{C})$$

We now turn our attention to a particular class of presheaves.

Instead of working on a general category C, to simplify the exposition we will work with categories which are "sets with an algebraic structure" (for example Grp, Ab, Ring, ... but not Top).

Definition 8.2.3. A **type of algebraic structure** is a category C together with a functor $F \colon \mathsf{C} \to \mathsf{Set}$ verifying:

1. F is faithful.
2. C has limits and F commutes with them.
3. C has direct limits and F commutes with them.
4. F reflects isomorphisms.

Example 8.2.2. The categories of pointed sets, (abelian) groups, monoids, rings R-modules, and Lie algebras – together with their respective forgetful functors – are types of algebraic structures.

We state some properties of this type of categories without proof.

Lemma 8.2.1 ([Stacks, Tag 007O])**.** Let (C, F) be a type of algebraic structure. Then:

1. C has a terminal object 1 and $F(1) = \{*\}$.
2. C has products and $F\left(\prod X_i\right) = \prod F\left(X_i\right)$.
3. C has pullbacks and $F(X \times_Z Y) = F(X) \times_{F(Z)} F(Y)$.
4. C has equalisers and if $\mathrm{eq} \to X$ is the equaliser of $f, g \colon X \rightrightarrows Y$ then $F(\mathrm{eq}) \to F(X)$ is the equaliser of $Ff, Fg \colon F(X) \rightrightarrows F(Y)$.
5. $X \to Y$ is a monomorphism if and only if $F(X) \to F(Y)$ is injective.
6. If $Ff \colon F(X) \to F(Y)$ is surjective then f is an epimorphism.
7. Given $X_1 \to X_2 \to \cdots$, $\varinjlim X_i$ exists and $F(\varinjlim X_i) = \varinjlim F(X_i)$. And more generally for any direct limit.

8. Direct limits commute with limits.

Therefore in everything that follows after this point we will implicitly assume that C is a type of algebraic structure every time we talk about sheaves.

Definition 8.2.4. Let X be a topological space. The category of **C-valued sheaves** over X is the full subcategory $\mathsf{Sh}_X(\mathsf{C})$ of $\mathsf{PSh}_X(\mathsf{C})$ whose objects verify, for every open set $U \subseteq X$ and every open cover $\{U_i\}_{i\in\mathcal{I}}$ of U

SH.1) **Identity**: Given two sections $s, t \in \mathscr{F}(U)$ such that for every $i \in \mathcal{I}$ $\quad s|_{U_i} = t|_{U_i}$ we have that $s = t$.

SH.2) **Gluability**: Given a collection of sections $\{s_i \mid s_i \in \mathscr{F}(U_i)\}_{i\in\mathcal{I}}$ that agree on pairwise intersections $(\forall i, j \in \mathcal{I} \quad s_i|_{U_i\cap U_j} = s_j|_{U_i\cap U_j})$ there exists a section $s \in \mathscr{F}(U)$ such that $\forall i \in \mathcal{I} \quad s|_{U_i} = s_i$.

Notation. From now on, to distinguish a presheaf from a sheaf we will denote the former by $\mathcal{F}, \mathcal{G}, \mathcal{H}, \ldots$ and the latter by $\mathscr{F}, \mathscr{G}, \mathscr{H}, \ldots$

Example 8.2.3. Smooth and continuous function form a sheaf of $\mathbb{R}$-modules over $X := \mathbb{R}$. However, bounded functions $\mathcal{B}$ only form a presheaf.
Indeed if we consider the open covering $\mathcal{U} := \{]n, n+1[\}_{n\in\mathbb{N}}$ of $\mathbb{R}$ and $f_n(x) := n$ we have that $f_n \in \mathcal{B}(]n, n+1[)$ for all $n \in \mathbb{N}$ but any *glueing* of these functions is clearly not bounded and thus SH.2 fails.

Remark 8.2.1. Axioms SH.1 and SH.2 can be interpreted as the following equaliser diagram:

$$\mathscr{F}(U) \longrightarrow \prod_{i\in\mathcal{I}} \mathscr{F}(U_i) \rightrightarrows \prod_{(i,j)\in\mathcal{I}\times\mathcal{I}} \mathscr{F}(U_i \cap U_j)$$

where $\{U_i\}_{i\in\mathcal{I}}$ is an open cover of U.
In particular taking the empty cover of $\varnothing \in \mathsf{Open}(X)$, since the empty product is terminal we have

$$\mathscr{F}(\varnothing) \longrightarrow 1 \rightrightarrows 1$$

i.e: $\mathscr{F}(\varnothing)$ is a terminal object of C.

Example 8.2.4 (Sheaves on a point)**.** Let $\{*\}$ be the one-point topological space. We are going see that there is an equivalence of categories

$$\mathsf{Sh}_{\{*\}}(\mathsf{C}) \simeq \mathsf{C}$$

Consider the functors

$$F\colon \mathsf{Sh}_{\{*\}}(\mathsf{C}) \to \mathsf{C} \qquad G\colon \mathsf{C} \to \mathsf{Sh}_{\{*\}}(\mathsf{C})$$

where $F(\mathscr{F}) := \mathscr{F}(\{*\})$ for all $\mathscr{F} \in \mathsf{Sh}_{\{*\}}(\mathsf{C})$ and for all $X \in \mathsf{C}$, $G(X)$ is the sheaf defined as

$$G(X)(U) := \begin{cases} X & U = \{*\} \\ 1 & U = \varnothing \end{cases}$$

with the obvious restriction maps. We have

$$(F \circ G)(X) = G \qquad (G \circ F)(\mathscr{F}) \cong \mathscr{F}$$

where the isomorphism comes from $\mathscr{F}(\varnothing)$ being terminal (but not necessarilly 1).

8.2.1 Stalks

The main difference between sheaves and presheaves is their local nature. We are going to see that there are "global" properties of sheaves that can be checked if we know all of the "local" information.

Before trying to abstract this, let's go back to our motivating example of smooth functions on $\mathbb{R}^n$. A germ of a smooth function at a point $p \in X$ is an object of the form (f, U) such that $p \in U$ open and $f \in \mathscr{O}(U)$, modulo the relation $(f, U) \sim (g, V)$ if there exists $p \in W \subseteq U \cap V$ open such that $f|_W = g|_W$. We call this set of germs the stalk of $\mathscr{O}$ at p and denote it by $\mathscr{O}_p$. It is easy to see that $\mathscr{O}_p$ has the structure of an $\mathbb{R}$-module. Note that here we have not used the two sheaf properties, so this definition works considering $\mathscr{O}$ either as a presheaf or as a sheaf.

Furthermore if $p \in U$ we get maps $\mathscr{O}(U) \to \mathscr{O}_p$, which hints us that we can model this as a colimit.

Definition 8.2.5. Let X be a topological space and let $\mathcal{F}$ be a C-valued presheaf. For all points $p \in X$ we define the **stalk** of $\mathcal{F}$ at p as

$$\mathcal{F}_p \coloneqq \varinjlim_{\substack{U \ni p \\ U \in \mathsf{Open}(X)}} \mathcal{F}(U)$$

Note that this comes with maps $\mathcal{F}(U) \to \mathcal{F}_p$ for all $U \subseteq X$ open such that $p \in U$. If $p \in U$ and $s \in \mathscr{F}(U)$ then the image of s in $\mathscr{F}_p$ is called the **germ** of s at p and denote it by s_p.
Since this is a direct limit following Example 3.2.5 we may describe $\mathcal{F}_p$ as

$$\mathcal{F}_p = \{(U, s) \mid p \in U, s \in \mathcal{F}(U)\} \Big/ \sim$$

where

$$(U, s) \sim (V, t) \iff \exists W \subseteq U \cap V \text{ open} \mid p \in W \wedge s|_W = t|_W$$

Using this characterization we may define the stalk of a morphism of presheaves $\phi : \mathcal{F} \to \mathcal{G}$ as

$$\begin{aligned} \phi_x \colon \mathcal{F}_x &\to \mathcal{G}_x \\ [(U, s)] &\mapsto [(U, \phi(s))] \end{aligned}$$

Therefore we have a stalk functor

$$-_p \colon \mathsf{PSh}_X(\mathsf{C}) \to \mathsf{C}$$

By using the same definition we can extend this to a functor

$$-_p \colon \mathsf{Sh}_X(\mathsf{C}) \to \mathsf{C}$$

Remark 8.2.2. By the properties of algebraic structures, stalks commute with finite limits.

The following lemma somewhat explains the agricultural terminology: sections of a sheaf can be seen as a collection of germs bundled together.

Lemma 8.2.2. Let $\mathscr{F}$ be a C-valued sheaf over a topological space X. For all open subsets $U \subseteq X$ the map $\mathscr{F}(U) \to \prod_{p\in U} \mathscr{F}_p$ is injective.
In other words, a sheaf is completely determined by its germs.

Proof. Let $s, t \in \mathscr{F}(U)$ such that for all $s_p = t_p$ in $\mathscr{F}_p$ for all $p \in U$.

By definition of stalk there exists an open neighbourhood $V_p \subseteq U$ of p such that $s|_{V_p} = t|_{V_p}$. Since this works for all $p \in U$, we have an open cover $\{V_p\}_{p\in U}$ of U.

Therefore since $\mathscr{F}$ is a sheaf by SH.1 we have $s = t$. □

Using this result we can now prove one of the most important theorems for sheaves. Roughly speaking, it states that knowing all of the "local" information is the same as knowing the "global" information.

Theorem 8.2.1. Let $\phi, \psi \colon \mathcal{F} \to \mathcal{G}$ be a morphism of presheaves where $\mathcal{F}$ is a C-valued presheaf and $\mathcal{G}$ is a C-valued sheaf. Then $\phi = \psi$ if and only if for all $p \in X$ the stalks at p are equal, i.e: $\phi_p = \psi_p$.

Proof. The direction [$\Longrightarrow$] is trivial.

For the [$\Longleftarrow$] direction suppose that $\phi_p = \psi_p$ for all $p \in X$ and consider the commutative diagram

$$\begin{array}{ccc} \mathcal{F}(U) & \overset{\phi}{\underset{\psi}{\rightrightarrows}} & \mathcal{G}(U) \\ {\scriptstyle i_{\mathcal{F}}}\downarrow & & \downarrow{\scriptstyle i_{\mathcal{G}}} \\ \prod_{p\in U} \mathcal{F}_p & \overset{\prod \phi_p}{\underset{\prod \psi_p}{\rightrightarrows}} & \prod_{p\in U} \mathcal{G}_p \end{array}$$

For all $s \in \mathcal{F}(U)$ we have

$$i_{\mathcal{G}}(\phi(U)(s)) = \big(\phi_p(s_p)\big)_{p\in U} = \big(\psi_p(s_p)\big)_{p\in U} = i_{\mathcal{G}}(\psi(U)(s))$$

Since $i_{\mathcal{G}}$ is injective we have

$$\phi(U)(s) = \psi(U)(s)$$

Therefore since this holds for all $s \in \mathcal{F}(U)$ and $U \subseteq X$ open we have that $\phi = \psi$. □

Example 8.2.5. Theorem 8.2.1 is false in the case of presheaves.
Indeed, if we consider $\mathcal{B}$ the presheaf of bounded functions on $\mathbb{R}$ and as morphisms the identity $\mathbb{1}_{\mathcal{B}} \colon \mathcal{B} \to \mathcal{B}$ and $\psi \colon \mathcal{B} \to \mathcal{B}$ defined as $\psi(U) = \mathbb{1}_U$ if U is bounded and the zero morphism otherwise since stalks are computed by looking at small neighbourhoods and every point $p \in \mathbb{R}$ admits infinitely small bounded neighbourhoods we have

$$(\mathbb{1}_{\mathcal{B}})_p = \psi_p = \mathbb{1}_{\mathcal{B}_p}$$

However, if we consider $f(x) := 1 \in \mathcal{B}(\mathbb{R})$ we get

$$\mathbb{1}_{\mathcal{B}}(\mathbb{R})(f) = 1 \neq 0 = \psi(\mathbb{R})(f)$$

The category of sheaves also has a nice characterization isomorphism, monomorphism and epimorphisms in terms of stalks that will be particularly useful once we study exact sequences of sheaves.

Proposition 8.2.1. Let $\phi\colon \mathscr{F} \to \mathscr{G}$ be a morphism of sheaves. Then ϕ is a isomorphism (resp. mono/epimorphism) if and only if for all $p \in X$ the induced stalk map $\phi_p\colon \mathscr{F}_p \to \mathscr{G}_p$ is an isomorphism (resp. mono/epimorphism).

Proof. We will check the isomorphism case.

$\implies$ Suppose that $\phi : \mathscr{F} \to \mathscr{G}$ is an isomorphism in $\mathsf{Sh}_X(\mathsf{C})$. Then there exists $\psi : \mathscr{G} \to \mathscr{F}$ such that

$$\psi \circ \phi = \mathbb{1}_{\mathscr{F}} \qquad \phi \circ \psi = \mathbb{1}_{\mathscr{G}}$$

Therefore since taking stalks is functorial for all $p \in X$ we have

$$\psi_p \circ \phi_p = \mathbb{1}_{\mathscr{F}_p} \qquad \phi_p \circ \psi_p = \mathbb{1}_{\mathscr{G}_p}$$

i.e: $\phi_p : \mathscr{F}_p \to \mathscr{G}_p$ is an isomorphism with inverse ψ_p.

$\impliedby$ Note that we only need to check the case where $\mathsf{C} = \mathsf{Set}$ since the functor $F\colon \mathsf{C} \to \mathsf{Set}$ reflects isomorphisms. To do this, we will see that for all open subsets $U \subseteq X$ the map $\phi(U)\colon \mathscr{F}(U) \to \mathscr{G}(U)$ is injective and surjective.
To see injectivity, let $s, t \in \mathscr{F}(U)$ such that $\phi(U)(s) = \phi(U)(t)$. For all $p \in U$ we have

$$\phi_p(s_p) = \big(\phi(U)(s)\big)_p = \big(\phi(U)(t)\big)_p = \phi_p(t_p)$$

Therefore since ϕ_p is injective $s_p = t_p$. Using sheaves are determined by its germs we get $s = t$ and thus $\phi(U)$ is injective.
To see surjectivity consider $t \in \mathscr{G}(U)$. For all $p \in U$ since ϕ_p is surjective there exists $\sigma_p \in \mathscr{F}_p$ such that $\phi_p(\sigma_p) = t_p$. By definition of stalk there exists an open neighbourhood $V_p \subseteq U$ of p and a section $s^{(p)} \in \mathscr{F}(V_p)$ such that $s^{(p)}_p = \sigma_p$.
By shrinking V_p if necessary we can assume that

$$\phi(V_p)(s^{(p)}) = t|_{V_p}$$

Now consider two such neighbourhoods V_p and V_q. On the intersection we have

$$\phi(V_p \cap V_q)(s^{(p)}|_{V_p \cap V_q}) = t|_{V_p \cap V_q} = \phi(V_p \cap V_q)(s^{(q)}|_{V_p \cap V_q})$$

Therefore since $\phi(V_p \cap V_q)$ is injective $s^{(p)}|_{V_p \cap V_q} = s^{(p)}|_{V_p \cap V_q}$. By the gluability axiom there exists a unique section $s \in \mathscr{F}(U)$ such that $s|_{V_p} = s^{(p)}$ for all $p \in U$.
Finally since $\phi(U)(s)|_{V_p} = \phi(V_p)(s^{(p)}) = t|_{V_p}$ for all $p \in U$ the uniqueness of the gluability axiom now on $\mathscr{G}$ implies that $\phi(U)(s) = t$ and thus $\phi(U)$ is surjective. □

8.3 Sheafification

We have established the definition of presheaves and sheaves. It is clear from the definitions that every sheaf is a presheaf. This defines a forgetful functor:

$$\iota\colon \mathsf{Sh}_X(\mathsf{C}) \hookrightarrow \mathsf{PSh}_X(\mathsf{C})$$

We want to associate to each presheaf a sheaf in a way that *respects* this forgetful functor.

In the language of category-theory, this is expressed by an adjunction.

Definition 8.3.1 (Sheafification)**.** Let X be a topological space and C a category. The **sheafification** (or associated sheaf) functor, denoted by $-^{\#}\colon \mathsf{PSh}_X(\mathsf{C}) \to \mathsf{Sh}_X(\mathsf{C})$, is the left adjoint to the inclusion functor $\iota\colon \mathsf{Sh}_X(\mathsf{C}) \hookrightarrow \mathsf{PSh}_X(\mathsf{C})$.
That is, for every presheaf $\mathcal{F} \in \mathsf{PSh}_X(\mathsf{C})$ and every sheaf $\mathcal{G} \in \mathsf{Sh}_X(\mathsf{C})$, there is a natural bijection:

$$\mathrm{Hom}_{\mathsf{Sh}_X(\mathsf{C})}(\mathcal{F}^{\#}, \mathcal{G}) \cong \mathrm{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(\mathcal{F}, \iota\mathcal{G})$$

The unit of this adjunction gives a canonical morphism $\theta\colon \mathcal{F} \mapsto \mathcal{F}^{\#}$ in $\mathsf{PSh}_X(\mathsf{C})$ which verifies that for any presheaf morphism $\phi\colon \mathcal{F} \to \mathcal{G}$ where $\mathcal{F}$ is a presheaf and $\mathcal{G}$ is a sheaf there exists a unique morphism $\psi\colon \mathcal{F}^{\#} \to \mathcal{G}$ such that $\psi \circ \theta = \phi$.

For this notion to be useful, we must prove that it exists. We will do this by constructing the sheafification of a presheaf $\mathcal{F}$, but before doing so we fix some terminology on constructing sheaves from a *bigger* one.

Definition 8.3.2. Let X be a topological space. A presheaf $\mathcal{F}$ is a **subpresheaf** of a presheaf $\mathcal{G}$ if for all open sets $U \subseteq X$ we have $\mathcal{F}(U) \subseteq \mathcal{G}(U)$ and the restriction maps of $\mathcal{G}$ induce the restriction maps of $\mathcal{F}$. If $\mathscr{F}$ and $\mathscr{G}$ are sheaves we say that $\mathscr{F}$ is a **subsheaf** of $\mathscr{G}$.

Construction (Godement construction of sheafification)**.** Let $\mathcal{F} \in \mathsf{PSh}_X(\mathsf{C})$.

For every open set $U \subseteq X$, let

$$\Lambda(\mathcal{F})(U) \coloneqq \prod_{p \in U} \mathcal{F}_p$$

Considering for $V \subseteq U$ the morphism

$$\begin{aligned} \mathrm{res}_{U,V}\colon \Lambda(\mathcal{F})(U) &\to \Lambda(\mathcal{F})(V) \\ \prod_{p \in U} \mathcal{F}_p &\mapsto \prod_{p \in V} \mathcal{F}_p \end{aligned}$$

as the restriction maps, we see that $\Lambda(\mathcal{F})$ is a C-valued presheaf.

Let's check that $\Lambda(\mathcal{F})$ verifies the sheaf axioms. To do this, consider an open set $U \subseteq X$.

SH.1) Let $s, t \in \mathcal{F}(U)$ be two sections such that there exists an open covering $\{U_i\}_{i \in \mathcal{I}}$ such that $\forall i \in \mathcal{I} \quad s|_{U_i} = t|_{U_i}$.
Note that for any $x \in U$ since there exists $j \in \mathcal{I}$ such that $x \in U_j$ we have that $s_x = t_x$. Therefore, since this holds for all $x \in U$, we have that $s = t$.

SH.2) Let $\{U_i\}_{i \in \mathcal{I}}$ be an open covering of U and let $\{s_i \mid s_i \in \Lambda(\mathcal{F})(U_i)\}_{i \in \mathcal{I}}$ be a family of sections such that for any $i, j \in \mathcal{I}$ we have $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$.
Define a section $s \in \Lambda(\mathcal{F})(U)$ by choosing, for each $x \in U$, an index $k \in \mathcal{I}$ such that $x \in U_k$ and setting $s_x = (s_k)_x$. Note that this is well-defined by the compatibility condition of the family. Furthermore, for all $i \in \mathcal{I}$ by construction we have that $s|_{U_i} = s_i$.

Note that there is a presheaf morphism $\theta\colon \mathcal{F} \to \Lambda(\mathcal{F})$ defined for any $s \in \mathcal{F}(U)$ by sending s to its germ at every point:

$$\theta(U)(s) = (s_p)_{p\in U} \in \prod_{p\in U} \mathcal{F}_p$$

where s_p denotes the image of s in the stalk $\mathcal{F}_p$.

We define the sheafification $\mathcal{F}^\#$ of $\mathcal{F}$ as the subpresheaf of $\Lambda(\mathcal{F})$ consisting of sections that are *locally compatible* with $\mathcal{F}$. Explicitly, for every open set $U \subseteq X$ an element $\sigma = (\sigma_p)_{p\in U} \in \Lambda(\mathcal{F})(U)$ belongs to $\mathcal{F}^\#(U)$ if and only if:

> For every $p \in U$, there exists an open neighborhood $V \subseteq U$ of p and a section $t \in \mathcal{F}(V)$ such that for all $q \in V$, $\sigma_q = t_q$ (as elements of $\mathcal{F}_q$).

We need to check that our definition of $\mathcal{F}^\#$ verifies the universal property of sheafification. To do this, consider a sheaf $\mathcal{G}$ and a morphism of presheaves $\phi\colon \mathcal{F} \to \mathcal{G}$. We wish to show that there exists a unique sheaf morphism $\psi\colon \mathcal{F}^\# \to \mathcal{G}$ such that $\phi \circ \theta = \psi$. Diagramatically

$$\begin{array}{ccc} \mathcal{F} & \xrightarrow{\phi} & \mathcal{G} \\ {\scriptstyle\theta}\downarrow & \nearrow_{\psi} & \\ \mathcal{F}^\# & & \end{array}$$

Let's check existence first by defining ψ explicitly. Let $U \subseteq X$ be an open set and $s \in \mathcal{F}^\#(U)$. By definition there exists an open cover $\{U_i\}_{i\in\mathcal{I}}$ of U and sections $t_i \in \mathcal{F}(V_i)$ such that $s|_{U_i} = \theta(t_i)$ for all $i \in \mathcal{I}$.

We want ψ to verify $\psi(s)|_{U_i} = \psi(s|_{U_i}) = \psi(\theta_{t_i})$. This forces us to define

$$\psi(\theta(t_i)) \coloneqq \phi(t_i) \in \mathcal{G}(U_i)$$

Let's see that $\{\phi(t_i)\}_{i\in\mathcal{I}}$ can be glued in $\mathcal{G}$. Note that for all $y \in U_i \cap U_j$ we have $(t_i)_y = (t_j)_y$. Therefore using the induced stalk morphism $\phi_y\colon \mathcal{F}_y \to \mathcal{G}_y$ we have

$$\phi_y((t_i)_y) = \phi_y((t_j)_y)$$

Therefore $(\phi(t_i))_y = (\phi(t_j))_y$ for all $y \in U_i \cap U_j$, meaning that

$$\phi(t_i)|_{U_i\cap U_j} = \phi(t_j)|_{U_i\cap U_j}$$

Applying SH.2 to this family we get that there is a unique section $t \in \mathcal{G}(U)$ such that $t|_{U_i} = \phi(t_i)$ for all $i \in \mathcal{I}$. Therefore we can define $\psi_U(s) \coloneqq t$, which verifies the universal property by construction.

To check uniqueness, since sheaf morphisms are determined by their induced maps on stalks and for all points $x \in X$ we have $\theta_x = \mathbb{1}$, ψ verifies $\psi_x = \phi_x$ for all $x \in X$, making it unique.

Remark 8.3.1. Another construction of sheafification is the *espace étalé* of $\mathcal{F}$. Roughly

speaking, the section of $\mathcal{F}^{\#}$ on an open set $U \subseteq X$ correspond to

$$\mathcal{F}^{\#}(U) = \left\{ s : U \to \coprod_{p \in P} \mathcal{F}_p \text{ continuous } \mid \pi \circ s = \mathbb{1}_U \right\}$$

where $\pi : \coprod_{p \in P} \mathcal{F}_p \to X$ is the obvious projection sending a germ $s_p \in \mathcal{F}_p$ to its base point $p \in X$ and $\coprod_{p \in P} \mathcal{F}_p$ has the topology generated by $\{s_p \in \mathcal{F}_p \mid p \in U\}_{U \in \mathsf{Open}(X)}$. More details about this construction can be found at [Vak25].

The following properties are direct.

Lemma 8.3.1. Let $\mathscr{F}$ be a sheaf. Then $(\iota\mathscr{F})^{\#} \cong \mathscr{F}$, i.e: $-^{\#} \circ \iota \cong \mathbb{1}_{\mathsf{Sh}_X(\mathsf{C})}$.

Proof. Consider the identity map $\mathbb{1} : \iota\mathscr{F} \to \mathscr{F}$. Since $\iota\mathscr{F}$ is a presheaf and $\mathscr{F}$ is a sheaf by definition of sheafification there exists a unique morphism $\psi : (\iota\mathscr{F})^{\#} \to \mathscr{F}$ such that $\psi \circ \theta = \mathbb{1}$. Note that

$$(\theta \circ \psi) \circ \theta = \theta \circ (\psi \circ \theta) = \theta$$

Applying now the definition of sheafification to $\theta : \mathscr{F} \to (\iota\mathscr{F})^{\#}$ we get a unique map $h : (\iota\mathscr{F})^{\#} \to (\iota\mathscr{F})^{\#}$ such that $h \circ \theta = \theta$. Therefore since $\mathbb{1} \circ \theta = \theta$ and $(\theta \circ \psi) \circ \theta = \theta$ we have that $\theta \circ \psi = \mathbb{1}$, which means that θ is an isomorphism. □

Lemma 8.3.2. Let X be a topological space and $\mathcal{F}$ a C-valued presheaf over X. Then for all $p \in X$ there is bijection of stalks

$$\mathcal{F}_p \cong (\mathcal{F}^{\#})_p$$

Proof. Direct from the construction of sheafification. □

8.4 Sheaves in an abelian category

We want to apply the machinery we have built in Part II to sheaves. In this section, X will be a fixed topological space and C will be an abelian category for which sheafification exists.

Note that by Proposition 6.0.1 $\mathsf{PSh}_X(\mathsf{C})$ is an abelian category. We are going to see that $\mathsf{Sh}_X(\mathsf{C})$ is an abelian category.

We will do this by the means of the following category-theoretic auxiliary lemma.

Lemma 8.4.1. Let $a : \mathsf{A} \to \mathsf{B}$ and $b : \mathsf{B} \to \mathsf{A}$ be two functors. Suppose that

1. A and B are additive categories, a, b are additive functors and there is an adjunction $b \dashv a$.
2. B is an abelian category and b is left-exact.
3. $b \circ a \cong \mathbb{1}_{\mathsf{A}}$.

Then A is abelian.

Proof. Since B is an abelian by Lemma 6.0.1 all finite limits and colimits exist en B.

Since b is left-adjoint by Example 6.1.3 it is right-exact and therefore, exact. Analogously we have that a is left-exact.

Let $\varphi\colon B_1 \to B_2$ be a morphism in B. Since B is abelian there exist $\ker_{\mathsf{B}}\varphi$ and $\operatorname{coker}_{\mathsf{B}}\varphi$.

Since exact functors preserve kernels and cokernels we have that

$$b(\ker_{\mathsf{B}}\varphi) \cong \ker_{\mathsf{A}}(b\varphi) \qquad b(\operatorname{coker}_{\mathsf{B}}\varphi) \cong \operatorname{coker}_{\mathsf{A}}(b\varphi)$$

Therefore we have that every morphism of the form $b\varphi$ in A admits a kernel and a cokernel.

Note that, since $b \circ a \cong \mathbb{1}_{\mathsf{A}}$, every morphism ψ in A is of the form $\psi = b\varphi$. Thus all kernels and cokernels exist in A and

$$\ker_{\mathsf{A}}\psi \cong b\ker_{\mathsf{B}}(a\psi) \cong \ker_{\mathsf{B}} b(a(\psi)) = \ker_{\mathsf{b}}\psi \tag{8.4.1}$$

$$\operatorname{coker}_{\mathsf{A}}\psi \cong b\operatorname{coker}_{\mathsf{B}}(a\psi) \tag{8.4.2}$$

It only remains to show that $\operatorname{Coim}_{\mathsf{A}}\psi \cong \operatorname{Im}_{\mathsf{A}}\psi$. Since A has all finite limits and colimits a is left exact.

By Lemma 6.1.1 we have that a transforms kernels into kernels. Therefore

$$\begin{aligned}
\operatorname{coim}_{\mathsf{A}}\psi &= \operatorname{coker}_{\mathsf{A}}\left(\ker_{\mathsf{A}}\psi \to A_1\right) && \text{(definition)}\\
&\cong b\operatorname{coker}_{\mathsf{B}}\left(a(\ker_{\mathsf{A}}\psi \to A_1)\right) && \text{(8.4.1)}\\
&\cong b\operatorname{coker}_{\mathsf{B}}\left(\ker_{\mathsf{B}}(a\psi) \to a(A_1)\right) && (a \text{ preserves kernels})\\
&= b\operatorname{Coim}_{\mathsf{B}}(a\psi) && \text{(definition)}\\
&\cong b\operatorname{Im}_{\mathsf{B}}(a\psi) && (\mathsf{B} \text{ is abelian})\\
&= b(\ker_{\mathsf{B}}\left(a(A_2) \to \operatorname{coker}_{\mathsf{B}}(a\psi)\right)) && \text{(definition)}\\
&\cong \ker_{\mathsf{A}}\left((b\circ a)(A_2) \to b(\operatorname{coker}_{\mathsf{B}}(a\psi))\right) && (b \text{ preserves kernels})\\
&\cong \ker_{\mathsf{A}}\left(A_2 \to b(\operatorname{coker}_{\mathsf{B}}(a\psi))\right) && b\circ a \cong \mathbb{1}_{\mathsf{A}}\\
&\cong \ker_{\mathsf{A}}\left(A_2 \to \operatorname{coker}_{\mathsf{A}}\psi\right) && \text{(8.4.1)}\\
&= \operatorname{Im}_{\mathsf{A}}\psi && \text{(definition)}
\end{aligned}$$

□

We need to check that we can apply this lemma in our case.

Lemma 8.4.2. $\mathsf{Sh}_X(\mathsf{C})$ is an additive category and the forgetful functor $\iota\colon \mathsf{Sh}_X(\mathsf{C}) \to \mathsf{PSh}_X(\mathsf{C})$ is additive.

Proof. Let's check first that $\mathsf{Sh}_X(\mathsf{C})$ satisfies additive category axioms. Recall that Ad.3 is equivalent to Ad.4 by Lemma 5.0.1.

We will use the characterization of sheaves in terms of equalisers.

Ad.1 Since $\operatorname{Hom}_{\mathsf{Sh}_X(\mathsf{C})}(\mathscr{F},\mathscr{G}) = \operatorname{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(\mathscr{F},\mathscr{G})$ and $\mathsf{PSh}_X(\mathsf{C})$ is additive, $\operatorname{Hom}_{\mathsf{Sh}_X}$ admits the structure of an abelian group.

Ad.2 Note that the zero object in $\mathsf{PSh}_X(\mathsf{C})$ is given by the constant presheaf $\underline{0}_X^{\mathrm{pre}}$. Note that the sequence

$$\underline{0}_X^{\mathrm{pre}}(U) = 0 \longrightarrow \prod_{i\in\mathcal{I}} \underline{0}_X^{\mathrm{pre}}(U_i) = 0 \rightrightarrows \prod_{(i,j)\in\mathcal{I}\times\mathcal{I}} \underline{0}_X^{\mathrm{pre}}(U_i \cap U_j) = 0$$

is trivially an equaliser for all open covers $U = \bigcup_{i\in\mathcal{I}} U_i$. Therefore $\underline{0}_X^{\mathrm{pre}}$ is a sheaf and thus $\mathsf{Sh}_X(\mathsf{C})$ admits a zero object.

Ad.4 Let $\mathscr{F}, \mathscr{G} \in \mathsf{Sh}_X(\mathsf{C})$. We define their product elementwise as

$$(\mathscr{F} \times \mathscr{G})(U) = \mathscr{F}(U) \times \mathscr{G}(U)$$

Since this coincides with the product in $\mathsf{PSh}_X(\mathsf{C})$ it is a presheaf. Furthermore for any open cover $U = \bigcup_{i \in \mathcal{I}} U_i$ we have that

$$(\mathscr{F} \times \mathscr{G})(U) = \mathscr{F}(U) \times \mathscr{G}(U) \to \prod_{i \in \mathcal{I}} (\mathscr{F} \times \mathscr{G})(U_i) = \prod_{i \in \mathcal{I}} (\mathscr{F}(U_i)) \times \prod_{i \in \mathcal{I}} (\mathscr{G}(U_i))$$

$$\rightrightarrows \prod_{(i,j) \in \mathcal{I} \times \mathcal{I}} (\mathscr{F} \times \mathscr{G})(U_i \cap U_j) = \prod_{i \in \mathcal{I}} \left(\mathscr{F}(U_i \cap U_j)\right) \times \prod_{i \in \mathcal{I}} \left(\mathscr{G}(U_i \cap U_j)\right)$$

is an equaliser since it can be seen as the "product" of two equalisers. Therefore $\mathscr{F} \times \mathscr{G}$ is a sheaf.

By the construction of the zero object and products in $\mathsf{Sh}_X(\mathsf{C})$ we have that ι preserves both. Therefore by Proposition 5.0.1 ι is additive. □

Proposition 8.4.1. Let C be an abelian category. The sheafification functor $-^{\#}\colon \mathsf{PSh}_X(\mathsf{C}) \to \mathsf{Sh}_X(\mathsf{C})$ is exact.

Proof. Since $-^{\#}$ is left adjoint by Example 6.1.3 it is right exact.

To see that it is left exact we will use Lemma 6.1.1. In particular we are going to see that sheafification commutes with finite limits.

Let $\mathcal{F}\colon \mathsf{I} \to \mathsf{PSh}_X(\mathsf{C})$ be a diagram in $\mathsf{PSh}_X(\mathsf{C})$ with I finite.

We need to see that

$$\left(\lim_{i \in \mathsf{I}} \mathcal{F}_i\right)^{\#} \cong \lim_{i \in \mathsf{I}} \mathcal{F}_i^{\#}$$

Note that since $\iota\colon \mathsf{Sh}_X(\mathsf{C}) \to \mathsf{PSh}_X(\mathsf{C})$ is right adjoint it preserves limits. Therefore limits of sheaves are computed exactly as limits of presheaves.

By definition of limit we have a morphism

$$\phi\colon \left(\lim_{i \in \mathsf{I}} \mathcal{F}_i\right)^{\#} \to \lim_{i \in \mathsf{I}} \mathcal{F}_i^{\#}$$

To see that this is an isomorphism we will see that the induced stalk maps ϕ_p are isomorphisms for all $p \in X$, i.e: we want to show that

$$\left(\left(\lim_{i \in \mathsf{I}} \mathcal{F}_i\right)^{\#}\right)_p \cong \left(\lim_{i \in \mathsf{I}} \mathcal{F}_i^{\#}\right)_p$$

On the one hand since stalks commute with sheafification we have

$$\left(\left(\lim_{i \in \mathsf{I}} \mathcal{F}_i\right)^{\#}\right)_p \cong \left(\lim_{i \in \mathsf{I}} \mathcal{F}_i\right)_p$$

Furthermore since stalks commute with finite limits

$$\left(\lim_{i \in \mathsf{I}} \mathcal{F}_i\right)_p \cong \lim_{i \in \mathsf{I}} (\mathcal{F}_i)_p$$

On the other hand since stalks commute with finite limits

$$\left(\lim_{i\in\mathsf{I}} \mathscr{F}_i^{\#}\right)_p \cong \lim_{i\in\mathsf{I}} \left(\mathscr{F}_i^{\#}\right)_p$$

Furthermore since stalks commute with sheafification we have

$$\lim_{i\in\mathsf{I}} \left((\mathscr{F}_i)^{\#}\right)_p \cong \lim_{i\in\mathsf{I}} (\mathscr{F}_i)_p$$

Therefore

$$\left(\left(\lim_{i\in\mathsf{I}} \mathscr{F}_i\right)^{\#}\right)_p \cong \lim_{i\in\mathsf{I}} (\mathscr{F}_i)_p \cong \left(\lim_{i\in\mathsf{I}} \mathscr{F}_i^{\#}\right)_p$$

□

We finally have everything we need to prove that $\mathsf{Sh}_X(\mathsf{C})$ is abelian.

Theorem 8.4.1. If C is abelian then the category of C-valued sheaves $\mathsf{Sh}_X(\mathsf{C})$ over a topological space X is abelian. Furthermore if $\varphi\colon \mathscr{F} \to \mathscr{G}$ is a morphism in $\mathsf{Sh}_X(\mathsf{C})$ its kernel is the same as its kernel as a map of presheaves $\ker \iota\varphi$ and its cokernel is the sheafification of its cokernel as a map of presheaves:

$$\ker \varphi = \ker \iota\varphi \qquad \operatorname{coker} \varphi = \left(\operatorname{coker} \iota\varphi\right)^{\#}$$

and if $p \in X$ then

$$\left(\ker \varphi\right)_p = \ker(\varphi_p) \qquad \left(\operatorname{coker} \varphi\right)_p = \operatorname{coker}(\varphi_p)$$

Proof. Direct from Lemma 8.4.1 with

$$\mathsf{A} := \mathsf{Sh}_X(\mathsf{C}) \quad \mathsf{B} := \mathsf{PSh}_X(\mathsf{B})$$
$$a := \iota\colon \mathsf{Sh}_X(\mathsf{C}) \hookrightarrow \mathsf{PSh}_X(\mathsf{C}) \text{ forgetful} \quad b := -^{\#}\colon \mathsf{PSh}_X(\mathsf{C}) \to \mathsf{Sh}_X(\mathsf{C}) \text{ sheafification}$$

We can indeed apply this lemma since

1. $\mathsf{Sh}_X(\mathsf{C})$ is additive by Lemma 8.4.2 and $\mathsf{PSh}_X(\mathsf{C})$ is additive by Lemma 5.0.3, ι is additive by Lemma 8.4.2 and $-^{\#}$ is additive by Proposition 5.0.2, and the adjunction $-^{\#} \dashv \iota$ is given by definition of sheafification.
2. $\mathsf{PSh}_X(\mathsf{C})$ is abelian by Proposition 6.0.1 and $-^{\#}$ is left exact by Proposition 8.4.1.
3. $-^{\#} \circ \iota = \mathbb{1}_{\mathsf{Sh}_X(\mathsf{C})}$ comes from Lemma 8.3.1.

The commutativity of stalks comes form Remark 8.2.2 and Lemma 8.3.2. □

Furthermore combining the above characterization of kernels and cokernels and Proposition 8.2.1 we have the following criterion to check exactness.

Corollary 8.4.1. Let C be an abelian category. A complex of C-valued sheaves over X

$$0 \to \mathscr{F} \to \mathscr{G} \to \mathscr{H} \to 0$$

is exact in $\mathsf{Sh}_X(\mathsf{C})$ if and only if for all $p \in X$

$$0 \to \mathscr{F}_p \to \mathscr{G}_p \to \mathscr{H}_p \to 0$$

is exact in C.

Chapter 9

Operations on sheaves

In the previous chapter, we established the foundational properties of sheaves as localized algebraic structures operating on a fixed topological space. To fully leverage this framework in geometry and topology, we must understand how sheaves interact when moving between different topological spaces. This chapter explores the primary geometric operations used to change the base space of a sheaf along a continuous map $f : X \to Y$

We begin by constructing the direct image (or pushforward) functor f_*, which allows us to push data from X to Y, and the inverse image (or pullback) functor f^{-1}, which pulls data back from Y to X. A major focus of this section is establishing the core natural adjunction between these two operations, showing that pulling back and pushing forward are dual categorical maneuvers.

As a particular case we will study how taking one of the spaces to be $\{*\}$ we recover classical sheaf constructions such as the stalk. We also study the case for open and closed immersions, where as we will see another adjoint functor exists.

Finally, we unify these operations by proving the existence of a canonical short exact sequence linking a sheaf, its extension by zero from an open set, and its pushforward from the complementary closed set. This fundamental sequence provides the exact sequence machinery required for studying relative sheaf cohomology in later chapters.

9.1 Continuous maps and (pre)sheaves

Consider a continuous map between topological spaces $f : X \to Y$.

We want to find a way of translating presheaves and sheaves from X to Y along f and viceversa. We will start in the presheaf case and move to the sheaf case using sheafification when needed.

First, let's see how we can push forward data from X to Y. This will be easy since for any open subset $V \subseteq Y$ we have that $f^{-1}(V)$ is open in X.

Definition 9.1.1 (Direct image presheaf)**.** Let $f : X \to Y$ be a continuous map. For any C-valued presheaf over X we define the **direct image** (or pushforward) of $\mathcal{F}$ via f as the C-valued presheaf $f_*\mathcal{F}$ over Y which, for any open subset $V \subseteq Y$

$$\left(f_*\mathcal{F}\right)(V) := \mathcal{F}\left(f^{-1}(V)\right)$$

and for any $V_1 \subseteq V_2 \subseteq Y$ the restriction is given by

$$\begin{array}{ccc} f_*(\mathcal{F}(V_2)) & = & \mathcal{F}(f^{-1}(V_2)) \\ \vdots & & \downarrow \\ f_*(\mathcal{F}(V_1)) & = & \mathcal{F}(f^{-1}(V_1)) \end{array}$$

Note that this defines a functor

$$f_* \colon \mathsf{PSh}_X(\mathsf{C}) \to \mathsf{PSh}_Y(\mathsf{C})$$

Since taking inverse images of open sets behaves nicely under composition we have that direct images compose as expected: if $f : X \to Y$ and $g : Y \to Z$ be two continuous maps then the direct images $(g \circ f)_*$ and $g_* \circ f_*$ are equal.

Furthermore using that $f^{-1}(\bigcup_i V_i) = \bigcup_i f^{-1}(V_i)$ we have directly that if $\mathscr{F}$ is a sheaf then $f_*\mathscr{F}$ is a sheaf.

This means that we have a functor

$$f_* \colon \mathsf{Sh}_X(\mathsf{C}) \to \mathsf{Sh}_Y(\mathsf{C})$$

Let's try now pulling back data from Y to X. Note that in this case if $U \subseteq X$ is open $f(U) \subseteq Y$ is not necessarily open in Y. Therefore, we have to *aproximate* the data by looking at all open sets in Y that contain $f(U)$.

Definition 9.1.2 (Inverse image presheaf)**.** Let $f : X \to Y$ be a continuous map between two topological spaces X, Y. For any C-valued presheaf $\mathcal{G}$ over Y we define the **inverse image** (or pullback) of $\mathcal{G}$ via f as the C-valued presheaf $f_{\text{pre}}^{-1}\mathcal{G}$ over X which, for any open subset $U \subseteq X$

$$\left(f_{\text{pre}}^{-1}\mathcal{G}\right)(U) \coloneqq \varinjlim_{\substack{V \supseteq f(U) \\ V \in \mathsf{Open}(Y)}} \mathcal{G}(V)$$

Since this is a direct limit following Example 3.2.5 we may describe $\mathcal{F}_p$ as

$$\left(f_{\text{pre}}^{-1}\mathcal{G}\right)(U) = \{(s, V) \mid f(U) \subseteq V \in \mathsf{Open}(X) \wedge s \in \mathcal{G}(V)\} \Big/ \sim$$

where

$$(s, V) \sim (t, W) \iff \exists Z \subseteq V \cap W \mid f(U) \subseteq Z \in \mathsf{Open}(Y) \wedge s|_Z = t|_Z$$

In this way, if $U_1 \subseteq U_2 \subseteq X$ are open subsets of X the restriction is given by

$$\mathrm{res}_{U_2,U_1}\left(f_{\text{pre}}^{-1}\mathcal{G}\right)([s, V]) = [(s, V)]$$

Note that this defines a functor $f_{\text{pre}}^{-1} \colon \mathsf{PSh}_Y(\mathsf{C}) \to \mathsf{PSh}_X(\mathsf{C})$.

Notation. Let X be a topological space and $f : Y \hookrightarrow X$ an inclussion of a set $Y \subseteq X$ endowed with the induced topology. If $\mathscr{G} \in \mathsf{Sh}_X(\mathsf{C})$ is a C-valued sheaf (resp. presheaf) over X we call $f^{-1}\mathscr{G} \in \mathsf{Sh}_Y(\mathsf{C})$ (resp. $j_{\text{pre}}^{-1}\mathscr{G}$) the **restriction** of $\mathscr{G}$ to Y and denote it by $\mathscr{G}|_Y$.

Since stalks are particular cases of direct limits, inverse images play nicely with them.

Lemma 9.1.1. Let $f : X \to Y$ be a continuous map. Let $\mathcal{G} \in \mathsf{PSh}_Y(\mathsf{C})$ be a C-valued presheaf over Y. Then for all $p \in X$ there is a bijection of stalks.

$$\left(f_{\mathrm{pre}}^{-1}\mathcal{G}\right)_p \cong \mathcal{G}_{f(p)}$$

Proof.

$$\begin{aligned}
\left(f_{\mathrm{pre}}^{-1}\mathcal{G}\right)_p &= \varinjlim_{\substack{U \ni p \\ U \in \mathsf{Open}(X)}} \left(f_{\mathrm{pre}}^{-1}\mathcal{G}\right)(U) \\
&= \varinjlim_{\substack{U \ni p \\ U \in \mathsf{Open}(X)}} \left(\varinjlim_{\substack{V \supseteq f(U) \\ V \in \mathsf{Open}(Y)}} \mathcal{G}(V) \right) \\
&= \varinjlim_{\substack{V \ni f(p) \\ V \in \mathsf{Open}(Y)}} \mathcal{G}(V) \\
&= \mathcal{G}_{f(p)}
\end{aligned}$$

□

Therefore since exactness can be checked at the level of stalks we have:

Corollary 9.1.1. Let C be a abelian category. For all continuous maps $f : X \to Y$ the presheaf inverse image $f_{\mathrm{pre}}^{-1} : \mathsf{PSh}_Y(\mathsf{C}) \to \mathsf{PSh}_X(\mathsf{C})$ is exact.

Although the definition of the inverse image may seem complicated at first sight it is natural in the sense that it makes f_* and f_{pre}^{-1} adjoint.

Theorem 9.1.1. Let $f : X \to Y$ be a continuous function between two topological spaces X, Y. Let C be a category. Then there exists an adjunction

$$f_{\mathrm{pre}}^{-1} \dashv f_* : \mathsf{PSh}_X(\mathsf{C}) \underset{f_*}{\overset{f_{\mathrm{pre}}^{-1}}{\rightleftarrows}} \mathsf{PSh}_Y(\mathsf{C})$$

Proof. We want to construct natural transformations

$$\begin{aligned}
\eta &: \mathbb{1}_{\mathsf{PSh}_Y(\mathsf{C})} \implies f_* \circ f_{\mathrm{pre}}^{-1} \\
\varepsilon &: f_{\mathrm{pre}}^{-1} \circ f_* \implies \mathbb{1}_{\mathsf{PSh}_X(\mathsf{C})}
\end{aligned}$$

making for all $\mathcal{F} \in \mathsf{PSh}_X(\mathsf{C})$ and $\mathcal{G} \in \mathsf{PSh}_Y(\mathsf{C})$ the following diagrams commute

$$\begin{array}{ccc}
f_{\mathrm{pre}}^{-1}(\mathcal{G}) & \xrightarrow{\mathbb{1}_{\mathcal{G}}} & f_{\mathrm{pre}}^{-1}(\mathcal{G}) \\
{\scriptstyle f_{\mathrm{pre}}^{-1}(\eta_{\mathcal{G}})}\downarrow & \nearrow_{\varepsilon_{f_{\mathrm{pre}}^{-1}(\mathcal{G})}} & \\
\left(f_{\mathrm{pre}}^{-1} \circ f_* \circ f_{\mathrm{pre}}^{-1}\right)(\mathcal{G}) & &
\end{array}
\qquad
\begin{array}{ccc}
f_*(\mathcal{G}) & \xrightarrow{\mathbb{1}_{\mathcal{F}}} & f_*(\mathcal{G}) \\
{\scriptstyle \eta_{f_*(\mathcal{F})}}\downarrow & \nearrow_{f_*(\varepsilon_{\mathcal{F}})} & \\
\left(f_* \circ f_{\mathrm{pre}}^{-1} \circ f_*\right)(\mathcal{F}) & &
\end{array}$$

Note that, by definition of the direct image presheaf and inverse image presheaf we

have, for $U \in \mathsf{Open}(X)$ and $V \in \mathsf{Open}(Y)$

$$\left(f_{\mathrm{pre}}^{-1} \circ f_*\right)(\mathcal{F})(U) = \varinjlim_{\substack{W \supseteq f(U) \\ W \in \mathsf{Open}(Y)}} (f_*\mathcal{F})(W) = \varinjlim_{\substack{W \supseteq f(U) \\ W \in \mathsf{Open}(Y)}} \mathcal{F}(f^{-1}(W))$$

$$\left(f_* \circ f_{\mathrm{pre}}^{-1}\right)(\mathcal{G})(V) = (f_{\mathrm{pre}}^{-1}\mathcal{G})(f^{-1}(V)) = \varinjlim_{\substack{W \supseteq (f \circ f^{-1})(V) \\ W \in \mathsf{Open}(Y)}} \mathcal{G}(W)$$

Since we have $(f \circ f^{-1})(V) \subseteq V$, we can define $\eta_{\mathcal{G}}(V)$ as the canonical injection into the direct limit

$$\eta_{\mathcal{G}}(V) \colon \mathcal{G}(V) \to \varinjlim_{\substack{W \supseteq (f \circ f^{-1})(V) \\ W \in \mathsf{Open}(Y)}} \mathcal{G}(W)$$

On the other hand, since for every $W \subseteq f(U)$ open we have $f^{-1}(W) \subseteq (f^{-1} \circ f)(U) \subseteq U$ there exists a restriction morphism

$$\mathrm{res}_{f^{-1}(W),U} \colon \mathcal{F}(f^{-1}(W)) \to \mathcal{F}(U)$$

Using these maps we obtain a morphism

$$\varepsilon_{\mathcal{F}}(U) \colon \varinjlim_{\substack{W \supseteq f(U) \\ W \in \mathsf{Open}(Y)}} \mathcal{F}(f^{-1}(W)) \to \mathcal{F}(U) \qquad \square$$

Using this adjunction we can see that the composition works as expected.

Lemma 9.1.2. Let $f \colon X \to Y$ and $g \colon Y \to Z$ be continuous maps. Then the inverse images $(g \circ f)_{\mathrm{pre}}^{-1}$ and $f_{\mathrm{pre}}^{-1} \circ g_{\mathrm{pre}}^{-1}$ coincide (up to isomorphism).

Proof. Using the adjunction $f_{\mathrm{pre}}^{-1} \dashv f_*$ we have, on the one hand

$$(g \circ f)_{\mathrm{pre}}^{-1} \dashv (g \circ f)_* = g_* \circ f_*$$

On the other hand note that

$$f_{\mathrm{pre}}^{-1} \circ g_{\mathrm{pre}}^{-1} \dashv g_* \circ f_*$$

Indeed

$$\begin{aligned} \mathrm{Hom}_{\mathsf{PSh}_X(\mathsf{C})}\left(\left(f_{\mathrm{pre}}^{-1} \circ g_{\mathrm{pre}}^{-1}\right)(\mathcal{H}), \mathcal{F}\right) &\cong \mathrm{Hom}_{\mathsf{PSh}_Y(\mathsf{C})}\left(g_{\mathrm{pre}}^{-1}(\mathcal{H}), f_*(\mathcal{F})\right) \\ &\cong \mathrm{Hom}_{\mathsf{PSh}_Z(\mathsf{C})}\left(\mathcal{H}, \left(g_* \circ f_*\right)(\mathcal{F})\right) \end{aligned}$$

Therefore since adjoints are unique we have $(g \circ f)_{\mathrm{pre}}^{-1} \cong f_{\mathrm{pre}}^{-1} \circ g_{\mathrm{pre}}^{-1}$. $\square$

Extending the inverse image to the sheaf case is not as easy as with the direct image since $f_{\mathrm{pre}}^{-1}\mathcal{F}$ is not necessarily a sheaf. Therefore we need to sheafificate.

Definition 9.1.3. Let $f \colon X \to Y$ be a continuous map. The **inverse image** (or

pullback) sheaf via f is the functor

$$f^{-1} := -^{\#} \circ f_{\mathrm{pre}}^{-1} \circ \iota : \mathsf{Sh}_Y(\mathsf{C}) \to \mathsf{Sh}_X(\mathsf{C})$$

where $-^{\#}$ is the sheafification functor, f_{pre}^{-1} is the inverse image of presheaves and $\iota : \mathsf{Sh}_Y(\mathsf{C}) \hookrightarrow \mathsf{PSh}_Y(\mathsf{C})$ is the forgetful functor.

Using that sheafification adjunction we still have the adjunction $f^{-1} \dashv f_*$.

Theorem 9.1.2. Let $f : X \to Y$ be a continuous map There exists an adjunction

$$f^{-1} \dashv f_* : \mathsf{Sh}_X(\mathsf{C}) \underset{f_*}{\overset{f^{-1}}{\leftrightarrows}} \mathsf{Sh}_Y(\mathsf{C})$$

between the direct and inverse images of sheaves.

Proof. Using the adjunctions $-^{\#} \dashv \iota$ and $f_{\mathrm{pre}}^{-1} \dashv f_*$ we have

$$\begin{aligned}
\mathrm{Hom}_{\mathsf{Sh}_X(\mathsf{C})}(f^{-1}(\mathscr{G}), \mathscr{F}) &= \mathrm{Hom}_{\mathsf{Sh}_X(\mathsf{C})}\left(\left(-^{\#} \circ f_{\mathrm{pre}}^{-1} \circ \iota\right)(\mathscr{G}), \mathscr{F}\right) \\
&\cong \mathrm{Hom}_{\mathsf{PSh}_X(\mathsf{C})}((f_{\mathrm{pre}}^{-1} \circ \iota)(\mathscr{G}), \iota(\mathscr{F})) \\
&= \mathrm{Hom}_{\mathsf{PSh}_Y(\mathsf{C})}(\iota(\mathscr{G}), (f_* \circ \iota)(\mathscr{F})) \\
&= \mathrm{Hom}_{\mathsf{PSh}_Y(\mathsf{C})}(\iota(\mathscr{G}), (\iota \circ f_*)(\mathscr{F})) \\
&= \mathrm{Hom}_{\mathsf{Sh}_Y(\mathsf{C})}(\mathscr{G}, f_*(\mathscr{F}))
\end{aligned}$$

□

Analogously as in Lemma 9.1.3 we see that composition behaves as expected.

Lemma 9.1.3. Let $f : X \to Y$ and $g : Y \to Z$ be continuous maps. Then the inverse images $(g \circ f)^{-1}$ and $f^{-1} \circ g^{-1}$ coincide (up to isomorphism).

Stalk computation also works fine.

Lemma 9.1.4. Let $f : X \to Y$ be a continuous map. Let $\mathscr{G} \in \mathsf{Sh}_Y(\mathsf{C})$ be a C-valued sheaf over Y. Then for all $p \in X$ there is a bijection of stalks.

$$\left(f^{-1}\mathscr{G}\right)_p \cong \mathscr{G}_{f(p)}$$

Proof.

$$\begin{aligned}
\left(f^{-1}\mathscr{G}\right)_p &= \left(\left(f_{\mathrm{pre}}^{-1}(\iota\mathscr{G})\right)^{\#}\right)_p && \text{(definition)} \\
&= \left(f_{\mathrm{pre}}^{-1}(\iota\mathscr{G})\right)_p && \text{(Lemma 8.3.2)} \\
&= (\iota\mathscr{G})_{f(p)} && \text{(Lemma 9.1.1)} \\
&= \mathscr{G}_{f(p)} && \text{(Lemma 8.3.1 and Lemma 8.3.2)}
\end{aligned}$$

□

Thus as in the presheaf case since exactness can be checked at the level of stalks we have:

Corollary 9.1.2. Let C be a abelian category. For all continuous maps $f : X \to Y$ the sheaf inverse image $f^{-1} : \mathsf{Sh}_Y(\mathsf{C}) \to \mathsf{Sh}_X(\mathsf{C})$ is exact.

9.2 Particular cases

9.2.1 Sheaves over a point

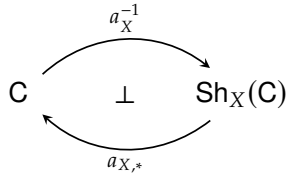
To gain a little bit more intuition for these operations, we are going to study them when one of our spaces is the one-point topological space $\{*\}$. Recall that in this case we have $\mathsf{Sh}_*(\mathsf{C}) \simeq \mathsf{C}$.

We will see that some of the most common sheaf constructions can be seen as particular cases of f_* and f^{-1} in this case.

Let's consider first the following continuous map:

$$\begin{aligned} a_X \colon X &\to \{*\} \\ x &\mapsto * \end{aligned}$$

This produces a pair of adjoint functors



Definition 9.2.1 (Global sections)**.** The **global sections** functor is

$$\Gamma(X, -) \coloneqq a_{X,*} \colon \mathsf{Sh}_X(\mathsf{C}) \to \mathsf{C}$$

Less abstractly, it sends a C-valued sheaf $\mathscr{F} \in \mathsf{Sh}_X(\mathsf{C})$ to $\Gamma(X, \mathscr{F}) = \mathscr{F}(X)$ and a morphism of sheaves $\phi : \mathscr{F} \to \mathscr{G}$ to $\Gamma(\phi, X) = \mathscr{F}(X) \xrightarrow{\phi(X)} \mathscr{G}(X)$.

Notation. If $Y \subseteq X$ is a subset of X and $\mathscr{F} \in \mathsf{Mod}(\underline{\mathbf{k}}_X)$ we set $\Gamma(Y, \mathscr{F}) \coloneqq \Gamma(Y, \mathscr{F}|_Y)$.

Definition 9.2.2 (Constant sheaf)**.** The **constant sheaf** functor is

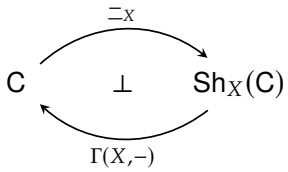
$$\underline{-}_X \coloneqq a_X^{-1} \colon \mathsf{C} \to \mathsf{Sh}_X(\mathsf{C})$$

Less abstractly, it sends an object $A \in \mathsf{C}$ to the sheaf $\underline{A}_X$ whose sections over an open set $U \subseteq X$ are the locally constant functions from U to A (viewing A as being equipped with the discrete topology).

Remark 9.2.1. The formulation $\underline{A}_X = a_X^{-1}(A)$ is equivalent to the standard construction of the constant sheaf via sheafification.
Indeed, the presheaf inverse image of A along a_X assigns to any open $U \subseteq X$ the colimit over open neighborhoods of $a_X(U) = \{*\}$. Since the only neighborhood is $\{*\}$ itself, this presheaf is exactly the constant presheaf $U \mapsto A$. The inverse image functor a_X^{-1} then applies sheafification, yielding the constant sheaf.

From the adjunction $f^{-1} \dashv f_*$.

Corollary 9.2.1. There is an adjunction



Using this adjunction by Example 6.1.3 we have that $\Gamma(X,-)$ is left exact. However as the next example shows it is in general not right exact.

Example 9.2.1. Let $X \coloneqq \mathbb{C}\setminus\{0\}$ and consider $\underline{\mathbb{Z}}_X$ the constant sheaf of integers, $\mathscr{O}_X$ the sheaf of holomorphic functions and $\mathscr{O}_X^*$ the sheaf of non-vanishing holomorphic functions on X.
Let $\exp\colon \mathscr{O}_X \to \mathscr{O}_X^*$ be the map that sends an holomorphic function f on an open set U to $e^{2\pi i f}$. We have that

$$0 \to \underline{\mathbb{Z}}_X \to \mathscr{O}_X \xrightarrow{\exp} \mathscr{O}_X^* \to 0$$

is a short exact sequence of sheaves.
Indeed it is exact at $\underline{\mathbb{Z}}_X$ and $\mathscr{O}_X$ since $e^{2\pi i f} = 1$ if and only if f is locally constant to an integer and thus $\ker\exp = \underline{\mathbb{Z}}_X$. Exactness at $\mathscr{O}_X^*$ since exactness can be checked locally and for all $p \in X$ there exists an holomorphic branch of the logarithm in a neighbourhood of p.
However, the induced sequence

$$0 \to \Gamma(X,\underline{\mathbb{Z}}_X) \to \Gamma(X,\mathscr{O}_X) \xrightarrow{\Gamma\exp} \Gamma(X,\mathscr{O}_X^*) \to 0$$

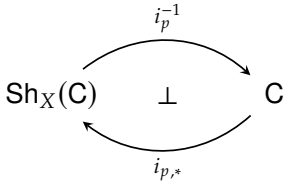
is not exact since if we consider $\mathbb{1}_X \in \mathscr{O}_X^*$ this would imply that there exists a function $g \in \mathscr{O}_X$ such that $e^{2\pi i f(z)} = g(z)$ for all z, i.e: g would be a global holomorphic branch of the logarithm which cannot exist.

This failure of exactness is exactly what motivates the study of sheaf cohomology.

Now given a point $p \in X$ let's consider

$$\begin{aligned} i_p\colon \{*\} &\to X \\ * &\mapsto p \end{aligned}$$

As before this gives rise to an adjunction



Definition 9.2.3 (Skyscraper sheaf)**.** The **skyscraper sheaf** functor is

$$\mathrm{sk}_p \coloneqq i_{p,*}\colon \mathsf{C} \to \mathrm{Sh}_X(\mathsf{C})$$

Less abstractly, it sends an object $A \in \mathsf{C}$ to the sheaf

$$(\mathrm{sk}_p\, A)(U) = \begin{cases} A & p \in U \\ 1 & p \notin U \end{cases}$$

for all open subsets $U \subseteq X$ with restriction maps for $V \subseteq U$

$$\mathrm{res}_{U,V} = \begin{cases} \mathbb{1}_A & p \in V \\ \mathbb{1}_A & p \notin U \\ A \to 1 & p \in U \wedge p \notin V \end{cases}$$

The last functor i_p^{-1} is one we have already seen.

Lemma 9.2.1. For all points $p \in X$ we have

$$-_p \cong i_p^{-1}$$

Proof. Identifying $\mathsf{Sh}_{\{*\}}(\mathsf{C}) \simeq \mathsf{C}$ if $\mathcal{F} \in \mathsf{PSh}_X(\mathsf{C})$ is a C-valued presheaf we have

$$\left(i_{p,\mathrm{pre}}^{-1}\mathcal{F}\right)(\{*\}) = \varinjlim_{\substack{V \supseteq i_p(\{*\}) \\ V \in \mathsf{Open}(X)}} \mathcal{F}(V) = \varinjlim_{\substack{V \ni p \\ V \in \mathsf{Open}(X)}} \mathcal{F}(V) = \mathcal{F}_p$$

Therefore if $\mathscr{F} \in \mathsf{Sh}_X(\mathsf{C})$ is a C-valued sheaf we have

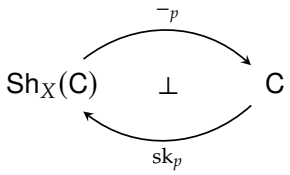
$$\begin{aligned} \left(i_p^{-1}\mathscr{F}\right)(\{*\}) &= \left(-^{\#} \circ i_{p,\mathrm{pre}}^{-1} \circ \iota\right)(\mathscr{F})(\{*\}) \\ &= \left((\iota(\mathscr{F}))^{\#}\right)_p \\ &= (\iota(\mathscr{F}))_p \\ &= \mathscr{F}_p \end{aligned}$$

□

Remark 9.2.2. From Corollary 9.1.2 we have that the stalk functor is exact, which gives another proof of exactness being checked at the level of stalks using Lemma 6.1.1.

As in the other case, directly from the the adjunction $f^{-1} \dashv f_*$.

Corollary 9.2.2. There is an adjunction



9.2.2 Open immersions and (pre)sheaves

Let's consider now an open immersion $j : U \hookrightarrow X$, where X is a topological space and $U \subseteq X$ is an open subset.

We begin by giving a more explicit description for both the sections and stalks of the inverse image.

Lemma 9.2.2. Let X be a topological space and $j : U \hookrightarrow X$ be an open immersion

1. Let $\mathcal{G} \in \mathsf{PSh}_X(\mathsf{C})$ be a C-valued presheaf over X.
   Then $j_{\mathrm{pre}}^{-1}\mathcal{G} \in \mathsf{PSh}_U(\mathsf{C})$ is given by, for any open subset $V \subseteq U$

$$j_{\mathrm{pre}}^{-1}\mathcal{G}(V) = \mathcal{G}(V)$$

Furthermore $j_{\text{pre}}^{-1} \circ j_* = \mathbb{1}_{\mathsf{PSh}_U(\mathsf{C})}$.

2. Let $\mathscr{G} \in \mathsf{Sh}_X(\mathsf{C})$ be a C-valued sheaf over X.
Then $j^{-1}\mathscr{G} \in \mathsf{Sh}_U(\mathsf{C})$ is given by, for any subset $V \subseteq U$

$$j^{-1}\mathscr{G}(V) = \mathscr{G}(V)$$

Furthermore $j^{-1} \circ j_* = \mathbb{1}_{\mathsf{Sh}_U(\mathsf{C})}$.

Proof. By definition we have

$$j_{\text{pre}}^{-1}\mathcal{G}(V) = \varinjlim_{\substack{W \supseteq j(V) \\ W \in \mathsf{Open}(U)}} \mathcal{G}(W) = \varinjlim_{\substack{W \supseteq V \\ W \in \mathsf{Open}(U)}} \mathcal{G}(W)$$

Therefore since V is open this last direct limit is exactly $\mathcal{G}(V)$.

Furthermore if $\mathcal{F} \in \mathsf{PSh}_U(\mathsf{C})$ for any open subset $V \subseteq U$ we have

$$\left(j_{\text{pre}}^{-1} \circ j_*\right)\mathcal{F}(V) = j_{\text{pre}}^{-1}\mathcal{F}(j^{-1}(V)) = \mathcal{F}(j^{-1}(V)) = \mathcal{F}(V)$$

The statement about sheaves follows directly since $V \mapsto \mathscr{G}(V)$ is clearly a sheaf if $\mathscr{G}$ is a sheaf. □

Lemma 9.2.3. Let X be a topological space and $j\colon U \hookrightarrow X$ be an open immersion Let $\mathscr{G} \in \mathsf{Sh}_X(\mathsf{C})$ be a C-valued sheaf over X. For all $p \in U$ we have

$$\left(j^{-1}\mathscr{G}\right)_p = \mathscr{G}_p$$

Proof.

$$\left(j^{-1}\mathscr{G}\right)_p = \varinjlim_{\substack{V \ni p \\ V \in \mathsf{Open}(U)}} j^{-1}\mathscr{G}(V) = \varinjlim_{\substack{V \ni p \\ V \in \mathsf{Open}(U)}} \mathscr{G}(V) = \varinjlim_{\substack{V \ni p \\ V \in \mathsf{Open}(X)}} \mathscr{G}(V) = \mathscr{G}_p \qquad \square$$

This characterization allows us to construct another adjoint functor $j_!$ (pronounced j lower "shriek") to j^{-1}, in this case with $j^{-1} \dashv j_!$. Let's do it first for the presheaf case.

Definition 9.2.4. Let X be a topological space and $j\colon U \hookrightarrow X$ be an open immersion. Let $\mathcal{F} \in \mathsf{PSh}_U(\mathsf{C})$ be a C-valued presheaf over U. Suppose that C has an initial object $e \in \mathsf{C}$.
The **extension** $j_!^{\text{pre}}\mathcal{F}$ of $\mathcal{F}$ by e is the C-valued presheaf over X given by, for any open subset $V \subseteq X$

$$j_!^{\text{pre}}\mathcal{F}(V) \coloneqq \begin{cases} e & V \not\subseteq U \\ \mathcal{F}(V) & V \subseteq U \end{cases}$$

The restriction maps are given by, for any $W \subseteq V \subseteq X$ open subsets

$$\text{res}_{V,W} \coloneqq \begin{cases} \text{res}_{V,W}^U & V \subseteq U \\ \mathbb{1}_e & V \not\subseteq U \\ e \to \mathcal{F}(W) & W \subseteq U \subseteq V \end{cases}$$

where $\text{res}_{V,W}^U$ is the restriction in $\mathsf{PSh}_U(\mathsf{C})$ and $e \to \mathcal{F}(W)$ the unique map from the initial object e.

Note that this defines a functor

$$j_!^{\mathrm{pre}} \colon \mathsf{PSh}_U(\mathsf{C}) \to \mathsf{PSh}_X(\mathsf{C})$$

Before proving that this is indeed a right-adjoint to j^{-1} let's see how it interacts with stalks.

Lemma 9.2.4. Let $j \colon U \hookrightarrow X$ be an open immersion. For any C-valued presheaf $\mathcal{F}$ the stalk at a point $p \in X$ of the extension $j_!^{\mathrm{pre}}\mathcal{F}$ is

$$\left(j_!^{\mathrm{pre}}\mathcal{F}\right)_p = \begin{cases} e & p \notin U \\ \mathcal{F}_p & p \in U \end{cases}$$

Proof. By definition of stalk

$$\left(j_!^{\mathrm{pre}}\mathcal{F}\right)_p = \varinjlim_{\substack{V \ni p \\ U \in \mathsf{Open}(U)}} j_!^{\mathrm{pre}}\mathcal{F}(V)$$

Therefore if $p \notin U$

$$\left(j_!^{\mathrm{pre}}\mathcal{F}\right)_p = \varinjlim_{\substack{V \ni p \\ U \in \mathsf{Open}(U)}} j_!^{\mathrm{pre}}\mathcal{F}(V) = \varinjlim_{\substack{V \ni p \\ U \in \mathsf{Open}(U)}} e = e$$

On the other hand if $p \in U$

$$\left(j_!^{\mathrm{pre}}\mathcal{F}\right)_p = \varinjlim_{\substack{V \ni p \\ U \in \mathsf{Open}(U)}} j_!^{\mathrm{pre}}\mathcal{F}(V) = \varinjlim_{\substack{V \ni p \\ U \in \mathsf{Open}(U)}} \mathcal{F}(V) = \mathcal{F}_p$$ □

Using that exactness can be checked at the level of stalks we get:

Corollary 9.2.3. If C is abelian then the extension by zero functor of presheaves $j_!^{\mathrm{pre}} \colon \mathsf{PSh}_U(\mathsf{C}) \to \mathsf{PSh}_X(\mathsf{C})$ is exact.

Now we are in conditions to prove the adjointness.

Theorem 9.2.1. Let X be a topological space and $j \colon U \hookrightarrow X$ be an open immersion. Suppose that C has an initial object $e \in \mathsf{C}$.
There is a adjunction

$$\mathsf{PSh}_U(\mathsf{C}) \underset{j_{\mathrm{pre}}^{-1}}{\overset{j_!^{\mathrm{pre}}}{\rightleftarrows}} \mathsf{PSh}_X(\mathsf{C}) \qquad \text{with } \mathsf{PSh}_U(\mathsf{C}) \perp \mathsf{PSh}_X(\mathsf{C})$$

Proof. Let $\mathcal{F} \in \mathsf{PSh}_U(\mathsf{C})$ and $\mathcal{G} \in \mathsf{PSh}_X(\mathsf{C})$. We want to construct an isomorphism

$$\Phi \colon \mathrm{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(j_!^{\mathrm{pre}}(\mathcal{F}), \mathcal{G}) \xrightarrow{\sim} \mathrm{Hom}_{\mathsf{PSh}_U(\mathsf{C})}(\mathcal{F}, j_{\mathrm{pre}}^{-1}(\mathcal{G}))$$

Let $\phi \colon j_!^{\mathrm{pre}}(\mathcal{F}) \to \mathcal{G}$ be a morphism of presheaves over X. We want to define a morphism of presheaves $\Phi(\phi) \colon \mathcal{F} \to j_{\mathrm{pre}}^{-1}(\mathcal{G})$ over U. Since j is an open map by

Lemma 9.2.2 we have $j_{\text{pre}}^{-1}\mathcal{F}(V) = \mathcal{F}(V)$ for all open subsets $V \subseteq X$. Therefore since ϕ provides a component map $\phi_V \colon j_!\mathcal{F}(V) \to \mathcal{G}(V)$ we can define

$$\Phi(\phi)_V \coloneqq \phi_V$$

Since ϕ is a presheaf morphism over X it commutes with the restriction maps for all open subsets of X. In particular, it commutes with the restriction maps for all open subsets of U and thus $\Phi(\phi)$ is a valid morphism of presheaves over U.

Now let $\psi \colon \mathcal{F} \to j_{\text{pre}}^{-1}(\mathcal{G})$ be a morphism of presheaves over U. We want to define a morphism of presheaves $\Psi(\psi) \colon j_!^{\text{pre}}(\mathcal{F}) \to \mathcal{G}$ over X. For any open set $W \subseteq X$ we define the component $\Psi(\psi)_W \colon j_!^{\text{pre}}\mathcal{F}(W) \to \mathcal{G}(W)$ componentwise as

$$\Psi(\psi)_W \coloneqq \begin{cases} \psi_W & W \subseteq U \\ e \to \mathcal{G}(W) & W \nsubseteq U \end{cases}$$

Since this commutes with the restriction maps it is a valid morphism of presheaves over X.

It only remains to see that Φ and Ψ are inverses. If $V \subseteq U$ is an open subset then

$$\left(\Phi \circ \psi\right)(\psi)_V = \Phi\left(\psi_V\right) = \psi_V$$

On the other hand if $W \subseteq X$ then we distinguish two cases. If $W \subseteq U$ then

$$\left(\Psi \circ \Phi\right)(\phi)_W = \Psi\left(\phi_W\right) = \phi_W$$

If $W \nsubseteq U$ then both $\left(\Psi \circ \Phi\right)(\phi)_W$ and ϕ_W are maps from the initial object e to $\mathcal{G}(W)$ and therefore coincide. □

Let's extend this definition to the sheaf case using sheafification.

Definition 9.2.5. Let X be a topological space and $j \colon U \hookrightarrow X$ be an open immersion. Suppose that C has an initial object $e \in \mathsf{C}$.
The sheaf **restriction** by e functor is

$$j_! \coloneqq -^{\#} \circ j_!^{\text{pre}} \circ \iota : \mathsf{Sh}_U(\mathsf{C}) \to \mathsf{Sh}_X(\mathsf{C})$$

Analogously to Lemma 9.2.4 and Corollary 9.1.2 we see that $j_!$ behaves as $j_!^{\text{pre}}$ in both sections and stalks and, in particular, we get that $j_!$ is exact.

Lemma 9.2.5. Let $j \colon U \hookrightarrow X$ be an open immersion. For any C-valued sheaf $\mathscr{F}$ over U the stalk at a point $p \in X$ of the extension $j_!\mathscr{F}$ is

$$\left(j_!\mathscr{F}\right)_p = \begin{cases} e & p \notin U \\ \mathscr{F}_p & p \in U \end{cases}$$

Corollary 9.2.4. If C is abelian then the extension by zero functor of sheaves $j_! \colon \mathsf{Sh}_U(\mathsf{C}) \to \mathsf{Sh}_X(\mathsf{C})$ is exact.

Finally, we see that the adjunction is still present in the sheaf case.

Theorem 9.2.2. Let X be a topological space and $j \colon U \hookrightarrow X$ be an open immersion. Suppose that C has an initial object $e \in \mathsf{C}$.

There is a adjunction

$$\mathsf{Sh}_U(\mathsf{C}) \underset{j^{-1}}{\overset{j_!}{\underset{\longleftarrow}{\overset{\longrightarrow}{\bot}}}} \mathsf{Sh}_X(\mathsf{C})$$

Proof. For all $\mathscr{F} \in \mathsf{Sh}_X(\mathsf{C})$ and $\mathscr{G} \in \mathsf{Sh}_U(\mathsf{C})$ we have

$$\begin{aligned}
\mathrm{Hom}_{\mathsf{Sh}_X(\mathsf{C})}\left(j_!(\mathscr{F}), \mathscr{G}\right) &\cong \mathrm{Hom}_{\mathsf{PSh}_X(\mathsf{C})}\left(\left(-^{\#} \circ j_!^{\mathrm{pre}} \circ \iota\right)(\mathscr{F}), \mathscr{G}\right) \\
&\cong \mathrm{Hom}_{\mathsf{PSh}_X(\mathsf{C})}\left(\left(j_!^{\mathrm{pre}} \circ \iota\right)(\mathscr{F}), \iota(\mathscr{G})\right) \\
&\cong \mathrm{Hom}_{\mathsf{PSh}_U(\mathsf{C})}\left(\iota(\mathscr{F}), \left(j_{\mathrm{pre}}^{-1} \circ \iota\right)(\mathscr{G})\right) \\
&\cong \mathrm{Hom}_{\mathsf{PSh}_U(\mathsf{C})}\left(\iota(\mathscr{F}), \left(\iota \circ j^{-1}\right)(\mathscr{G})\right) \\
&\cong \mathrm{Hom}_{\mathsf{Sh}_U(\mathsf{C})}\left(\mathscr{F}, j^{-1}(\mathscr{G})\right)
\end{aligned}$$

□

9.2.3 Closed immersions and (pre)sheaves

Let's now examine the opposite case, where we have a closed immersion $i\colon Z \hookrightarrow X$, where X is a topological space and $Z \subseteq X$ is an closed subset.

We begin by noting the following fact about stalks.

Lemma 9.2.6. Let $\mathscr{F}$ be a C-valued sheaf. The stalks of $i_*\mathscr{F}$ at a point $p \in X$ is

$$(i_*\mathscr{F})_p = \begin{cases} 1 & p \notin Z \\ \mathscr{F}_p & p \in Z \end{cases}$$

Proof. If $p \notin Z$ then there exist arbitrarily small open neighbourhoods U of p that do not intersect Z. Therefore since $\mathscr{F}$ is a sheaf $\mathscr{F}(i^{-1}(U)) = 1$, proving the first case.

If $p \in Z$ then any neighbourhood of p is of the form $U \cap Z$ with U a neighbourhood of p in X. □

Remark 9.2.3. If $j\colon U \hookrightarrow X$ is an open immersion and we consider $i\colon X \setminus U \to X$ the closed immersion of the complement we have a natural isomorphism $i^{-1} \cong j_*$ since the stalks are the same at all points.

Taking as inspiration the above characterization, we define what the support of a C-valued sheaf is.

Definition 9.2.6. Let $\mathscr{F}$ be a C-valued sheaf over X. The **support** of $\mathscr{F}$ is

$$\operatorname{supp} \mathscr{F} \coloneqq \{p \in X \mid \mathscr{F}_p \neq 1\}$$

If $s \in \mathscr{F}(X)$ is a section then its support is

$$\operatorname{supp} s \coloneqq \{p \in X \mid s_p \neq 1 \text{ in } \mathscr{F}_p\}$$

This definition let's us describe the direct image more cleanly.

Proposition 9.2.1. The direct image sheaf functor

$$i_*\colon \mathsf{Sh}_Z(\mathsf{C}) \to \mathsf{Sh}_X(\mathsf{C})$$

is fully faithfull. Furthermore, its essential image consists of those sheaves whose

support is contained in Z.

Proof. Let $\mathscr{F} \in \mathsf{Sh}_Z(\mathsf{C})$. Consider the map

$$i^{-1}i_*\mathscr{F} \to \mathscr{F}$$

Since by Lemma 9.2.6 this is an isomorphism for all $p \in Z$ by Proposition 8.2.1 it is a isomorphism. Therefore $i^{-1}i_* \cong \mathbb{1}_{\mathsf{Sh}_Z(\mathsf{C})}$, which means that i_* is fully faithfull.

To see the essential image part, we are going to see that $\mathscr{G} \in \operatorname{Im} i_*$ if and only if $\mathscr{G}_p = 1$ for all $p \in X \setminus Z$. Note that the direction $\implies$ is direct from Lemma 9.2.6. To see $\impliedby$, suppose that $\mathscr{G}$ verifies $\mathscr{G}_p = 1$ for all $p \in X \setminus Z$. Then the map

$$\mathscr{G} \to i_*i^{-1}\mathscr{G}$$

is an isomorphism on all stalks and thus by Proposition 8.2.1 it is an isomorphism. □

With this in mind we can construct another adjoint functor $i^!$ to i_*, in this case with $i_* \dashv i^!$.

Definition 9.2.7. Let X be a topological space and $Z \subseteq X$ a closed subset. For any C-valued sheaf $\mathscr{F}$ over X, the **extraordinary inverse image** is the subsheaf $i^!\mathscr{F}$ of $\mathscr{F}$ defined as, for any open subset $U \subseteq X$

$$\left(i^!\mathscr{F}\right)(U) := \{s \in \mathscr{F}(U) \mid \operatorname{supp} s \subseteq Z \cap U\}$$

Note that this is the largest subsheaf of $\mathscr{F}$ whose support is contained in Z.
By Proposition 9.2.1 we can view this as a functor

$$i^! \colon \mathsf{Sh}_X(\mathsf{C}) \to \mathsf{Sh}_Z(\mathsf{C})$$

Theorem 9.2.3. Let X be a topological space and $i \colon Z \hookrightarrow X$ be a closed immersion. If C is abelian, there is a adjunction

$$\mathsf{Sh}_Z(\mathsf{C}) \underset{i^!}{\overset{i_*}{\rightleftarrows}} \mathsf{Sh}_X(\mathsf{C}) \qquad (\perp)$$

Proof. We want to show that, for any C-valued sheaf $\mathscr{G} \in \mathsf{Sh}_Z(\mathsf{C})$ and $\mathscr{F} \in \mathsf{Sh}_X(\mathsf{C})$, there is a natural bijection

$$\operatorname{Hom}_{\mathsf{Sh}_X(\mathsf{C})}\left(i_*\mathscr{G}, \mathscr{F}\right) \cong \operatorname{Hom}_{\mathsf{Sh}_Z(\mathsf{C})}\left(\mathscr{G}, i^!\mathscr{F}\right)$$

Consider a morphism $\phi \colon i_*\mathscr{G} \to \mathscr{F}$. We want to construct a morphism $\psi \colon \mathscr{G} \to i^!\mathscr{F}$.

For any open subset $U \subseteq X$ we have a morphism in C

$$\phi_U \colon \mathscr{G}(U \cap Z) \to \mathscr{F}(U)$$

Let $W \subseteq Z$ be an open set and write $W = U \cap Z$ for some $U \subseteq X$ open. We want to show that the image of ϕ_U lands on $i^!\mathscr{F}(U)$. Let $s \in \mathscr{G}(U \cap Z)$ and consider its restriction to the open set $U \setminus Z$ of X. By the commutativity of the restriction maps with ϕ we get

$$\left(\phi_U(s)\right)|_{U\setminus Z} = \phi_{U\setminus Z}(s|_{(U\setminus Z)\cap Z})$$

Since $(U \setminus Z) \cap Z = \varnothing$ we have

$$\big(\phi_U(s)\big)\,|_{U\setminus Z} = \phi_{U\setminus Z}(s|_{(U\setminus Z)\cap Z}) = \phi_{U\setminus Z}(0) = 0$$

which means that

$$\operatorname{supp}(\phi_U(s)) \subseteq U \cap Z = W$$

Therefore since the image of ϕ_U is contained in $i^!\mathscr{F}(U)$ it naturally induces a map

$$\psi_W \colon \mathscr{G}(W) \to i^!\mathscr{F}(W)$$

Since by construction these maps commute with the restrictions, these maps define a morphism $\psi \colon \mathscr{G} \to i^!\mathscr{F}$.

Conversely, suppose now that we have a morphism $\psi \colon \mathscr{G} \to i^!\mathscr{F}$. We now want to construct a morphism $\phi \colon i_*\mathscr{G} \to \mathscr{F}$.

For any open subset $V \subseteq X$ let $H = V \cap Z$. We have a morphism in C

$$\psi_H \colon \mathscr{G}(H) \to i^!\mathscr{F}(H)$$

By composing ψ_H with the inclussion $i^!\mathscr{F}(H) \hookrightarrow \mathscr{F}(V)$ we obtain a map $\phi_V : \mathscr{G}(V \cap Z) \to \mathscr{F}(V)$. Therefore since these maps commute with the restrictions they assemble into a morphism $\phi \colon i_*\mathscr{G} \to \mathscr{F}$. □

Note that for closed immersions we have adjunctions $i^{-1} \dashv i_* \dashv i^!$. Therefore directly by Example 6.1.3 we have:

Corollary 9.2.5. If C is abelian then the direct image sheaf functor

$$i_* \colon \mathsf{Sh}_Z(\mathsf{C}) \to \mathsf{Sh}_X(\mathsf{C})$$

is exact.

9.2.4 A canonical exact sequence

We end this chapter by connecting open and closed immersions via a short exact sequence that will become useful when we study relative sheaf cohomology in chapter 14.

Theorem 9.2.4. Let X be a topological space and $U \subseteq X$ be an open subset. Consider the open and closed immersions $j \colon U \hookrightarrow X$ and $i \colon X \setminus U \hookrightarrow X$.
If C is abelian for all C-valued sheaves $\mathscr{F} \in \mathsf{Sh}_X(\mathsf{C})$ there exists a short exact sequence

$$0 \longrightarrow j_!j^{-1}\mathscr{F} \longrightarrow \mathscr{F} \longrightarrow i_*i^{-1}\mathscr{F} \longrightarrow 0$$

Furthermore for any morphism of sheaves $\mathscr{F} \to \mathscr{G}$ we get a morphism of short exact sequences

$$\begin{array}{ccccccccc}
0 & \longrightarrow & j_!j^{-1}\mathscr{F} & \longrightarrow & \mathscr{F} & \longrightarrow & i_*i^{-1}\mathscr{F} & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & j_!j^{-1}\mathscr{G} & \longrightarrow & \mathscr{G} & \longrightarrow & i_*i^{-1}\mathscr{G} & \longrightarrow & 0
\end{array}$$

i.e: this construction is functorial.

Proof. Using the characterization of the stalks at Lemma 9.1.4, Lemma 9.2.3, Lemma 9.2.4 and Lemma 9.2.6 we have

$$\left(j_! j^{-1}\mathscr{F}\right)_p = \begin{cases} 0 & p \in X \setminus U \\ \left(j^{-1}\mathscr{F}\right)_p & p \in U \end{cases} = \begin{cases} 0 & p \in X \setminus U \\ \mathscr{F}_p & p \in U \end{cases}$$

$$\left(i_* i^{-1}\mathscr{F}\right)_p = \begin{cases} 0 & p \in X \setminus U \\ \left(i^{-1}\mathscr{F}\right)_p & p \in U \end{cases} = \begin{cases} 0 & p \in X \setminus U \\ \mathscr{F}_p & p \in U \end{cases}$$

Therefore for all $p \in X$ the sequence

$$0 \longrightarrow \left(j_! j^{-1}\mathscr{F}\right)_p \longrightarrow \mathscr{F}_p \longrightarrow \left(i_* i^{-1}\mathscr{F}\right)_p \longrightarrow 0$$

is exact and therefore so is

$$0 \longrightarrow j_! j^{-1}\mathscr{F} \longrightarrow \mathscr{F} \longrightarrow i_* i^{-1}\mathscr{F} \longrightarrow 0$$

The functoriallity comes from the naturality of the adjunctions $j_! \dashv j^{-1} \dashv j_*$ and $i^{-1} \dashv i_* \dashv i^!$. □

Notation. Let $\mathscr{F}$ be a C-valued sheaf over X. If $j\colon U \hookrightarrow X$ is an open immersion we call $j_! j^{-1}\mathscr{F} \in \mathsf{Sh}_X(\mathsf{C})$ the **extension** of $\mathscr{F}$ by U and denote it by $\mathscr{F}_U$. Analogously if $i\colon Z \hookrightarrow X$ is a closed immersion we call $i_* i^{-1}\mathscr{F}$ the extension of $\mathscr{F}$ by Z and denote it $\mathscr{F}_Z$.

Therefore with this notation Theorem 9.2.4 states that the following is an exact sequence for all open subsets $U \subseteq X$

$$0 \longrightarrow \mathscr{F}_U \longrightarrow \mathscr{F} \longrightarrow \mathscr{F}_{X\setminus U} \longrightarrow 0$$

PART IV

Sheaf cohomology

Introduction

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Chapter 10

Sheaf cohomology

While evaluating mathematical structures over localized patches is a natural starting point in geometry, compiling these local snapshots into unique global data presents a fundamental challenge. We have already seen that left-exact functors, such as global sections, systematically fail to preserve surjectivity when transitioning from local neighborhoods to an entire space. To precisely catalog and measure these global obstructions, we apply the abstract machinery of homological algebra to the category of sheaves.

In this chapter, we rigorously establish the foundation for sheaf cohomology as the right derived functor of the global sections functor. After verifying that the category of sheaves of modules over a commutative ring contains enough injective objects to permit derived resolutions, we analyze how these cohomological invariants transform under domain restrictions and extensions.

Notation. Given a ring R, we denote the category of sheaves of R-modules over a topological space X by $\mathsf{Mod}(\underline{R}_X)$.

In what follows, let $\mathbf{k}$ be a fixed commutative ring. Note that if we let $\mathbf{k} = \mathbb{Z}$ since $\mathsf{Mod}(\mathbb{Z}) \cong \mathsf{Ab}$ we have $\mathsf{Mod}(\underline{\mathbb{Z}}_X) \cong \mathsf{Sh}_X(\mathsf{Ab})$ and thus our work also applies to sheaves of abelian groups.

10.1 Injective sheaves and cohomology

Note that $\mathsf{Mod}(\mathbf{k})$ is an abelian category. Therefore by Theorem 8.4.1 $\mathsf{Mod}(\underline{\mathbf{k}}_X)$ is also abelian.

If we want to derive left-exact functors we need to show that $\mathsf{Mod}(\underline{\mathbf{k}}_X)$ has enough injectives.

Lemma 10.1.1. The category $\mathsf{Mod}(\underline{\mathbf{k}}_X)$ of sheaves of $\mathbf{k}$-modules over X has enough injectives.

Proof. We will use the following claim

> The category $\mathsf{Mod}(R)$ of modules over a ring R has enough injectives.

For a proof, see [Alu09, Corollary VIII.6.4].

Let $\mathscr{F} \in \mathsf{Mod}(\underline{\mathbf{k}}_X)$ and consider its stalk $\mathscr{F}_p \in \mathsf{Mod}(\mathbf{k})$ at a point $p \in X$.

Since $\mathsf{Mod}(\mathbf{k})$ has enough injectives we can choose an object I_p and a monomorphism

$$i_p \colon \mathscr{F}_p \rightarrowtail I_p$$

Consider the sheaf

$$\mathscr{I} \coloneqq \prod_{p \in X} \mathrm{sk}_p\, I_p$$

where $\mathrm{sk}_p\, I_p$ is the skyscraper sheaf supported at I_p. More explicitly, for every open set $U \subseteq X$ the sheaf $\mathscr{I}$ is defined as

$$\mathscr{I}(U) = \prod_{p \in U} I_p$$

with the obvious restriction maps.

Recall that by Corollary 9.2.2 the skyscraper functor is left-adjoint to the stalk functor, which is exact. Therefore by Proposition 7.1.2 $\mathscr{I}$ is an injective object of $\mathsf{Mod}(\underline{\mathbf{k}}_X)$. □

This means that we can derive all right-exact functors from the category of sheaves of $\mathbf{k}$-modules.

Having seen this, we now can finally define sheaf cohomology. Recall that by Corollary 9.2.1 $\Gamma(X, -)$ is right adjoint. Therefore by Example 6.1.3 $\Gamma(X, -)$ is left exact.

Definition 10.1.1 (Sheaf Cohomology). Let $\mathscr{F}$ be a sheaf of $\mathbf{k}$-modules over a topological space X. The i-th **cohomology** of X with coefficients in $\mathscr{F}$ is the i-th right derived functor of the global sections functor:

$$\mathrm{H}^i(X; \mathscr{F}) \coloneqq \mathrm{R}^i\, \Gamma(X, -)(\mathscr{F})$$

Notation. If $Y \subseteq X$ is a subset of X we set

$$\mathrm{H}^i(Y; \mathscr{F}) \coloneqq \mathrm{R}^i\, \Gamma(Y; -)(\mathscr{F}) = \mathrm{R}^i\, \Gamma(Y; -)(\mathscr{F}|_Y)$$

for all sheaves $\mathscr{F} \in \mathsf{Mod}(\underline{\mathbf{k}}_X)$.

We are going to see first how sheaf cohomology behaves when restricting and extending sheaves to subsets of X. Before we do so, we need to understand how injective sheaf of $\mathbf{k}$-modules behave under direct and inverse images.

Lemma 10.1.2. Let $\mathscr{I} \in \mathsf{Mod}(\underline{\mathbf{k}}_X)$ be an injective sheaf of $\mathbf{k}$-modules.

1. If $f \colon X \to Y$ is a continuous map then $f_*\mathscr{I}$ is injective.
2. If $j_U \colon U \hookrightarrow X$ is an open immersion then $\mathscr{I}|_U = j_U^{-1}\mathscr{I}$ is injective.

Proof.

1. Since f^{-1} is exact and by the adjunction $f^{-1} \dashv f_*$ we have

$$\mathrm{Hom}_{\mathsf{Mod}(\underline{\mathbb{k}}_X)}(f^{-1}(-), \mathscr{I}) \cong \mathrm{Hom}_{\mathsf{Mod}(\underline{\mathbb{k}}_Y)}(-, f_*\mathscr{I})$$

by Proposition 7.1.1 we have that $f_*\mathscr{I}$ is injective.

2. Follows from the first and $j_U^{-1} \cong i_{X\setminus U,*}$ where $i_{X\setminus U}\colon X \setminus U \hookrightarrow X$. This isomorphism follows directly from Theorem 8.2.1. □

Proposition 10.1.1. Let X be a topological space, $\mathscr{F} \in \mathsf{Mod}(\underline{\mathbf{k}}_X)$ a sheaf of $\mathbf{k}$-modules and $i \geq 0$.

1. If $U \subseteq X$ is open we have $\mathrm{H}^i(U;\mathscr{F}) \cong \mathrm{H}^i(U;\mathscr{F}|_U)$.
2. If $Z \subseteq X$ is closed we have $\mathrm{H}^i(Z;\mathscr{F}_Z) \cong \mathrm{H}^i(Z;\mathscr{F}|_Z)$.

Proof. Let $i_Y\colon Y \hookrightarrow X$ be the inclussion of $Y \subseteq X$ and $a_Y\colon Y \to \{*\}$.

1. We have
$$\mathrm{H}^i(U;\mathscr{F}) \cong \mathrm{R}^i(a_{U,*} \circ i_U^{-1})(\mathscr{F}) \cong \left(\mathrm{R}^i\, a_{U,*}\right)(i_U^{-1}\mathscr{F}) \cong \mathrm{H}^i(U;\mathscr{F}|_U)$$
where the second isomorphism comes from Theorem 7.2.3 since i_U^{-1} is exact and sends injectives to injectives.
2. We have
$$\mathrm{H}^i(Z;\mathscr{F}_Z) \cong \left(\mathrm{R}^i\, a_{X,*}\right)((i_{Z,*} \circ i_Z^{-1})(\mathscr{F})) \cong \left(\mathrm{R}^i(a_X \circ i_{S,*})\right)(i_Z^{-1}(\mathscr{F})) \cong \mathrm{H}^i(Z;\mathscr{F}|_Z)$$
where the second isomorphism comes from Theorem 7.2.3 since $i_{S,*}$ is exact and sends injectives to injectives. □

We also have the following

Lemma 10.1.3. Let X be a topological space and $\mathscr{F} \in \mathsf{Mod}(\underline{\mathbf{k}}_X)$. If $K \subseteq X$ is compact and relatively Hausdorff[a] then
$$\varinjlim_{\substack{U \supset K \\ U \in \mathsf{Open}(X)}} \mathrm{H}^i(U;\mathscr{F}) \cong \mathrm{H}^i(K;\mathscr{F})$$

[a] $K \subseteq X$ is relatively Hausdorff if two distinct points in K admit disjoint neighbourhoods in X. Note that if $K = X$ then we recover that X is Hausdorff.

Proof. Suppose first that $i = 0$. In this case we want to show that there is an isomorphism induced by the restriction
$$\varinjlim_{\substack{U \supset K \\ U \in \mathsf{Open}(X)}} \Gamma(U,\mathscr{F}) \xrightarrow{\sim} \Gamma(K,\mathscr{F})$$
Let $s \in \Gamma(K,\mathscr{F}|_K)$. There exists a finite family of open subsets $\{U_i\}_{i=1}^n$ covering K and sections $s_i \in \Gamma(U_i,\mathscr{F})$ such that $s_i|_{K\cap U_i} = s|_{K\cap U_i}$. Furthermore we can find another family of open sets $\{V_i\}_{i=1}^n$ covering K such that $K \cap \mathrm{ad}(V_i) \subset U_i$. We want to glue together the sections s_i on a neighbourhood of K.

By induction over n we can assume $n = 2$. If $K_i = K \cap \mathrm{ad}(V_i)$ then $s_1|_{K_1\cap K_2} = s_2|_{K_1\cap K_2}$. Let W be an open neighbourhood of K such that $s_1|_K = s_2|_K$ and let W_i be an open subset of U_i such that $K_i \setminus W \subset W_i$ and $W_1 \cap W_2 = \varnothing$. Set $U_i' = W_i \cup W$. Then $s_1|_{U_1'\cap U_2'} = s_2|_{U_1'\cap U_2'}$ and this defines $t \in \Gamma(U_1' \cup U_2',\mathscr{F})$ with $t|_K = s$ finishing the base case.

Now, consider two functors $F, G\colon \mathsf{Mod}(\underline{\mathbf{k}}_X) \to \mathsf{Mod}(\mathbf{k})$

$$F(\mathscr{F}) \coloneqq \varinjlim_{\substack{U \supset K \\ U \in \mathsf{Open}(X)}} \Gamma(U, \mathscr{F}) \qquad G(\mathscr{F}) \coloneqq \Gamma(K, \mathscr{F})$$

From the first step we have $F \cong G$. Therefore for all $i \geq 0$ we have an isomorphism

$$\mathrm{R}^i F(\mathscr{F}) \cong \mathrm{R}^i G(\mathscr{F})$$

To compute these functors, choose an injective resolution $\mathscr{F} \to \mathscr{I}^\bullet$ of $\mathscr{F}$. On the one hand

$$\mathrm{R}^i G(\mathscr{F}) = \mathrm{H}^i(\Gamma(K; I^\bullet)) = \mathrm{H}^i(K; \mathscr{F})$$

On the other

$$\mathrm{R}^i F(\mathscr{F}) = \mathrm{H}^i\left(\varinjlim_{\substack{U \supset K \\ U \in \mathsf{Open}(X)}} \Gamma(U, \mathscr{I}^\bullet)\right) \cong \varinjlim_{\substack{U \supset K \\ U \in \mathsf{Open}(X)}} \mathrm{H}^i\left(\Gamma(U, \mathscr{I}^\bullet)\right) = \varinjlim_{\substack{U \supset K \\ U \in \mathsf{Open}(X)}} \mathrm{H}^i\left(U; \mathscr{F}\right)$$

Therefore

$$\varinjlim_{\substack{U \supset K \\ U \in \mathsf{Open}(X)}} \mathrm{H}^i\left(U; \mathscr{F}\right) \cong \mathrm{R}^i F(\mathscr{F}) \cong \mathrm{R}^i G(\mathscr{F}) = \mathrm{H}^i(K; \mathscr{F}) \qquad \square$$

10.1.1 Mayer-Vietoris

The functorial behavior of restriction and extension established above suggests that we can compute the global cohomology of a space by systematically fracturing it into simpler, overlapping sub-domains. In classical algebraic topology, the standard technique for computing invariants via open or closed decompositions is governed by the Mayer-Vietoris sequence.

It is worth noting that the following results work replacing $\Gamma(X, -)$ with any other left-exact functor such as f_* for any continuous map $f : X \to Y$.

Theorem 10.1.1 (Open Mayer-Vietoris). Let X be a topological space. Suppose that $X = U \cup V$ for some open sets U, V. Then for every sheaf of $\mathbf{k}$-modules $\mathscr{F} \in \mathsf{Mod}(\underline{\mathbf{k}}_X)$ there exists a long exact sequence in cohomology:

$$\begin{array}{l}
0 \longrightarrow \mathrm{H}^0(X;\mathscr{F}) \longrightarrow \mathrm{H}^0(U;\mathscr{F}) \oplus \mathrm{H}^0(V;\mathscr{F}) \longrightarrow \mathrm{H}^0(U\cap V;\mathscr{F}) \\
\quad \longrightarrow \mathrm{H}^1(X;\mathscr{F}) \longrightarrow \mathrm{H}^1(U;\mathscr{F}) \oplus \mathrm{H}^1(V;\mathscr{F}) \longrightarrow \mathrm{H}^1(U\cap V;\mathscr{F}) \\
\quad \longrightarrow \mathrm{H}^2(X;\mathscr{F}) \longrightarrow \mathrm{H}^2(U;\mathscr{F}) \oplus \mathrm{H}^2(V;\mathscr{F}) \longrightarrow \cdots
\end{array}$$

Proof. Consider the sequence of sheaves of $\mathbf{k}$-modules

$$0 \longrightarrow \mathscr{F} \xrightarrow{s \mapsto (s|_U, s|_V)} \mathscr{F}|_U \oplus \mathscr{F}|_V \xrightarrow{(s,t) \mapsto s|_{U\cap V} - t|_{U\cap V}} \mathscr{F}|_{U\cap V} \longrightarrow 0$$

Let $p \in X = U \cup V$. If $p \in U$ but $p \notin V$ the induced sequence on the stalks is

$$0 \longrightarrow \mathscr{F}_p \longrightarrow \mathscr{F}_p \oplus 0 \longrightarrow 0 \longrightarrow 0$$

which is exact. Analogously if $p \in V$ but $p \notin U$ we also get an exact sequence.

Now, if $p \in U \cap V$ we have

$$0 \longrightarrow \mathscr{F}_p \xrightarrow{s \mapsto (s_x, s_x)} \mathscr{F}_p \oplus \mathscr{F}_p \xrightarrow{(s,t) \mapsto s-t} \mathscr{F}_p \longrightarrow 0$$

which is also exact.

Therefore

$$0 \longrightarrow \mathscr{F} \longrightarrow \mathscr{F}|_U \oplus \mathscr{F}|_V \longrightarrow \mathscr{F}|_{U \cap V} \longrightarrow 0$$

is exact in $\mathsf{Mod}(\underline{\mathbf{k}}_X)$. Applying Theorem 7.2.1 with $\Gamma(X, -)$ and using Proposition 10.1.1 we get the result. □

Theorem 10.1.2 (Closed Mayer-Vietoris)**.** Let X be a topological space. Suppose that $X = Z \cup W$ for some closed sets Z, W. Then for every sheaf of $\mathbf{k}$-modules $\mathscr{F} \in \mathsf{Mod}(\underline{\mathbf{k}}_X)$ there exists a long exact sequence in cohomology:

$$\begin{aligned} 0 \longrightarrow \mathrm{H}^0(X;\mathscr{F}) \longrightarrow \mathrm{H}^0(X;\mathscr{F}_Z) \oplus \mathrm{H}^0(X;\mathscr{F}_W) \longrightarrow \mathrm{H}^0(X;\mathscr{F}_{Z\cap W}) \\ \mathrm{H}^1(X;\mathscr{F}_Z) \longrightarrow \mathrm{H}^1(X;\mathscr{F}_Z) \oplus \mathrm{H}^1(X;\mathscr{F}_W) \longrightarrow \mathrm{H}^1(X;\mathscr{F}_{Z\cap W}) \\ \mathrm{H}^2(X;\mathscr{F}) \longrightarrow \mathrm{H}^2(X;\mathscr{F}_U) \oplus \mathrm{H}^2(X;\mathscr{F}_V) \longrightarrow \cdots \end{aligned}$$

Proof. Analogous to the open version with

$$0 \longrightarrow \mathscr{F} \longrightarrow \mathscr{F}_Z \oplus \mathscr{F}_W \longrightarrow \mathscr{F}_{Z\cap W} \longrightarrow 0 \quad \square$$

10.2 Acyclic resolutions

In Lemma 10.1.1 we have seen that, even though they exist, injectives sheaves are complicated objecs. Recall that cohomology can be computed using $\Gamma(X, -)$-acyclic resolutions.

In this section we introduce a class of $\Gamma(X, -)$-acyclic objects that will help us later.

Definition 10.2.1 (Flasque sheaf)**.** A sheaf of $\mathbf{k}$-modules $\mathscr{F}$ is **flasque** (or **flabby**) if for any inclussion of open subsets $V \subseteq U$ the restriction map $\mathrm{res}_{U,V} : \mathscr{F}(U) \to \mathscr{F}(V)$ is surjective.

Proposition 10.2.1. Flasque sheaves are $\Gamma(X, -)$-acyclic.

Proof. Let $\mathscr{F}$ be a flasque sheaf. Since $\mathsf{Mod}(\underline{\mathbf{k}}_X)$ has enough injectives we have a short exact sequence

$$0 \longrightarrow \mathscr{F} \longrightarrow \mathscr{I} \longrightarrow \mathscr{G} \longrightarrow 0$$

where $\mathscr{I}$ is an injective object. We will first show that, for any open subset $U \subseteq X$, the map $\mathscr{I}(U) \to \mathscr{G}(U)$ is surjective.

Let $s \in \mathscr{G}(U)$. Let $V, W \subseteq U$ be two open subsets and $t_V \in \mathscr{I}(V), t_W \in \mathscr{I}(W)$ be two sections lifting s to $\mathscr{I}$. Consider

$$t_V|_{V\cap U} - t_W|_{V\cap W} \in \mathscr{F}(V \cap W)$$

Since $\mathscr{F}$ is flasque there exists a section $x_V \in \mathscr{F}(V)$ such that $x_V|_{V\cap W} = t_V - t_W$. Therefore if $t'_V = t_V - x_V$ we have $t'_V|_{V\cap W} = t_W|_{V\cap W}$, i.e: there exists a section $t \in \mathscr{I}(V \cup W)$ such that $t|_V = t'|_V$ and $t'|_W = t|_W$. Clearly t is sent to $s|_{V\cap W}$.

Consider now a pair (W, t) which is maximal (for the obvious order relation), where W is an open subset of U and t is a section of $\mathscr{I}$ lifting s to W. Since s lifts locally on $\mathscr{I}$ and using the previous result we see that $W = U$.

But then using the long exact sequence in cohomology of our sequence and the fact that for $i > 0 \quad \mathrm{H}^i(X, \mathscr{I}) = 0$ imply that $\mathrm{H}^1(X, \mathscr{F}) = 0$ we get

$$\forall k - 1 \geq 1 \quad \mathrm{H}^k(X, \mathscr{F}) = \mathrm{H}^{k-1}(X, \mathscr{G})$$

Furthermore surjectivity of the maps $\mathscr{I}(U) \twoheadrightarrow \mathscr{G}(U)$ implies that $\mathscr{G}$ is flasque.

It follows by induction that

$$\forall k > 0 \quad \mathrm{H}^k(X, \mathscr{F}) = 0$$ □

Using flasque sheaves we can construct a canonical (and functorial) resolution of a sheaf of $\mathbf{k}$-modules to compute its cohomology, the so called Godement resolution. We only give here a sketch of its construction and refer the reader to [Stacks, Section 0FKR] for more details.

Construction (Godement resolution). Let $\mathscr{F}$ be a sheaf of $\mathbf{k}$-modules. The **Godement resolution** of $\mathscr{F}$ is constructed by considering the inclussion of $\mathscr{F}$ into the *Godement sheaf* $\mathscr{F}_{\mathrm{God}}$

$$U \mapsto \bigoplus_{p\in U} \mathscr{F}_p =: \mathscr{F}_{\mathrm{God}}(U)$$

Note that $\mathscr{F}_{\mathrm{God}}$ is flasque.

We then inject the quotient $\mathscr{F}_{\mathrm{God}} / \mathscr{F}$ into $\left(\mathscr{F}_{\mathrm{God}} / \mathscr{F}\right)_{\mathrm{God}}$ and so on.

Example 10.2.1. Let's try to compute sheaf cohomology using the Godement resolution.
Let $X = \{p, q\}$ with the *Sierpiński* topology $\mathcal{T} = \{\varnothing, \{p\}, X\}$. Consider the constant sheaf supported at the integers $\mathscr{F} = \underline{\mathbb{Z}}$.
To compute the cohomology of $\mathscr{F}$, we are going to construct the complex of global sections:

$$0 \longrightarrow \Gamma(X, \mathscr{F}_{\mathrm{God}}) \xrightarrow{\mathrm{d}^0} \Gamma\left(X, \left(\mathscr{F}_{\mathrm{God}} / \mathscr{F}\right)_{\mathrm{God}}\right) \xrightarrow{\mathrm{d}^1} \cdots$$

Note that the stalks of $\mathscr{F}$ are:

$$\mathscr{F}_p = \mathscr{F}(\{p\}) = \mathbb{Z}, \qquad \mathscr{F}_q = \mathscr{F}(X) = \mathbb{Z}.$$

Consider the Godement sheaf associated to $\mathscr{F}$:

$$\mathscr{F}_{\text{God}}(U) = \bigoplus_{x \in U} \mathscr{F}_x.$$

We have that:

$$\Gamma(X, \mathscr{F}_{\text{God}}) = \mathscr{F}_{\text{God}}(X) = \mathscr{F}_p \oplus \mathscr{F}_q = \mathbb{Z} \oplus \mathbb{Z}.$$

This means that we have a map d^{-1} as follows:

$$\mathrm{d}^{-1} : \underbrace{\Gamma(X, \mathscr{F})}_{\mathbb{Z}} \to \underbrace{\Gamma(X, \mathscr{F}_{\text{God}})}_{\mathbb{Z}\oplus\mathbb{Z}}, \quad \mathrm{d}^{-1}(k) = (k, k).$$

Let $\mathscr{Q}^1 = {}^{\mathscr{F}_{\text{God}}}\big/_{\mathscr{F}}$ be the cokernel sheaf. If we compute its stalks, we have that:

$$(\mathscr{Q}^1)_p = {}^{\mathscr{F}_{\text{God},p}}\big/_{\mathscr{F}_p} = {}^{\mathbb{Z}}\big/_{\mathbb{Z}} = 0,$$
$$(\mathscr{Q}^1)_q = {}^{\mathscr{F}_{\text{God},q}}\big/_{\mathscr{F}_q} = {}^{\mathbb{Z}\oplus\mathbb{Z}}\big/_{\langle (k,k) \rangle} \cong \mathbb{Z}.$$

Therefore, the global sections of the Godement sheaf of $\mathscr{Q}^1$ are:

$$\Gamma\left(X, (\mathscr{Q}^1)_{\text{God}}\right) = (\mathscr{Q}^1)_p \oplus (\mathscr{Q}^1)_q = 0 \oplus \mathbb{Z} = \mathbb{Z}.$$

This produces a differential map:

$$\mathrm{d}^0 : \underbrace{\Gamma(X, \mathscr{F}_{\text{God}})}_{\mathbb{Z}\oplus\mathbb{Z}} \to \underbrace{\Gamma\left(X, (\mathscr{Q}^1)_{\text{God}}\right)}_{\mathbb{Z}}, \quad \mathrm{d}^0(m, n) = m - n.$$

Let $\mathscr{Q}^2$ be the cokernel of the map $\mathscr{F}_{\text{God}} \to (\mathscr{Q}^1)_{\text{God}}$. If we compute its stalks, we have:

$$(\mathscr{Q}^2)_p = {}^{0}\big/_{0} = 0,$$
$$(\mathscr{Q}^2)_q = {}^{\mathbb{Z}}\big/_{\mathbb{Z}} = 0.$$

Therefore, we have that:

$$\Gamma\left(X, (\mathscr{Q}^2)_{\text{God}}\right) = 0 \oplus 0 = 0.$$

This means that our resolution of global sections is:

$$0 \longrightarrow \mathbb{Z}\oplus\mathbb{Z} \xrightarrow[(m,n)\mapsto m-n]{d^0} \mathbb{Z} \longrightarrow 0$$

Therefore, the cohomology groups of X with coefficients in $\underline{\mathbb{Z}}$ are:

$$\mathrm{H}^0(X, \underline{\mathbb{Z}}) = \ker(d^0) = \{(m, n) \in \mathbb{Z}\oplus\mathbb{Z} \mid m - n = 0\} \cong \mathbb{Z},$$
$$\mathrm{H}^1(X, \underline{\mathbb{Z}}) = \operatorname{coker}(d^0) = {}^{\mathbb{Z}}\big/_{\operatorname{im}(d^0)} = {}^{\mathbb{Z}}\big/_{\mathbb{Z}} = 0,$$
$$\mathrm{H}^i(X, \underline{\mathbb{Z}}) = 0 \quad \forall i > 1.$$

This example shows that computing sheaf cohomology using the Godement resolution is an almost imposible task in general. In the following chapters we will see how we can solve this problem.

Chapter 11

Cohomology and homotopy

If two topological spaces are continuously deformed into one another, we intuitively expect their underlying algebraic properties to remain invariant. In this chapter, we formalize this intuition by showing that sheaf cohomology is a homotopy invariant. To do this rigorously without losing the local data of our sheaves, we must generalize standard homotopy to relative topology over a base parameter space S.

We do this adapting the results of [KS90, Section 2.7].

This results will also be useful when computing the cohomology of simple spaces as we will see later.

Notation. We will denote $I := [0, 1]$ the unit interval.

We begin first by defining what we mean for a homotopy relative to a base space.

Definition 11.0.1. Let X and Y be topological spaces. A map $f : X \to Y$ is **continuous over a topological space** S if it is continuous and there exists continuous maps $p_X : X \to S$ and $p_Y : Y \to S$ making the following diagram commute

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ & {\scriptstyle p_X} \searrow \quad \swarrow {\scriptstyle p_Y} & \\ & S & \end{array}$$

Note that this relative framework is generalization of the non-relative case, since any standard topological scenario can be viewed through a relative lens simply by taking a product with the base space as demonstrated below.

Example 11.0.1. Any continuous map $f : X \to Y$ can be made continuous over S trivially by considering

$$\begin{array}{ccc} X \times S & \xrightarrow{(f, \mathbb{1}_S)} & Y \times S \\ & {\scriptstyle p_X} \searrow \quad \swarrow {\scriptstyle p_Y} & \\ & S & \end{array}$$

where p_X and p_Y are the obvious projections.

With relative maps defined, we can now define relative homotopies. A homotopy over S is a continuous deformation where every intermediate step is itself a continuous

map over S. This ensures that points do not stray out of their assigned fibers during the deformation.

Definition 11.0.2. Let X, Y and S be topological spaces.

1. Let $f_0, f_1 \colon X \rightrightarrows Y$ be two continuous maps over S. We say that f_0 and f_1 are **homotopic** over S if there exists a continuous map $h \colon X \times I \to Y$ over S such that $f_0 = h(-, 0)$ and $f_1 = h(-, 1)$. We denote this by $f_0 \sim_S f_1$.
2. A continuous maps over S $f \colon X \to Y$ is a **homotopy equivalence** over S if there exists a continuous maps over S $g \colon Y \to X$ such that $g \circ f \sim_S \mathbb{1}_X$ and $f \circ g \sim_S \mathbb{1}_Y$. In this case we write $X \sim_S Y$.
3. A topological space Z is **contractible** over S if the map $p_Z \colon Z \to S$ is a homotopy equivalence.

Example 11.0.2. If $S = \{*\}$ is the one-point topological space we recover the usual definition of homotopy of topological spaces.

Remark 11.0.1. If $f_0, f_1 \colon X \rightrightarrows Y$ are homotopic over S we have a commutative diagram

$$\begin{array}{ccc} X & & \\ {\scriptstyle j_i}\downarrow & \searrow^{f_t} & \\ X \times I & \xrightarrow{h} & Y \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle p_Y} \\ X & \xrightarrow[p_X]{} & S \end{array}$$

where $j_t(z) = (z, t)$ for $t \in \{0, 1\}$.

Because homotopies are parameterized by the unit interval, we must understand how sheaf cohomology behaves on I before establishing the global invariance theorem.

Lemma 11.0.1. Let $\mathscr{F} \in \mathsf{Mod}(\underline{\mathbf{k}}_I)$. Then

1. For $i > 1$ we have $\mathrm{H}^i(I; \mathscr{F}) = 0$.
2. If $\mathscr{F}(I) \to \mathscr{F}_t$ is an epimorphism for all $t \in I$ then $\mathrm{H}^1(I; \mathscr{F}) = 0$.

Proof. Let $j \geq 1$ and $s \in \mathrm{H}^j(I; \mathscr{F})$. We want to show that $s = 0$.

For $0 \leq t_1 < t_2 \leq 1$ consider the morphism

$$f_{t_1,t_2} : \mathrm{H}^i(I; \mathscr{F}) \to \mathrm{H}^i([t_1, t_2]; \mathscr{F}|_{[t_1,t_2]})$$

and let

$$J := \{t \in [0, 1] \mid f_{0,t}(s) = 0\}$$

Since $\mathrm{H}^j(\{0\}, \mathscr{F}) = 0$ we have that $0 \in J$. Furthermore if $t \in J$ then for any $t' \leq t$ the map $f_{0,t'}$ factors as

$$\begin{array}{ccc} \mathrm{H}^i(I; \mathscr{F}) & \xrightarrow{f_{0,t'}} & \mathrm{H}^i([0, t']; \mathscr{F}) \\ {\scriptstyle f_{0,t}}\downarrow & \nearrow & \\ \mathrm{H}^i([0, t]; \mathscr{F}) & & \end{array}$$

Therefore since $f_{0,t}(s) = 0$ we have that $f_{0,t'}(s) = 0$, i.e: $[0,t] \subseteq J$ and thus J is a sub-interval of $[0,1]$ starting at 0.

To prove that $J = [0,1]$ we will see that it is both open and closed.

Open Suppose $t_0 \in J$. By Lemma 10.1.3 we have that

$$\mathrm{H}^j([0,t_0];\mathscr{F}) \cong \varinjlim_{t>t_0} \mathrm{H}^j([0,t];\mathscr{F})$$

Therefore there exists $[0, t_0 + \varepsilon[$ such that $f_{0,t}(s) = 0$ for all $t \in [0, t_0 + \varepsilon[$, and thus J is open.

Closed Let $t_0 \coloneqq \sup J$. We want to see that $t_0 \in J$. For $t \in [0,t_0]$ consider the Mayer-Vietoris sequence associated with $[0,t_0] = [0,t] \cup [t,t_0]$:

$$\cdots \longrightarrow \mathrm{H}^i([0,t_0];\mathscr{F}) \longrightarrow \mathrm{H}^i([0,t];\mathscr{F}) \oplus \mathrm{H}^i([t,t_0];\mathscr{F}) \longrightarrow \mathrm{H}^i(\{t\},\mathscr{F}) \longrightarrow \cdots$$

If $i > 1$ then since $\mathrm{H}^i(\{t\};\mathscr{F}) = \mathrm{H}^{i-1}(\{t\};\mathscr{F}) = 0$ we have an isomorphism

$$\mathrm{H}^i([0,t_0];\mathscr{F}) \cong \mathrm{H}^i([0,t];\mathscr{F}) \oplus \mathrm{H}^i([t,t_0];\mathscr{F})$$

Similarly, if $\mathscr{F}(I) \twoheadrightarrow \mathscr{F}_t$ is an epimorphism we have

$$\mathrm{H}^0([0,t_0];\mathscr{F}) \twoheadrightarrow \mathrm{H}^0(\{t\};\mathscr{F}) \longrightarrow \mathrm{H}^1([0,t_0];\mathscr{F}) \xrightarrow{\sim} \mathrm{H}^1([0,t];\mathscr{F}) \oplus \mathrm{H}^1([t,t_0];\mathscr{F})$$

In both cases for $f_{0,t_0}(s)$ to be zero it suffices for its components in $\mathrm{H}^i([0,t];\mathscr{F})$ and $\mathrm{H}^i([t,t_0];\mathscr{F})$ to vanish.
On the one hand since $t < t_0 = \sup J$ we can choose $t \in J$, so $f_{0,t}(s) = 0$. On the other by Lemma 10.1.3 we have

$$\varinjlim_{t<t_0} \mathrm{H}^i([t,t_0];\mathscr{F}) \cong \mathrm{H}^i(\{t_0\};\mathscr{F}) = 0$$

Therefore $f_{0,t_0}(s) = 0$ and thus $t_0 \in J$.

Since J is non-empty, open and closed and $[0,1]$ is connected necessarily $J = [0,1]$.
Therefore $s = f_{0,1}(s) = 0$. □

We also need a mechanism to compare sheaves across different spaces over S. Given a relative map $f : X \to Y$, we cannot directly compare the derived functors $\mathrm{R}^i p_{X,*}$ and $\mathrm{R}^i p_{Y,*}$ because they act on different sheaf categories ($\mathsf{Mod}(\underline{\mathbf{k}}_X)$ and $\mathsf{Mod}(\underline{\mathbf{k}}_Y)$ respectively). To remedy this, we use the inverse image functor p^{-1} to pull coefficients up from the base space S.

Lemma 11.0.2. Let $f : X \to Y$ be a continuous map over S.
For all $i \geq 0$ there exists a natural transformation

$$f^{\#,i} \coloneqq \mathrm{R}^i p_{Y,*} \circ p_Y^{-1} \to \mathrm{R}^i p_{X,*} \circ p_X^{-1}$$

Proof. Let $\mathscr{F} \in \mathsf{Mod}(\underline{\mathbf{k}}_S)$ be a sheaf over S.

Consider an injective resolution $p_Y^{-1}(\mathscr{F}) \to \mathscr{I}^\bullet$ of the inverse image $p_Y^{-1}(\mathscr{F}) \in \mathsf{Mod}(\underline{\mathbf{k}}_Y)$ Using the unit of the adjunction $f_* \dashv f^{-1}$ we get a natural map of complexes in Y

$$\mathscr{I}^\bullet \to \left(f_* \circ f^{-1}\right)(\mathscr{I}^\bullet)$$

Applying the direct image $p_{Y,*}$ to this, since $p_Y \circ f = p_X$ we get a morphism of complexes in S

$$p_{Y,*}(\mathscr{I}^\bullet) \to \left(p_{X,*} \circ f^{-1}\right)(\mathscr{I}^\bullet)$$

Note that, since f^{-1} is exact, $f^{-1}(\mathscr{I}^\bullet)$ is an exact resolution of $p_X^{-1}(\mathscr{F})$. However, it need not be made of injectives. To fix this, consider a proper injective resolution $p_X^{-1}(\mathscr{F}) \to \mathscr{J}^\bullet$. By Theorem 7.1.2 there exists a morphism (unique up to homotopy of complexes) lifting the identity

$$f^{-1}(\mathscr{I}^\bullet) \to \mathscr{J}^\bullet$$

Applying $p_{X,*}$ we get a morphism in S

$$\left(p_{X,*} \circ f^{-1}\right)(\mathscr{I}^\bullet) \to p_{X,*}\left(\mathscr{J}^\bullet\right)$$

Therefore composing with the previous map we get

$$p_{Y,*}(\mathscr{I}^\bullet) \to \left(p_{X,*} \circ f^{-1}\right)(\mathscr{I}^\bullet) \to p_{X,*}\left(\mathscr{J}^\bullet\right)$$

Applying the i-th right derived functor we get a map

$$\left(\mathrm{R}^i p_{Y,*} \circ p_Y^{-1}\right)(\mathscr{F}) \to \left(\mathrm{R}^i p_{X,*} \circ p_X^{-1}\right)(\mathscr{F})$$ □

> **Remark 11.0.2.** If $S = \{*\}$ then $\mathsf{Sh}_*(\mathsf{C}) \simeq \mathsf{C}$ and $p_Z^{-1}\underline{M} = \underline{M}$ for all $M \in \mathsf{C}$ and $Z \in \{X, Y\}$.
> In this case the morphism of Lemma 11.0.2 is
>
> $$f^{\#,i}(\underline{M})\colon \mathrm{H}^i(Y;\underline{M}) \to \mathrm{H}^i(X;\underline{M})$$
>
> We denote this map by f^* when there is no confusion with the direct image of sheaves.

We now arrive at the central result of this chapter where we show that relative homotopies preserve sheaf cohomology. This means that continuously deforming a map over S does not alter its effect on the cohomology of sheaves pulled up from S.

> **Theorem 11.0.1.** Let $f_0, f_1\colon X \rightrightarrows Y$ be two homotopic maps over S. Then $f_0^{\#,i} = f_1^{\#,i}$ for all $i \geq 0$.

Proof. Let $\mathscr{F} \in \mathsf{Sh}_S(\mathsf{C})$. Choose injective resolutions $p_Y^{-1}(\mathscr{F}) \to \mathscr{I}^\bullet$ and $p_X^{-1}(\mathscr{F}) \to \mathscr{J}^\bullet$.

Since $f_t = h \circ j_t$ we can factor $\mathscr{I}^\bullet \to \left(f_{t,*} \circ f_t^{-1}\right)(\mathscr{I}^\bullet)$ through

$$\mathscr{I}^\bullet \to \left(h_* \circ h^{-1}\right)(\mathscr{I}^\bullet) \to \left(\left(h \circ j_t\right)_* \circ \left(h \circ j_t\right)^{-1}\right)(\mathscr{I}^\bullet)$$

Pushing this factorization via $p_{Y,*}$ we get

$$p_{Y,*}(\mathscr{I}^\bullet) \to \left(p_{Y,*} \circ h_* \circ h^{-1}\right)(\mathscr{I}^\bullet) \to \left(p_{Y,*} \circ \left(h \circ j_t\right)_* \circ \left(h \circ j_t\right)^{-1}\right)(\mathscr{I}^\bullet)$$

On the one hand since $p_Y \circ h = p_X \circ p$ the middle term becomes

$$\left(p_{Y,*} \circ h_* \circ h^{-1}\right)(\mathscr{I}^\bullet) = \left(\left(p_Y \circ h\right)_* \circ h^{-1}\right)(\mathscr{I}^\bullet) = \left(\left(p_X \circ p\right)_* \circ h^{-1}\right)(\mathscr{I}^\bullet)$$

On the other since $f_t = h \circ j_t$ and $p_Y \circ f_t = p_X$ we have

$$\left(p_{Y,*} \circ \left(h \circ j_t\right)_* \circ \left(h \circ j_t\right)^{-1}\right)\left(\mathscr{I}^\bullet\right) = \left(\left(p_Y \circ f_t\right)_* \circ f_t^{-1}\right)\left(\mathscr{I}^\bullet\right) = \left(p_{X,*} \circ f_t^{-1}\right)\left(\mathscr{I}^\bullet\right)$$

As in Lemma 11.0.2 since $f_t^{-1}(\mathscr{I}^\bullet)$ is an exact (but not necessarilly injective) resolution of $p_X^{-1}(\mathscr{F})$ there exists a unique (up to homotopy) lifting of the identity

$$f_t^{-1}(\mathscr{I}^\bullet) \to \mathscr{J}^\bullet$$

where $p_X^{-1}(\mathscr{F}) \to \mathscr{J}^\bullet$ is an injective resolution of $p_X^{-1}(\mathscr{F})$. Applying $p_{X,*}$ we obtain a map

$$\left(p_{X,*} \circ f_t^{-1}\right)\left(\mathscr{I}^\bullet\right) \to p_{X,*}\left(\mathscr{J}^\bullet\right)$$

Therefore we obtain a factorization of $f^{\#,t}$

$$p_{Y,*}\left(\mathscr{I}^\bullet\right) \xrightarrow{g_1} \left(p_{X,*} \circ p_* \circ h^{-1}\right)\left(\mathscr{I}^\bullet\right) \xrightarrow{g_2} p_{X,*}\left(\mathscr{J}^\bullet\right)$$

Denote by

$$\alpha_t \coloneqq \mathrm{H}^\bullet g_2 \colon \mathrm{H}^\bullet\left(\left(p_{X,*} \circ p_* \circ h^{-1}\right)\left(\mathscr{I}^\bullet\right)\right) \to \mathrm{H}^\bullet\left(p_{X,*}\left(\mathscr{J}^\bullet\right)\right)$$

the induced map on cohomology of g_2. Since g_1 does not depend on t, we only need to see that $\alpha_0 = \alpha_1$.

The map α_t is induced by

$$\left(p_* \circ h^{-1}\right)\left(\mathscr{I}^\bullet\right) \to \underbrace{\left(j_t^{-1} \circ h^{-1}\right)}_{f_t^{-1}}\left(\mathscr{I}^\bullet\right) \to \mathscr{J}^\bullet$$

Note that $h^{-1}(\mathscr{I}^\bullet)$ is an exact sequence resolving $\left(h^{-1} \circ p_Y^{-1}\right)(\mathscr{F}) \cong \left(p^{-1} \circ p_X^{-1}\right)(\mathscr{F})$. Since by Lemma 11.0.1 higher cohomology ($k > 0$) vanishes, the cohomology of α_t is concentrated in degree zero, which is

$$\left(p_* \circ p^{-1} \circ p_X^{-1}\right)(\mathscr{F}) \to \left(j_t^{-1} \circ p^{-1} \circ p_X^{-1}\right)(\mathscr{F})$$

However, note that we have isomorphisms

$$p_X^{-1}(\mathscr{F}) \xrightarrow{\sim} \left(p_* \circ p^{-1} \circ p_X^{-1}\right)(\mathscr{F}) \xrightarrow{\sim} \left(j_t^{-1} \circ p^{-1} \circ p_X^{-1}\right)(\mathscr{F}) \xrightarrow{\sim} p_X^{-1}(\mathscr{F})$$

whose composition is the identity on p_X^{-1}, which means that α_t is induced by the identity. Indeed:

- The first one comes from the unit of the adjunction $p^{-1} \dashv p_*$.
- Applying p_* to the unit of the adjunction $j_t^{-1} \dashv j_{t,*}$ we get

$$p_* \xrightarrow{\sim} p_* \circ j_{t,*} \circ j_t^{-1} = \left(p \circ j_t\right)_* \circ j_t^{-1} = \mathbb{1}_* \circ j_t^{-1} = j_t^{-1}$$

- The last one is direct from the composition of the inverse image

$$j_t^{-1} \circ p^{-1} = \left(p \circ j_t\right)^{-1} = \mathbb{1}^{-1} = \mathbb{1}$$

Therefore since α_t does not depend on t we get $\alpha_0 = \alpha_1$ and thus $f_0^{\#,i} = f_1^{\#,i}$. □

Corollary 11.0.1. Let $f : X \to Y$ be a homotopy equivalence over S. Then $f^{\#,i}$ is a natural isomorphism for all $i \geq 0$.

Proof. By definition there exists a continuous map over S $g : Y \to X$ such that $g \circ f \sim_S \mathbb{1}_X$ and $f \circ g \sim_S \mathbb{1}_Y$.

Since $f^{\#,i}$ is a natural transformation we have

$$(g \circ f)^{\#,i} = f^{\#,i} \circ g^{\#,i} \qquad (f \circ g)^{\#,i} = g^{\#,i} \circ f^{\#,i}$$

Therefore since homotopic maps induce the same transformation

$$f^{\#,i} \circ g^{\#,i} = (\mathbb{1}_X)^{\#,i} = \mathbb{1} \qquad g^{\#,i} \circ f^{\#,i} = (\mathbb{1}_Y)^{\#,i} = \mathbb{1}$$ □

We end this chapter by giving some results that will help us compute the sheaf cohomology of simple spaces in the next chapter.

Corollary 11.0.2. Let X be contractible over S. Then for all $\mathscr{F} \in \mathsf{Mod}(\underline{\mathbf{k}}_S)$ the adjunction unit map

$$\mathscr{F} \to \left(\mathrm{R}^0 p_{X,*} \circ p_X^{-1}\right)(\mathscr{F})$$

is an isomorphism and for all $i > 0$

$$\left(\mathrm{R}^i p_{X,*} \circ p_X^{-1}\right)(\mathscr{F}) = 0$$

Proof. We can view S as a space over itself via the identity map $\mathbb{1}_S : S \to S$. By hypothesis the map $p_X : X \to S$ is a homotopy equivalence over S. Therefore by Corollary 11.0.1 $p_X^{\#,i}$ is a natural isomorphism

$$\mathrm{R}^i \mathbb{1}_{S,*} \circ \mathbb{1}_S^{-1} \xrightarrow{\sim} \mathrm{R}^i p_{X,*} \circ p_X^{-1}$$

For $i = 0$ we have $\mathrm{R}^0 \mathbb{1}_{S,*} = \mathbb{1}$. Therefore

$$\mathbb{1}_S \xrightarrow{\sim} \mathrm{R}^0 p_{X,*} \circ p_X^{-1}$$

On the other hand since $\mathbb{1}_{S,*}$ is exact $\mathrm{R}^i \mathbb{1}_{S,*} = 0$ for all $i > 0$. Therefore

$$0 \xrightarrow{\sim} \mathrm{R}^i p_{X,*} \circ p_X^{-1}$$ □

Definition 11.0.3. Let X be a topological space over S with $p_X : X \to S$. Let $i : A \hookrightarrow X$ be a subspace with $p_A := p_X|_A$.
We say that A is a **deformation retract** of X over S if there exists a continuous map over S $r : X \to A$ with $r \circ i = \mathbb{1}_A$ and a homotopy over S $F : X \times I \to A$ between $\mathbb{1}_X$ and $i \circ r$. The map r is called the **retraction** of X into A.
If $F(a, t) = a$ for all $t \in I$ and $a \in A$, we say that A is a **strong deformation retract** of X over S.

Corollary 11.0.3. Let $i : A \hookrightarrow X$ be a strong deformation retract of X over S. Then the induced map

$$i^{\#,n} : \mathrm{R}^n p_{X,*} \circ p_X^{-1} \to \mathrm{R}^n p_{A,*} \circ p_A^{-1}$$

is a natural isomorphism for all $n \geq 0$.

Proof. Let $r\colon X \to A$ be the retraction of X into A. Since $r \circ i = \mathbb{1}_A$ we have for all $n \geq 0$

$$i^{\#,n} \circ r^{\#,n} = (i \circ r)^{\#,n} = \mathbb{1}_A^{\#,n} = \mathbb{1}_{\mathrm{R}^i p_{A,*} \circ p_A^{-1}}$$

On the other hand since $i \circ r \sim_S \mathrm{id}_X$ by Theorem 11.0.1 we have

$$r^{\#,n} \circ i^{\#,n} = (r \circ i)^{\#,n} = \mathbb{1}_X^{\#,n} = \mathbb{1}_{\mathrm{R}^i p_{X,*} \circ p_X^{-1}}$$

Therefore $r^{\#,n}$ and $i^{\#,n}$ are inverses of each other and thus natural isomorphisms. □

Chapter 12

Some examples of cohomology

We focus our attention now to the computation of the cohomology of some topological spaces. Our main tools will be the Mayer-Vietoris sequence and the homotopy invariance of cohomology.

Example 12.0.1 ($\mathbb{S}^1$). Let $\mathscr{F} \in \mathsf{Mod}(\underline{\mathbf{k}}_S)$ be a sheaf of $\mathbf{k}$-modules over S and $X \coloneqq \mathbb{S}^1 \times S$. Consider the open cover of $\mathbb{S}^1 = H^+ \cup H^-$ where H^+ is the upper hemisphere and H^- the lower one, both extended slightly past the equator. Let $U = H^+ \times S$ and $V = H^- \times S$. Since H^+ and H^- are homeomorphic to open intervals, which are contractible, both U and V are contractible over S. Furthermore $U \cap V = (W_1 \times S) \coprod (W_2 \times S)$ consists of two disjoint pieces, each of these contractible over S.
Therefore since cohomology of contractible spaces is concentrated in degree 0 we only have

$$\begin{aligned} \mathrm{R}^0 p_{U,*}\mathscr{F}|_U &\cong \mathscr{F} \\ \mathrm{R}^0 p_{V,*}\mathscr{F}|_V &\cong \mathscr{F} \\ \mathrm{R}^0 p_{U\cap V,*}\mathscr{F}|_{U\cap V} &\cong \mathscr{F} \oplus \mathscr{F} \end{aligned}$$

Writing the Mayer-Vietoris sequence since the higher terms vanish we end up with

$$0 \longrightarrow \mathrm{R}^0 p_{X,*}\mathscr{F}|_X \longrightarrow \mathscr{F} \oplus \mathscr{F} \xrightarrow{\alpha} \mathscr{F} \oplus \mathscr{F} \xrightarrow{\delta} \mathrm{R}^1 p_{X,*}\mathscr{F}|_X \longrightarrow 0$$

Therefore directly we get $\mathrm{R}^i p_{X,*}\mathscr{F}_X = 0$ for all $i \geq 2$.
Note that $\alpha \colon \mathscr{F} \oplus \mathscr{F} \to \mathscr{F} \oplus \mathscr{F}$ maps a pair of sections (s_U, s_V) into their difference on the overlap. Since the section s_U is defined globally on U its restriction to W_1 and to W_2 is identical and analogously for s_V. Therefore we have

$$\alpha(s_U, s_V) = (s_U - s_V, s_U - s_V)$$

and thus $\ker \alpha = \{(s, s) \mid s \in \mathscr{F}\}$ is the diagonal which means that

$$\mathrm{R}^0 p_{X,*}\mathscr{F}|_X \cong \mathscr{F}$$

On the other hand by exactness δ is surjective, meaning that

$$\mathrm{R}^1 p_{X,*}\mathscr{F}|_X \cong \operatorname{coker} \alpha \cong {}^{\mathscr{F} \oplus \mathscr{F}}\big/_{\operatorname{Im} \alpha} = {}^{\mathscr{F} \oplus \mathscr{F}}\big/_{\langle (s,s) \rangle} \cong \mathscr{F}$$

Therefore

$$\mathrm{R}^i p_{\mathbb{S}^1 \times S,*}\mathscr{F}|_{\mathbb{S}^1 \times S} = \begin{cases} \mathscr{F} & i = 0 \\ \mathscr{F} & i = 1 \\ 0 & i \geq 2 \end{cases}$$

In particular taking $S = \{*\}$ since $\mathscr{F} \cong \underline{M}$ for some $M \in \mathsf{Mod}(\mathbf{k})$ we get

$$\mathrm{H}^i(\mathbb{S}^1; \underline{M}) = \begin{cases} M & i = 0 \\ M & i = 1 \\ 0 & i \geq 2 \end{cases}$$

Example 12.0.2 ($\mathbb{S}^n$)**.** Let $\mathscr{F} \in \mathsf{Mod}(\underline{\mathbf{k}}_S)$. We are going to see that cohomology of $X = \mathbb{S}^n \times S$ over S by induction over n is

$$\mathrm{R}^i p_{X,*}\mathscr{F}|_X = \begin{cases} \mathscr{F} & i = 0 \\ \mathscr{F} & i = n \\ 0 & i \notin \{0, n\} \end{cases}$$

We have already computed the base case $n = 1$ in Example 12.0.1, so we can suppose $n \geq 2$.
Consider the open cover $\mathbb{S}^n = H^+ \cup H^-$ where H^+ is the open upper hemisphere and H^- is the lower open hemisphere, both extending slightly past the equator. We have that $U \coloneqq H^+ \times S$ and $V \coloneqq H^- \times S$ are contractible over S. Furthermore since $H^+ \cap H^-$ is a thickened equatorial band it can be *squeezed* into the equator itself, which it is a $(n-1)$-sphere. This means that $H^+ \cap H^-$ strong-deformation retracts onto $\mathbb{S}^{n-1}$ and thus, $U \cap V$ strong-deformation retracts onto $\mathbb{S}^{n-1} \times S$.
Since $\mathbb{S}^{n-1}$ is connected the first terms of the Mayer-Vietoris sequence are

$$0 \longrightarrow \mathrm{R}^0 p_{X,*}\mathscr{F}|_X \longrightarrow \mathscr{F} \oplus \mathscr{F} \xrightarrow{\alpha} \mathscr{F} \xrightarrow{\delta} \mathrm{R}^1 p_{X,*}\mathscr{F}|_X \longrightarrow 0 \oplus 0 \longrightarrow \cdots$$

In this case we have $\alpha(s_U, s_V) = s_U - s_V$ and thus $\mathrm{R}^0 p_{X,*}\mathscr{F}|_X \cong \mathscr{F}$. Furthermore since α is clearly surjective and δ is surjective because the sequence is exact we have $\mathrm{R}^1 p_{X,*}\mathscr{F}|_X \cong 0$.
If we focus now on the higher terms of the Mayer-Vietoris sequence we have, for $i \geq 1$

$$\cdots \longrightarrow \mathrm{R}^i p_{U,*} \oplus \mathrm{R}^i p_{V,*} \longrightarrow \mathrm{R}^i p_{U\cap V,*} \xrightarrow{\delta} \mathrm{R}^{i+1} p_{X,*} \longrightarrow \mathrm{R}^{i+1} p_{U,*} \oplus \mathrm{R}^{i+1} p_{V,*} \longrightarrow \cdots$$

Since U and V are contractible over S these terms are zero by Corollary 11.0.2 and thus we get an isomorphism

$$\mathrm{R}^i p_{U\cap V,*} \xrightarrow{\sim} \mathrm{R}^{i+1} p_{X,*}$$

Substituting $U \cap V$ we get

$$\mathrm{R}^{i+1} p_{X,*}\mathscr{F}|_{\mathbb{S}^n \times S} \cong \mathrm{R}^i p_{X,*}\mathscr{F}|_{\mathbb{S}^{n-1} \times S} \cong \begin{cases} \mathscr{F} & i \in \{0, n-1\} \\ \mathscr{F} & i \notin \{0, n-1\} \end{cases}$$

Therefore

$$\mathrm{R}^i p_{\mathbb{S}^n \times S,*}\mathscr{F}|_{\mathbb{S}^n \times S} = \begin{cases} \mathscr{F} & i = 0 \\ \mathscr{F} & i = n \\ 0 & i \notin \{0, n\} \end{cases}$$

In particular taking $S = \{*\}$ since $\mathscr{F} \cong \underline{M}$ for some $M \in \mathsf{Mod}(\mathbf{k})$ we get

$$\mathrm{H}^i(\mathbb{S}^n; \underline{M}) = \begin{cases} M & i = 0 \\ M & i = n \\ 0 & i \notin \{0, n\} \end{cases}$$

Example 12.0.3 ($\mathbb{R}^n \setminus \{0\}$)**.** We are going to see first that $X := (\mathbb{R}^n \setminus \{0\}) \times S$ is a strong deformation retract of $A := \mathbb{S}^{n-1} \times S$. To do this, consider the inclusion $i \colon A \hookrightarrow X$ and the retraction $r \colon X \to A, r(v, s) := \left(\frac{v}{\|v\|}, s\right)$. As the homotopy we can take

$$\begin{aligned} H \colon X \times [0,1] &\to A \\ (v, s, t) &\mapsto \left((1-t)v + t\frac{v}{\|v\|}, s\right) \end{aligned}$$

Therefore by Corollary 11.0.3 for all $\mathscr{F} \in \mathsf{Mod}(\underline{\mathbf{k}}_S)$ we have and $i \geq 0$

$$\mathrm{R}^i p_{X,*}\mathscr{F}|_X \cong \mathrm{R}^i p_{A,*}\mathscr{F}|_X$$

In particular if $n \geq 2$ we have

$$\mathrm{R}^i p_{(\mathbb{R}^n\setminus\{0\})\times S,*}\mathscr{F}|_{(\mathbb{R}^n\setminus\{0\})\times S} \cong \begin{cases} \mathscr{F} & i = 0 \\ \mathscr{F} & i = n-1 \\ 0 & i \notin \{0, n-1\} \end{cases}$$

The case $n = 1$ is different since $\mathbb{R} \setminus \{0\}$ has two connected components. We have a deformation retract $\mathbb{R} \setminus \{0\} \to \mathbb{S}^0 \cong \{\pm 1\}$. Therefore

$$\mathrm{R}^i p_{(\mathbb{R}\setminus\{0\})\times S,*}\mathscr{F}|_{(\mathbb{R}\setminus\{0\})\times S} \cong \begin{cases} \mathscr{F} \oplus \mathscr{F} & i = 0 \\ 0 & i \neq 0 \end{cases}$$

In particular taking $S = \{*\}$ since $\mathscr{F} \cong \underline{M}$ for some $M \in \mathsf{Mod}(\mathbf{k})$ we get

$$\mathrm{H}^i(\mathbb{R} \setminus \{0\}; \underline{M}) = \begin{cases} M \oplus M & i = 0 \\ 0 & i \neq 0 \end{cases}$$

$$\mathrm{H}^i(\mathbb{R}^n \setminus \{0\}; \underline{M}) = \begin{cases} M & i = 0 \\ M & i = n-1 \\ 0 & i \notin \{0, n-1\} \end{cases} \qquad \forall n > 1$$

Example 12.0.4 ($\mathbb{RP}^2$)**.** Let $\mathscr{F} \in \mathsf{Mod}(\underline{\mathbf{k}}_S)$ and $X := \mathbb{RP}^2 \times S$.
Consider an open cover $\mathbb{RP}^2 = U_0 \cup V_0$ where U_0 is an open disk around a point, V_0 is an open neighbourhood of the *equator* $\mathbb{RP}^1 \subset \mathbb{RP}^2$ (which is homeomorphic to an open Möbius band). In this case $U_0 \cap V_0$ is homeomorphic to an open annulus. If we let $U := U_0 \times S$ and $V := V_0 \times S$ then $X = U \cup V$ and we can use Mayer-Vietoris to compute the cohomology.
On the one hand since U_0 is contractible U is contractible over S and thus by

Corollary 11.0.2

$$\mathrm{R}^i p_{U,*}\mathscr{F}|_U \cong \begin{cases} \mathscr{F} & i = 0 \\ 0 & i > 0 \end{cases}$$

On the other since V_0 and $U_0 \cap V_0$ strongly deformation retract into their inner circle $\mathbb{S}^1$ both V and $U \cap V$ strongly deformation retract over S into $\mathbb{S}^1 \times S$ and thus by Corollary 11.0.3 and Example 12.0.1

$$\mathrm{R}^i p_{V,*}\mathscr{F}_V = \mathrm{R}^i p_{U\cap V} \cong \begin{cases} \mathscr{F} & i \in \{0,1\} \\ 0 & i > 1 \end{cases}$$

Therefore the associated Mayer-Vietoris sequence is

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathrm{R}^0 p_{X,*}\mathscr{F}|_X & \longrightarrow & \mathscr{F}\oplus\mathscr{F} & \xrightarrow{\alpha} & \mathscr{F} \\ & & & & & \overset{\delta_0}{\searrow} & \\ & & \mathrm{R}^1 p_{X,*}\mathscr{F}|_X & \longrightarrow & 0\oplus\mathscr{F} & \xrightarrow{\beta} & \mathscr{F} \\ & & & & & \overset{\delta_1}{\searrow} & \\ & & \mathrm{R}^2 p_{X,*}\mathscr{F}|_X & \longrightarrow & 0 & & \end{array}$$

The map α is given by $(s_1, s_2) \mapsto s_1 - s_2$ and thus its kernel is the diagonal making

$$\mathrm{R}^0 p_{X,*}\mathscr{F}|_X \cong \mathscr{F}$$

Furthermore since α is surjective we get $\operatorname{Im}\alpha = \mathscr{F}$ and thus $\delta_0 = 0$.
The map β is induced by the inclussion $U \cap V \hookrightarrow V$. Therefore since the annulus $U_0 \cap V_0$ wraps twice around the inner circle of V_0 we have that β is multiplication by two. Therefore by exactness

$$\mathrm{R}^i p_{\mathbb{RP}^2\times S,*}\mathscr{F}|_{\mathbb{RP}^2\times S} \cong \begin{cases} \mathscr{F} & i = 0 \\ \ker(\times 2\colon \mathscr{F} \to \mathscr{F}), & i = 1 \\ \operatorname{coker}(\times 2\colon \mathscr{F} \to \mathscr{F}) = \mathscr{F}/_{2\mathscr{F}} & i = 2 \\ 0 & i > 2 \end{cases}$$

In particular taking $S = \{*\}$ since $\mathscr{F} \cong \underline{M}$ for some $M \in \mathsf{Mod}(\mathbf{k})$ we get

$$\mathrm{H}^i(\mathbb{RP}^2; \underline{M}) \cong \begin{cases} \underline{M} & i = 0 \\ \ker\left(\times 2\colon \underline{M} \to \underline{M}\right), & i = 1 \\ \underline{M}/_{2\underline{M}} & i = 2 \\ 0 & i > 2 \end{cases}$$

Example 12.0.5 ($\mathbb{RP}^n$). Let $\mathscr{F} \in \mathsf{Mod}(\underline{\mathbf{k}}_S)$. We are going to see that the cohomology of $X \coloneqq \mathbb{RP}^n \times S$ over S is

$$\mathrm{R}^i p_{X,*}\mathscr{F}|_X \cong \begin{cases} \mathscr{F} & i = 0 \\ \ker(\times 2\colon \mathscr{F} \to \mathscr{F}) & i \in \{1,\dots,n-1\}, i \text{ odd} \\ \mathscr{F}/_{2\mathscr{F}} & i \in \{1,\dots,n\}, i \text{ even} \\ \mathscr{F} & i = n \text{ odd} \\ 0 & i > n \end{cases}$$

We have already seen the base case $n = 2$, therefore we can assume the result for $n - 1$ and prove it for n.
Consider the open cover of X given by

$$U := (\text{Open } n\text{-disk}) \times S \qquad V := (\text{Tubular neighbourhood of } \mathbb{RP}^{n-1}) \times S$$

Since U is contractible over S we have

$$\mathrm{R}^i p_{U,*}\mathscr{F}|_U \cong \begin{cases} \mathscr{F} & i = 0 \\ 0 & i > 0 \end{cases}$$

Since V strongly deformation retracts over S to $\mathbb{RP}^{n-1} \times S$

$$\mathrm{R}^i p_{V,*}\mathscr{F}|_V \cong \mathrm{R}^i p_{\mathbb{RP}^{n-1}\times S,*}\mathscr{F}|_{\mathbb{RP}^{n-1}\times S}$$

Since $U \cap V$ strongly deformation retracts over S to $\mathbb{S}^{n-1} \times S$

$$\mathrm{R}^i p_{U\cap V,*}\mathscr{F}|_{U\cap V} \cong \begin{cases} \mathscr{F} & i \in \{0, n-1\} \\ 0 & i \notin \{0, n-1\} \end{cases}$$

Plugging this into the Mayer-Vietoris sequence we have

- $i = 0$: Is the same as in the $\mathbb{RP}^2$ so
$$\mathrm{R}^0 p_{X,*}\mathscr{F}|_X \cong \mathscr{F}$$
- $i \in \{1, \dots, n-2\}$: In this range we have
$$0 \to \mathrm{R}^i p_{X,*}\mathscr{F}|_X \to \mathrm{R}^i p_{\mathbb{RP}^{n-1}\times S,*}\mathscr{F}|_{\mathbb{RP}^{n-1}\times S} \to 0$$
And thus we have an isomorphism
$$\mathrm{R}^i p_{X,*}\mathscr{F}|_X \cong \mathrm{R}^i p_{\mathbb{RP}^{n-1}\times S,*}\mathscr{F}|_{\mathbb{RP}^{n-1}\times S}$$
which is our inductive step.
- $i \in \{n-1, n\}$: In this case
$$0 \to \mathrm{R}^{n-1} p_{X,*}\mathscr{F}|_X \to \mathrm{R}^{n-1} p_{\mathbb{RP}^{n-1}\times S,*}\mathscr{F}|_{\mathbb{RP}^{n-1}\times S} \xrightarrow{\beta} \mathscr{F} \xrightarrow{\delta_{n-1}} \mathrm{R}^n p_{X,*}\mathscr{F}|_X \to 0$$
The map β depends on the parity of n:
  - If n is even then the map $\mathbb{S}^{n-1} \to \mathbb{RP}^{n-1}$ has degree two and by induction
  $$\mathrm{R}^{n-1} p_{\mathbb{RP}^{n-1}\times S,*}\mathscr{F}|_{\mathbb{RP}^{n-1}\times S} \cong \mathscr{F}$$
  Therefore $\beta\colon \mathscr{F} \to \mathscr{F}$ is multiplication by two and thus
  $$\mathrm{R}^{n-1} p_{X,*}\mathscr{F}|_X \cong \ker(\times 2\colon \mathscr{F} \to \mathscr{F})$$
  $$\mathrm{R}^n p_{X,*}\mathscr{F}|_X \cong \mathscr{F}/_{2\mathscr{F}}$$
  - If n is odd the map $\mathbb{S}^{n-1} \to \mathbb{RP}^{n-1}$ has degree zero and by induction
  $$\mathrm{R}^{n-1} p_{\mathbb{RP}^{n-1}\times S,*}\mathscr{F}|_{\mathbb{RP}^{n-1}\times S} \cong \mathscr{F}/_{2\mathscr{F}}$$
  Therefore $\beta\colon \mathscr{F}/_{2\mathscr{F}} \to \mathscr{F}$ is the zero map and thus
  $$\mathrm{R}^{n-1} p_{X,*}\mathscr{F}|_X \cong \mathscr{F}/_{2\mathscr{F}}$$
  $$\mathrm{R}^n p_{X,*}\mathscr{F}|_X \cong \mathscr{F}$$

Chapter 13

de Rham Cohomology

In this chapter, we bridge the gap between differential geometry and the abstract machinery of sheaf cohomology. Historically, de Rham cohomology was developed to capture topological data about a smooth manifold using the analytical properties of differential forms. Our primary goal here is to re-interpret this classical construction through a modern algebraic lens.

By treating spaces of differential forms as sheaves, we will establish a fundamental isomorphism showing that classical de Rham cohomology is merely a realization of sheaf cohomology with coefficients in the locally constant sheaf $\underline{\mathbb{R}}$.

Throughout this chapter we will work with sheaves of $\mathbb{R}$-modules. However, the same results follow *mutatis mutandis* if we work only with sheaves of abelian groups. We will assume that all manifolds are real. The complex case works analogously replacing $\mathbb{R}$ with $\mathbb{C}$ where appropiate.

13.1 Differential forms and the de Rham complex

We start this chapter by giving a quick recap of differential forms in smooth manifolds. We refer the reader to [GP74, Chapter 4] and [Lee12, Chapter 14] for more details.

To talk about differential forms globally and locally, we first need to assemble their point-wise linear-algebraic counterparts into a smooth geometric object. This is accomplished by taking the exterior power of the cotangent bundle.

Definition 13.1.1. Let M be a smooth manifold of dimension m.
The **bundle of k-forms** $\Lambda^k(\mathrm{T}^*M)$ is the smooth vector bundle defined with:

- *Total space*: $\Lambda^k(\mathrm{T}^*M) = \coprod_{p\in M} \Lambda^k(\mathrm{T}^*_pM)$.
- *Projection map*: $\pi : \Lambda^k(\mathrm{T}^*M) \to M,\ \pi(p,\omega) = p$.
- *Fiber space*: For every $p \in M$ the fiber $E_p = \pi^{-1}(\{p\}) = \Lambda^k(\mathrm{T}^*_pM)$ is a real vector space of dimension $\binom{m}{k}$.

To define the local trivialization, consider a local chart $(U, \{x^i\}_{i=1}^m)$ of M. Then the differentials $\{\mathrm{d}x^{i_1} \wedge \cdots \wedge \mathrm{d}x^{i_k}\}_{1\le i_1<\cdots<i_k\le m}$ form a basis for each fiber with base point in U and the map $\phi^{-1} : \pi^{-1}(U) \to U \times \mathbb{R}^{\binom{m}{k}}$ is given by sending a form ω to its coefficients in this base.

Notation. To simplify working with these local bases, we adopt the standard multi-index notation for the remainder of the text. Let $I = (i_1, \dots, i_k)$ be a multi-index. We will write

$$\mathrm{d}x^I = \mathrm{d}x^{i_1} \wedge \cdots \mathrm{d}x^{i_k}$$

In this way the basis of the local trivialization is $\{\mathrm{d}x^I\}_I$.

With the underlying vector bundle established, we can now define differential forms dynamically as smoothly varying choices of tensors across the manifold. In the language of sheaf theory, this naturally corresponds to taking the sheaf of sections of our bundle.

Definition 13.1.2 (Smooth k-form)**.** Let M be a smooth manifold of dimension m.
The **sheaf of k-forms** Ω^k over M is the sheaf of sections of the bundle of k-forms $\Lambda^k(\mathrm{T}^*M)$.
More concretely, a k-form over an open set U is a map $\omega : U \to \Lambda^k(\mathrm{T}^*M)$ such that $\pi \circ \omega = \mathbb{1}_U$.

Example 13.1.1. Let M be a smooth manifold of dimension m.
Note that $\Lambda^0(\mathrm{T}^*M) = M \times \mathbb{R}$ is the trivial vector bundle. Therefore a 0-form is a map $\omega : M \to M \times \mathbb{R}$ such that $\pi \circ \omega = \mathbb{1}_M$, i.e: ω is a smooth function.
This means that $\Omega^0(M) \cong C^\infty(M)$.

The core driving force behind de Rham theory is the existence of a differential operator that links these sheaves together by increasing the degree of the forms. Locally, this operator behaves like a directional extension of the standard total derivative.

Definition 13.1.3 (Exterior derivative)**.** Let ω be a k-form over a smooth manifold M. If $(U, \{x^i\}_{i=1}^m)$ is a local chart then there exist functions $\alpha_I \in C^\infty(M)$ such that

$$\omega = \sum_I \alpha_I \mathrm{d}x^I$$

where $I = (i_1, \dots, i_k)$ is a multi-index and $\mathrm{d}x^I = \mathrm{d}x^{i_1} \wedge \cdots \mathrm{d}x^{i_k}$.
We can define the **exterior derivative** of ω as the $k+1$-form

$$\mathrm{d}\omega = \sum_I \mathrm{d}\alpha_I \wedge \mathrm{d}x^I = \sum_I \left(\sum_{j=1}^m \frac{\partial \alpha_I}{\partial x^j} \mathrm{d}x^j \right) \wedge \mathrm{d}x^I$$

For these operators to chain together into a cochain complex, the composition of two successive derivatives must vanish. This works thanks to the symmetry of partial derivatives and the antisymmetry of the wedge product.

Lemma 13.1.1. Let M be a smooth manifold of dimension m.
Then for any k-form the exterior derivative verifies $\mathrm{d}^2\omega = 0$, i.e: the sequence

$$\Omega^0(M) \xrightarrow{\mathrm{d}} \Omega^1(M) \xrightarrow{\mathrm{d}} \Omega^2(M) \xrightarrow{\mathrm{d}} \cdots$$

is a complex of sheaves of $\mathbb{R}$-modules.

Proof. It suffices to show that for any smooth function $f \in C^\infty(M) \cong \Omega^0(M)$ we have $\mathrm{d}^2 f = 0$. Indeed, if $\mathrm{d}^2 f = 0$ then for any basic form $f\mathrm{d}x^I$ we have

$$\mathrm{d}(f\mathrm{d}x^I) = \mathrm{d}(\mathrm{d}f \wedge \mathrm{d}x^I) = \mathrm{d}^2 f \wedge \mathrm{d}x^I - \mathrm{d}f \wedge \mathrm{d}^2 x^I = 0$$

Let $f \in C^\infty(M)$ be a smooth function. Its exterior derivative is

$$\mathrm{d}f = \sum_{i=1}^{m} \frac{\partial f}{\partial x^i} \mathrm{d}x^i$$

Differentiating again we get

$$\mathrm{d}^2 f = \sum_{j=1}^{m} \frac{\partial}{\partial x^j} \left(\sum_{i=1}^{m} \frac{\partial f}{\partial x^i} \right) \wedge \mathrm{d}x^j = \sum_{i,j=1}^{m} \frac{\partial^2 f}{\partial x^i \partial x^j} \mathrm{d}x^i \wedge \mathrm{d}x^j$$

By Clairaut's theorem we have that $\frac{\partial^2 f}{\partial x^i \partial x^j} = \frac{\partial^2 f}{\partial x^j \partial x^i}$ and by the antisymmetry of the wedge product $\mathrm{d}x^i \wedge \mathrm{d}x^j = -\mathrm{d}x^j \wedge \mathrm{d}x^j$. Therefore all the terms cancel leaving

$$\mathrm{d}^2 f = 0 \qquad \square$$

Example 13.1.2. Let M be a smooth manifold of dimension m.
Suppose that $k > m$. Since the fibers of the bundle of k-forms $\Lambda^k(\mathrm{T}^*M)$ have dimension $\binom{m}{k}$ this means that at least one index i in the multi-index I will be repeated. Therefore since $\mathrm{d}x^i \wedge \mathrm{d}x^i = 0$ we have that $\mathrm{d}x^I = 0$, i.e: $\Lambda^k(\mathrm{T}^*M) = 0$.
In particular $\Omega^k(M) = 0$ for all $k > m$.

We can now introduce the de Rham complex and its associated cohomology.

Definition 13.1.4 (de Rham complex)**.** Let M be a smooth manifold of dimension m. The **de Rham complex** of M is the complex of sheaves of $\mathbb{R}$-modules

$$\Omega^\bullet(M) := \Omega^0(M) \xrightarrow{\mathrm{d}} \Omega^1(M) \xrightarrow{\mathrm{d}} \cdots \xrightarrow{\mathrm{d}} \Omega^m(M) \xrightarrow{\mathrm{d}} 0$$

where $\Omega^k(M)$ is the sheaf of k-forms over M and d is the exterior derivative.

Definition 13.1.5. Let M be a smooth manifold. We define the **de Rham cohomology** of M as the cohomology of the de Rham complex and denote it by $\mathrm{H}^\bullet_{\mathrm{dR}}(M, \mathbb{R})$.

13.2 Fine sheaves

It is known that all smooth manifolds (with or without boundary) admit a partition of unity (see [Lee12, Theorem 2.23]). In this section, we are going to see how we can use this analytical tool to compute sheaf cohomology.

First we need to extend the concept of a partition of unity to a sheaf.

Definition 13.2.1 (Partition of unity)**.** Let $\mathscr{F}$ be a sheaf of $\mathbf{k}$-modules over X. A **partition of unity** of $\mathscr{F}$ subordinate to a locally finite open covering $\mathcal{U} = \{U_i\}_{i \in \mathcal{I}}$ of X is a collection of endomorphisms

$$\{f_i : \mathscr{F} \to \mathscr{F}\}_{i \in \mathcal{I}}$$

satisfying

1. For all $i \in \mathcal{I}$
$$\operatorname{supp} f_i \subset U_i$$
2. $$\sum_{i \in \mathcal{I}} f_i = \mathbb{1}_{\mathscr{F}}$$

> Note that this sum is well defined since for any point $x \in X$ there exists an open neighbourhood where all but finitely many of the maps f_i are the zero maps, i.e: the sum is locally finite.

Now we can introduce the objects of study of this section: fine sheaves. Roughly speaking these a sheaves such that whose structure allows them to be split and glued together using these localized operators.

Definition 13.2.2 (Fine sheaf)**.** A sheaf of **k**-modules $\mathscr{F}$ is **fine** if for every open cover $\mathscr{U} = \{U_i\}_{i\in\mathcal{I}}$ of X there exists a partition of unity $\{f_i\}_{i\in\mathcal{I}}$ subordinate to this covering.

Proposition 13.2.1. Fine sheaves are $\Gamma(X, -)$ acyclic.

Proof. Let $\mathscr{F}$ be a fine sheaf. Using the Godement resolution we have that $\mathscr{F}$ admits a flasque resolution $\mathscr{F} \to \mathscr{I}^\bullet$.

Using Proposition 10.2.1 by Theorem 7.2.2 we have that

$$\mathrm{H}^k(X;\mathscr{F}) = \ker\left(\Gamma(X,\mathscr{I}^k) \to \Gamma(X,\mathscr{I}^{k+1})\right) \Big/ \mathrm{Im}\left(\Gamma(X,\mathscr{I}^{k-1}) \to \Gamma(X,\mathscr{I}^k)\right)$$

Let $\alpha \in \ker\left(\Gamma(X,\mathscr{I}^k) \to \Gamma(X,\mathscr{I}^{k+1})\right)$ with $k > 0$. Since $\mathscr{I}^\bullet$ is locally exact in degree greater than 0 we have that α comes from $\mathscr{I}^{k-1}$, i.e: there exists an open cover $\mathscr{U} = \{U_i\}_{i\in\mathcal{I}}$ of X such that $\alpha|_{U_i} = \mathrm{d}\beta_i$ with $\beta_i \in \mathscr{I}^{k-1}(U_i)$.

Consider a partition of unity $\{f_i\}_{i\in\mathcal{I}}$ subordinate to $\mathscr{U}$ and let

$$\beta \coloneqq \sum_{i\in\mathcal{I}} f_i\beta_i$$

where the sum is locally finite. Note that $f_i\beta_i$ is a section of $\mathscr{I}^{k-1}(X)$ with values in $\mathscr{I}^{k-1}_p(X)$:

$$(f_i\beta_i)_p = \begin{cases} 0 & p \notin U_i \\ f_i\beta_i & p \in U_i \end{cases}$$

Since f_i has support in U_i this defines a section in $\mathscr{I}^{k-1}(U_i)$.

We have that $\mathrm{d}\beta = \alpha$ since $\alpha = \sum_{i\in\mathcal{I}} f_i\alpha|_{U_i}$. Therefore $\ker\left(\Gamma(X,\mathscr{I}^k) \to \Gamma(X,\mathscr{I}^{k+1})\right) = \mathrm{Im}\left(\Gamma(X,\mathscr{I}^{k-1}) \to \Gamma(X,\mathscr{I}^k)\right)$, which means

$$\forall k > 0 \quad \mathrm{H}^k(X,\mathscr{F}) = 0$$ □

An example of fine sheaves is one we have just studied: sheaves of k-forms.

Lemma 13.2.1. Let M be a smooth manifold. Then for all k the sheaf of k-forms Ω^k is fine.

Proof. Let $\mathscr{U} = \{U_i\}_{i\in\mathcal{I}}$ be an arbitrary open cover of M.

Since M is paracompact there exists a partition of unity $\{f_i\}_{i\in\mathcal{I}} \subset \mathcal{C}^\infty(M)$ subordinate to $\mathscr{U}$. This means that

$$\begin{aligned} \forall i \in \mathcal{I} \quad & \operatorname{supp} f_i \subseteq U_i \\ \forall p \in M \quad & \sum_{i\in\mathcal{I}} f_i(p) = 1 \end{aligned}$$

and this last sum is finite for each p.

For all indices $i \in \mathcal{I}$ consider the map $\eta_i \colon \Omega^k \to \Omega^k$ defined as, for all $U \subseteq M$ open

$$\begin{aligned} \eta_i(U) : \Omega^k(U) &\to \Omega^k(U) \\ \omega &\mapsto f_i \cdot \omega \end{aligned}$$

Note that since $\operatorname{supp} f_i \subseteq U_i$ we have $\operatorname{supp} \eta_i \subseteq U_i$. Furthermore for any k-form ω

$$\sum_{i \in \mathcal{I}} \eta_i(\omega) = \sum_{i \in \mathcal{I}} f_i \cdot \omega = \left(\sum_{i \in \mathcal{I}} f_i \right) \cdot \omega = 1 \cdot \omega = \omega$$

Therefore $\sum_{i \in \mathcal{I}} \eta_i = \mathbb{1}_{\Omega^k}$, making $\{\eta_i\}_{i \in \mathcal{I}}$ a partition of unity subordinate to $\mathscr{U}$. □

Therefore if we can prove that there exists a sheaf of $\mathbb{R}$-modules $\mathscr{F} \in \mathsf{Mod}(\underline{\mathbb{R}}_M)$ such that $\Omega^\bullet$ is a resolution of $\mathscr{F}$ we would have that sheaf cohomology with values in $\mathscr{F}$ and de Rham cohomology coincide.

As we are going to see, this is almost a direct consequence of the classical Poincaré lemma of multi-variable calculus if we consider $\mathscr{F} = \underline{\mathbb{R}}_M$.

Lemma 13.2.2 (Poincaré lemma — [Lee12, Theorem 11.49]). Let $U \subseteq \mathbb{R}^n$ be a contractible open set and $k \geq 1$. If $\omega \in \Omega^k(U)$ is a k-form such that $\mathrm{d}\omega = 0$ then there exists a $(k-1)$-form $\eta \in \Omega^{k-1}(U)$ such that $\mathrm{d}\eta = \omega$.

First, let's check that $\Omega^\bullet$ is an exact complex of sheaves.

Lemma 13.2.3. The complex of sheaves $0 \to \Omega^0 \to \Omega^1 \to \cdots$ is exact.

Proof. For $k \geq 1$ consider the relevant part of the complex

$$\Omega^{k-1} \xrightarrow{\mathrm{d}^{k-1}} \Omega^k \xrightarrow{\mathrm{d}^k} \Omega^{k+1}$$

We need to show that for any $p \in M$ the complex

$$\left(\Omega^{k-1}\right)_p \xrightarrow{\mathrm{d}_p^{k-1}} \left(\Omega^k\right)_p \xrightarrow{\mathrm{d}_p^k} \left(\Omega^{k+1}\right)_p$$

is exact, i.e: $\operatorname{Im} \mathrm{d}_p^{k-1} = \ker \mathrm{d}_p^k$.

For the $[\subseteq]$ inclusion, let $\mathrm{d}\eta \in \operatorname{Im} \mathrm{d}^{k-1} \subseteq \Omega^k$. By Lemma 13.1.1 we have that $\mathrm{d}(\mathrm{d}\eta) = 0$, i.e: $\mathrm{d}\eta \in \ker \mathrm{d}^k$.

For the $[\supseteq]$ inclusion, let $[\omega]_p \in \ker \mathrm{d}_p^k$, i.e: $[\omega]_p \in \left(\Omega^k\right)_p$ verifies $\mathrm{d}_p^k [\omega]_p = 0$. By definition there exists an open neighbourhood U of p and a k-form ω such that $\mathrm{d}^k \omega|_U = 0$.

Since M is a smooth manifold of dimension m we can choose a local chart (V, ψ) around p such that $V \subseteq U$ and $\psi(V)$ is an open ball in $\mathbb{R}^m$. Since open balls are contractible, by Poincaré's lemma there exists a $(k-1)$-form $\eta \in \Omega^{k-1}(V)$ such that $\mathrm{d}^{k-1}\eta = \omega|_V$.

Therefore there exists $[\eta]_p \in \left(\Omega^{k-1}\right)_p$ such that $\mathrm{d}_p^{k-1}[\eta]_p = [\omega]_p$, i.e: $[\omega]_p \in \operatorname{Im} d_p^{k-1}$. □

Having seen this, we can finally state and prove the main theorem of the chapter.

Theorem 13.2.1. Let M be a smooth manifold. Then

$$\mathrm{H}^\bullet(M, \underline{\mathbb{R}}_M) \cong \mathrm{H}^\bullet_{\mathrm{dR}}(M, \mathbb{R})$$

Proof. Consider the resolution

$$0 \longrightarrow \underline{\mathbb{R}} \longrightarrow \Omega^\bullet$$

where $\Omega^\bullet$ is the de Rham complex of M.

By Lemma 13.2.3 we know that $\Omega^\bullet$ is exact, therefore we only need to show that this resolution is exact at Ω^0, i.e: that for all $p \in M$

$$0 \longrightarrow \underline{\mathbb{R}}_p \xrightarrow{\iota_p} \Omega^0_p \xrightarrow{\mathrm{d}_p} \Omega^1_p \longrightarrow 0$$

is a short exact sequence, i.e: $\operatorname{Im} \iota_p = \ker \mathrm{d}_p$.

- $\subseteq$ Let $[c]_p \in \underline{\mathbb{R}}_p$. By definition this is a locally constant function $c : U \to \mathbb{R}$ on some open neighbourhood U of p.
  Since c is locally constant we have that $\mathrm{d}c|_U = 0$, i.e $\mathrm{d}_p[c]_p = 0$.
- $\supseteq$ Let $[f]_p \in \Omega^0_p$ such that $\mathrm{d}_p[f]_p = 0$. This means that there exists an open neighbourhood U of p and a function $f \in \Omega^0(V) \cong C^\infty(M)$ such that $\mathrm{d}f|_U = 0$. Therefore f is locally constant in each of the connected components of U, i.e: $[f]_p \in \underline{\mathbb{R}}_p$.

Therefore since by Lemma 13.2.1 the resolution is made of fine sheaves and fine sheaves are $\Gamma(X, -)$ acyclic by definition of sheaf cohomology

$$\mathrm{H}^\bullet(M, \underline{\mathbb{R}}_M) \cong \mathrm{H}^\bullet_{\mathrm{dR}}(M, \mathbb{R})$$ □

As a consequence of Example 13.1.2 we get the following finiteness result for sheaf cohomology.

Corollary 13.2.1. Let M be a smooth manifold of dimension m. Then for all $k > m$

$$\mathrm{H}^k(M, \underline{\mathbb{R}}_M) = 0$$

To end this chapter we revisit the computation of the cohomology of the circle $\mathbb{S}^1$ to verify that our results indeed hold.

Example 13.2.1 (The cohomology of $\mathbb{S}^1$)**.** Consider the circle $\mathbb{S}^1$ as a smooth manifold of dimension 1.
We are going to compute its sheaf cohomology with coefficients in $\underline{\mathbb{R}}$, $\mathrm{H}^\bullet(\mathbb{S}^1, \underline{\mathbb{R}})$, using the de Rham complex

$$\Omega^0(\mathbb{S}^1) \xrightarrow{\mathrm{d}^0} \Omega^1(\mathbb{S}^1) \xrightarrow{\mathrm{d}^1} 0$$

On the one hand

$$\mathrm{H}^0(\mathbb{S}^1, \underline{\mathbb{R}}_M) = \ker \mathrm{d}^0 = \{f \in \Omega^0(\mathbb{S}^1) \cong C^\infty(\mathbb{S}^1) \mid \mathrm{d}f = 0\}$$

Since $\mathbb{S}^1$ is connected, the only functions with null derivative are the constant

functions. Therefore

$$\mathrm{H}^0(\mathbb{S}^1, \underline{\mathbb{R}}_M) = \mathbb{R}$$

On the other

$$\mathrm{H}^1(\mathbb{S}^1, \underline{\mathbb{R}}_M) = {\ker \mathrm{d}^1}/{\operatorname{Im} \mathrm{d}^0} = {\{\omega \in \Omega^1(\mathbb{S}^1) \mid \mathrm{d}\omega = 0\}}/{\{\mathrm{d}f \mid f \in \Omega^0(\mathbb{S}^1) \cong C^\infty(\mathbb{S}^1)\}}$$

Note that since $\Omega^2(M) = 0$ all 1-forms $\omega \in \Omega^1(\mathbb{S}^1)$ verify $\mathrm{d}\omega = 0$. Therefore

$$\{\omega \in \Omega^1(\mathbb{S}^1) \mid \mathrm{d}\omega = 0\} = \Omega^1(\mathbb{S}^1)$$

Consider now the *integration* map

$$\begin{aligned} I : \Omega^1(\mathbb{S}^1) &\to \mathbb{R} \\ \omega &\mapsto \int_{\mathbb{S}^1} \omega \end{aligned}$$

Let $\mathrm{d}\theta$ be the differential of the angle. Since $I(\mathrm{d}\theta) = 2\pi$ we have that I is surjective. Furthermore by Stoke's theorem we have that $I(\omega) = 0$ if and only if $w = \mathrm{d}f$ for some $f \in C^\infty(\mathbb{S}^1)$.
Therefore by the first isomorphism theorem we have that

$$\mathrm{H}^1(\mathbb{S}^1, \underline{\mathbb{R}}_M) = {\Omega^1(\mathbb{S}^1)}/{\ker I} \cong \operatorname{Im} I = \mathbb{R}$$

In summary, we have that

$$\mathrm{H}^k(\mathbb{S}^1, \underline{\mathbb{R}}_M) = \begin{cases} \mathbb{R} & k \in \{0, 1\} \\ 0 & k > 1 \end{cases}$$

which agrees with the our previous computation using only sheaf cohomology.

Chapter 14

Cohomology theories

In this chapter, we step back from explicit calculations to examine the broader structural properties of sheaf cohomology, placing it within the universal framework of algebraic topology.

Historically, Eilenberg and Steenrod showed that various geometric invariants (such as singular, simplicial, or cellular cohomology) could be characterized uniquely by a small set of simple axioms. Our goal here is to compare the definition of sheaf cohomology against this classic axiomatic standard. By verifying that sheaf cohomology satisfies these universal criteria, we will justify why it can be viewed as a generalization of classical cohomology to spaces.

In this chapter we restrict ourselves to sheaves of abelian groups.

14.1 Cohomology theories and CW-Complexes

Definition 14.1.1. A **topological pair** is a tuple (X, A) where X is a topological space and A is a subspace of X. A morphism of topological pairs $(X, A) \to (Y, B)$ is a continuous map $f : X \to Y$ such that $f(A) \subseteq B$.
We denote by Top_2 the category of topological pairs.

We start by giving the definition of a cohomology theory following [ES45].

Definition 14.1.2. A **cohomology theory** (on Top) consists of

1. A sequence of contravariant functors $\mathrm{H}^n : \mathsf{Top}_2 \to \mathsf{Ab}$ for all $n \in \mathbb{Z}$.
   We denote by $\mathrm{H}^n(X) := \mathrm{H}^n(X, \varnothing)$. If $f : (X, A) \to (Y, B)$ is a continuous map we denote $f^* : \mathrm{H}^n(Y, B) \to \mathrm{H}^n(X, A)$ the induced map (where n is to be understood from the context).
2. A sequence of natural transformations $\delta^n : \mathrm{H}^n(A) \to \mathrm{H}^{n+1}(X, A)$.

Satisfying the following axioms:

1. **Homotopy**: If $f_0, f_1 : (X, A) \to (Y, B)$ are homotopic then its induced maps on H^n are equal
$$f_0^* = f_1^* : \mathrm{H}^n(Y, B) \to \mathrm{H}^n(X, A)$$
2. **Excission**: If $(X, A) \in \mathsf{Top}_2$ and $U \subseteq A$ is such that $\mathrm{ad}(U) \subset \mathrm{int}(A)$ then the inclussion map $i : (X \setminus U, A \setminus U) \to (X, A)$ induces an isomorphism in H^n:
$$i^* : \mathrm{H}^n(X, A) \xrightarrow{\sim} \mathrm{H}^n(X \setminus U, A \setminus U)$$

3. **Exactness**: Each pair $(X, A) \in \mathsf{Top}_2$ induces a long exact sequence via the inclussions $i\colon A \to X$ and $j\colon X \to (X, A)$:

$$\cdots \longrightarrow \mathrm{H}^{n-1}(X) \xrightarrow{i^*} \mathrm{H}^{n-1}(A) \xrightarrow{\delta^{n-1}} \mathrm{H}^n(X, A) \xrightarrow{j^*} \mathrm{H}^n(X) \xrightarrow{i^*} \mathrm{H}^n(A) \xrightarrow{\delta^n} \mathrm{H}^{n+1}(X, A) \xrightarrow{j^*} \mathrm{H}^{n+1}(X) \longrightarrow \cdots$$

We also require the following axiom due to Milnor [Mil]

4. **Additivity**: If $X = \coprod_{i\in I} X_i$ is the disjoint union of topological spaces then

$$\mathrm{H}^n(X) \cong \prod_{i\in I} \mathrm{H}^n(X_i)$$

We say that a cohomology theory is ordinary if it further verifies

5. **Dimension**: If $\{*\}$ is the one-point topological space then for all $n \neq 0$

$$\mathrm{H}^n(\{*\}) = 0$$

In this case, $\mathrm{H}^0(\{*\})$ is called the coefficient group of the theory.

For sufficiently nice spaces, one can show that there exists only one ordinary cohomology theory with a given coefficient group.

Definition 14.1.3 (CW-complex). A **CW-complex** is a topological space X constructed inductively in the following way:

1. Start with a discrete set X^0, called the 0-cells of X.
2. Inductively, form the n-skeleton X^n from X^{n-1} by attaching n-cells e^n_α via maps $\varphi_\alpha\colon \mathbb{S}^{n-1} \to X^{n-1}$.
   This means that X^n is the quotient space of $X^{n-1} \coprod_\alpha \mathbb{D}^n_\alpha$ under the identifications $x \sim \varphi_\alpha(x)$ for $x \in \partial\mathbb{D}^{n-1}_\alpha$. The cell e^n_α is the homeomorphic image of $\mathbb{D}^n_\alpha \setminus \partial\mathbb{D}^{n-1}_\alpha$ under the quotient map.
3. Set $X \coloneqq \bigcup_{n\in\mathbb{N}} X^n$ with the weak topology: a set $A \subseteq X$ is open if $A \cap X^n$ is open in X^n for all n.

We will not prove any properties of CW-complexes, refering the reader to [Hat02, Appendix], but even though their purely combinatorial nature they are enough to capture the topological properties of compact manifolds.

Proposition 14.1.1 ([Hat02, Corollary A.12]). A compact manifold is homotopy equivalent to a CW-complex.

This brings us to the core structural uniqueness theorem: over these cellular environments, all ordinary theories with identical core coefficients collapse into a single one.

Theorem 14.1.1 (Uniqueness of cohomology theories — [Hat02, Theorem 4.59]). Let $\{\mathrm{H}^n, \delta^n\}_{n\in\mathbb{N}}$ and $\left\{\widetilde{\mathrm{H}}^n, \widetilde{\delta}^n\right\}_{n\in\mathbb{N}}$ be two ordinary cohomology theories in the category of CW pairs with the same coefficient group. Then for all $n \geq 0$ there is a natural

isomorphism

$$\mathrm{H}^n(X, A) \cong \widetilde{\mathrm{H}}^n(X, A)$$

The most famous cohomology theory is **singular cohomology** $\mathrm{H}_{\mathrm{sing}}$. In fact the axioms of cohomology theories were modelled after its behaviour and the proof of Theorem 14.1.1 works by proving that any ordinary cohomology theory is isomorphic to singular cohomology (given that they have the same coefficient group).

We will not go into any details of its definition or properties, refering the reader to [Hat02, Chapter 3] for more details.

Remark 14.1.1. Another strategy to prove that sheaf cohomology is an ordinary cohomology theory is to prove that it coincides with singular cohomology (or any other ordinary cohomology theory) by finding a suitable resolution as we did for the de Rham case.
This however would require knowledge about the chosen cohomology theory (and of course proving that it is indeed an ordinary cohomology theory). For the interested reader we refer to [Bre97, Chapter III].

14.2 Relative sheaf cohomology

The definition of a cohomology theory is done in the category of topological pairs Top_2. However, our definition of sheaf cohomology only accepts a single space as an input.

To fix this, we introduce the following version of cohomology.

Definition 14.2.1 (Relative Sheaf Cohomology)**.** Let X be a topological space and $A \subseteq X$ a closed subset. Let $j\colon X \setminus A \hookrightarrow X$ be the open immersion of the complement. For any sheaf of $\mathbf{k}$-modules $\mathscr{F}$ over X, the **relative cohomology** of the pair (X, A) with coefficients in $\mathscr{F}$ is:

$$\mathrm{H}^i(X, A; \mathscr{F}) \coloneqq \mathrm{H}^i(X; j_! j^{-1} \mathscr{F})$$

If $A = \varnothing$ we recover the usual sheaf cohomology

$$\mathrm{H}^i(X, \varnothing; \mathscr{F}) = \mathrm{H}^i(X; \mathscr{F})$$

Remark 14.2.1. Relative sheaf cohomology is the right derived functor

$$\Gamma(X; j_! j^{-1} \mathscr{F}) \coloneqq \{s \in \Gamma(X, \mathscr{F}) \mid s|_A = 0\}$$

Now we are ready to prove that relative sheaf cohomology is a cohomology theory by verifying the axioms one by one.

Theorem 14.2.1 (Exactness)**.** Let $A \subseteq X$ be a closed subset and $i\colon A \hookrightarrow X$ the closed immersion. For any sheaf of $\mathbf{k}$-modules $\mathscr{F}$, there exists a long exact sequence in cohomology:

$$\cdots \longrightarrow \mathrm{H}^i(X, A; \mathscr{F}) \longrightarrow \mathrm{H}^i(X; \mathscr{F}) \longrightarrow \mathrm{H}^i(A; \mathscr{F}_A) \longrightarrow \mathrm{H}^{i+1}(X, A; \mathscr{F}) \longrightarrow \cdots$$

Proof. By Theorem 9.2.4 we have a short exact sequence of sheaves:

$$0 \to j_! j^{-1} \mathscr{F} \to \mathscr{F} \to i_* i^{-1} \mathscr{F} \to 0$$

Applying the long exact sequence of derived functors to $\Gamma(X, -)$ and using **??** for the term $\mathrm{H}^i(X; i_* i^{-1}\mathscr{F}) \cong \mathrm{H}^i(A; \mathscr{F}_A)$, we obtain the result. □

Theorem 14.2.2 (Excision). Let $A \subseteq X$ be a closed subset. Let $Z \subseteq A$ be a subset such that $\mathrm{ad}(Z) \subseteq \mathrm{int}(A)$. Then for any sheaf $\mathscr{F} \in \mathsf{Mod}(\underline{\mathbf{k}}_X)$:

$$\mathrm{H}^i(X, A; \mathscr{F}) \cong \mathrm{H}^i(X \setminus Z, A \setminus Z; \mathscr{F})$$

Proof. Let $U \coloneqq X \setminus A$ and $j \colon U \hookrightarrow X$ the open immersion. By definition we have

$$\mathrm{H}^i(X, A; \mathscr{F}) = \mathrm{H}^i(X; \mathscr{G})$$

where $\mathscr{G} \coloneqq j_! j^{-1} \mathscr{F}$.

Without loss of generality suppose that Z is closed (if not consider $\mathrm{ad}(Z)$). Let $V = X \setminus Z$ and $W = \mathrm{int}(A)$. Since Z is closed V is open. Furthermore since $Z \subseteq \mathrm{int}(A) = W$ we have an open cover $X = V \cup W$.

By Mayer-Vietoris we have a long exact sequence

$$\cdots \longrightarrow \mathrm{H}^{i-1}(V \cap W; \mathscr{G}|_{V\cap W}) \to \mathrm{H}^i(X; \mathscr{G}) \longrightarrow \mathrm{H}^i(V; \mathscr{G}|_V) \oplus \mathrm{H}^i(W; \mathscr{G}|_W) \longrightarrow \mathrm{H}^i(V \cap W; \mathscr{G}|_{V\cap W}) \to \mathrm{H}^{i+1}(X; \mathscr{G}) \longrightarrow \cdots$$

Note that the stalks of $\mathscr{G}$ are

$$\mathscr{G}_p = \begin{cases} \mathscr{F}_p & p \in U \\ 0 & p \notin U \end{cases}$$

Therefore since $W \subseteq A$ and $A \cap U = \varnothing$ we have $\mathscr{G}_p = 0$ for all $p \in W$. In particular $\mathscr{G}|_W = 0$ and thus $\mathscr{G}|_{V\cap W} = 0$. Plugging this into the above long exact sequence we get isomorphisms

$$\mathrm{H}^i(X; \mathscr{G}) \xrightarrow{\sim} \mathrm{H}^i(V; \mathscr{G}|_V)$$

To finish, we need to check that $\mathrm{H}^i(V; \mathscr{G}) = \mathrm{H}^i(X \setminus Z, A \setminus Z, \mathscr{F})$. Note that

$$V \setminus (A \setminus Z) = (X \setminus Z) \setminus (A \setminus Z) = X \setminus A = U$$

Consider the open immersion $j' \colon U \hookrightarrow V$. Since $\mathscr{G}$ was formed by extending $\mathscr{F}|_U$ by zero to X, restricting it to V (which fully contains U) yields the same extension by zero relative to V:

$$\mathscr{G}|_V = j'_! (\mathscr{F}|_U) = j'_! j'^{-1} (\mathscr{F}|_V)$$

Therefore

$$\mathrm{H}^i(V; \mathscr{G}|_V) = \mathrm{H}^i\left(V; j'_! j'^{-1} \mathscr{F}|_V\right) = \mathrm{H}^i(X \setminus Z, A \setminus Z; \mathscr{F})$$ □

Theorem 14.2.3 (Homotopy). Let $f_0, f_1 \colon (X, A) \rightrightarrows (Y, B)$ be two homotopic maps of pairs. Let $\underline{M}_Y \in \mathsf{Mod}(\underline{\mathbf{k}}_Y)$ be a constant sheaf of $\mathbf{k}$-modules over Y. Then for all

$i \geq 0$, the induced maps on relative cohomology coincide:

$$f_0^* = f_1^* \colon \mathrm{H}^i(Y, B; \underline{M}_Y) \to \mathrm{H}^i(X, A; \underline{M}_X)$$

Proof. Let $H \colon (X \times I, A \times I) \to (Y \times I, B \times I)$ be the homotopy between f_0 and f_1. We have

$$f_0^* = j_0^* \circ H \qquad f_1^* = j_1^* \circ H$$

Therefore we need to prove

$$j_0^* = j_1^* \colon \mathrm{H}^i\left(X \times I, A \times I; \underline{M}_{X \times I}\right) \to \mathrm{H}^i\left(X, A; \underline{M}_X\right)$$

Note that $p_{(X,A)} \circ j_0 = \mathbb{1}_{(X,A)} = p_{(X,A)} \circ j_1$. Furthermore

$$i_0 \circ p_{(X,A)} \colon (X \times I, A \times I) \to (X \times I, A \times I)$$

is homotopic to the identity.

Therefore by Theorem 11.0.1 we get isomorphisms

$$p_X^* \colon \mathrm{H}^i(X; \underline{M}_X) \xrightarrow{\sim} \mathrm{H}^i\left(X \times I; \underline{M}_{X \times I}\right)$$
$$p_A^* \colon \mathrm{H}^i(A; \underline{M}_A) \xrightarrow{\sim} \mathrm{H}^i\left(A \times I; \underline{M}_{A \times I}\right)$$

By the exactness axiom we have long exact sequences

$$\begin{array}{ccccccccccc}
\cdots & \longrightarrow & \mathrm{H}^{i-1}(A; \underline{M}) & \longrightarrow & \mathrm{H}^i(X, A; \underline{M}) & \longrightarrow & \mathrm{H}^i(X; \underline{M}) & \longrightarrow & \mathrm{H}^i(A; \underline{M}) & \longrightarrow & \cdots \\
 & & \downarrow{\scriptstyle p_A^*}{\scriptstyle \sim} & & & & \downarrow{\scriptstyle p_X^*}{\scriptstyle \sim} & & \downarrow{\scriptstyle p_A^*}{\scriptstyle \sim} & & \\
\cdots & \longrightarrow & \mathrm{H}^{i-1}(A \times I; \underline{M}) & \longrightarrow & \mathrm{H}^i(X \times I, A \times I; \underline{M}) & \longrightarrow & \mathrm{H}^i(X \times I; \underline{M}) & \longrightarrow & \mathrm{H}^i(A \times I; \underline{M}) & \longrightarrow & \cdots
\end{array}$$

Therefore by the five lemma

$$p_{(X,A)}^* \colon \mathrm{H}^i(X, A; \underline{M}) \xrightarrow{\sim} \mathrm{H}^i(X \times I, A \times I; \underline{M})$$

By functoriallity of cohomology we have

$$j_0^* \circ p_{(X,A)}^* = \mathbb{1} = j_1^* \circ p_{(X,A)}^*$$

Therefore since $p_{(X,A)}^*$ is an isomorphism

$$j_0^* = j_1^*$$

□

Theorem 14.2.4 (Additivity). Let $X = \coprod_{\alpha \in \Lambda} X_\alpha$ be a disjoint union of topological spaces. Then for any sheaf of $\mathbf{k}$-modules $\mathscr{F}$ on X:

$$\mathrm{H}^i(X; \mathscr{F}) \cong \prod_{\alpha \in \Lambda} \mathrm{H}^i(X_\alpha; \mathscr{F}|_{X_\alpha})$$

Proof. By definition of the disjoint union topology, each X_α is an open subset of X.

Let $\mathscr{F} \in \mathsf{Mod}(\underline{\mathbf{k}}_X)$. Since $X_\alpha \cap X_\beta = \varnothing$ if $\alpha \neq \beta$ we have

$$\Gamma(X, \mathscr{F}) \cong \prod_{\alpha \in \Lambda} \Gamma(X_\alpha, \mathscr{F}|_{X_\alpha})$$

Furthermore if $\mathscr{F} \to \mathscr{I}^\bullet$ is an injective resolution since j_α^{-1} is right adjoint to $j_{\alpha,!}$, which is exact, by Proposition 7.1.2 $I^\bullet|_{X_\alpha}$ is an injective resolution of $\mathscr{F}|_{X_\alpha}$.

Therefore

$$\begin{aligned}
\mathrm{H}^i(X; \mathscr{F}) &\cong \mathrm{R}^i \Gamma(X, \mathscr{I}^\bullet) \\
&\cong \mathrm{R}^i \left(\prod_{\alpha \in \Lambda} \Gamma(X_\alpha, \mathscr{I}^\bullet|_{X_\alpha}) \right) \\
&\cong \prod_{\alpha \in \Lambda} \mathrm{R}^i \Gamma(X_\alpha, \mathscr{I}^\bullet|_{X_\alpha}) \\
&\cong \prod_{\alpha \in \Lambda} \mathrm{H}^i(X_\alpha, \mathscr{F}|_{X_\alpha})
\end{aligned}$$

□

Theorem 14.2.5 (Dimension). Let $\{*\}$ be the one point topological space. For all sheaves of $\mathbf{k}$-modules $\mathscr{F} \in \mathsf{Mod}(\underline{\mathbf{k}}_X)$ and $n \neq 0$

$$\mathrm{H}^n(\{*\}; \mathscr{F}) \cong 0$$

Proof. We have

$$\mathrm{H}^n(\{*\}; \mathscr{F}) = \mathrm{R}^n \Gamma(\{*\}, \mathscr{F})$$

Note that

$$\Gamma(\{*\}, \mathscr{F}) = \mathscr{F}(\{*\}) = \mathscr{F}_*$$

Therefore since the stalk functor is exact for all $n \neq 0$

$$\mathrm{H}^n(\{*\}; \mathscr{F}) = \mathrm{R}^n \mathscr{F}_* = 0$$

□

Having fully verified all five axioms invoking the uniqueness of cohomology theorieswe obtain complete equivalences between abstract sheaf cohomology and classical cohomology groups.

In particular from Proposition 14.1.1 and the homotopy invariance of cohomology we have the following.

Corollary 14.2.1. Let X be a space homotopic to a CW-complex. For any constant sheaf of abelian groups $\underline{G} \in \mathsf{Mod}(\underline{\mathbb{Z}}_X)$ we have an isomorphism

$$\mathrm{H}^i(X; \underline{G}) \cong \mathrm{H}^i_{\mathrm{sing}}(X; G)$$

Furthermore, we can combined this with the work we did in the last chapter obtaining a proof that de Rham cohomology and singular cohomology coincide for compact smooth manifolds.

Corollary 14.2.2. Let X be a compact smooth manifold. We have isomorphisms

$$\mathrm{H}^i_{\mathrm{sing}}(X;\mathbb{R}) \cong \mathrm{H}^i(X;\underline{\mathbb{R}}) \cong \mathrm{H}_{\mathrm{dR}}(X;\mathbb{R})$$

In particular de Rham cohomology H_{dR} is an ordinary cohomology theory.

Bibliography

[Alu09] Paolo Aluffi. *Algebra: Chapter 0*. Graduate Studies in Mathematics v. 104. Providence, R.I: American Mathematical Society, 2009. 713 pp. (cited on p. 101).

[SGA4] Michael Artin, Alexander Grothendieck, and Jean-Louis Verdier. *Théorie Des Topos et Cohomologie Étale Des Schémas (SGA 4)*. Vol. 269, 270, 305. Lecture Notes in Mathematics. Berlin Heidelberg: Springer Berlin Heidelberg.

[Bre97] Glen E. Bredon. *Sheaf Theory*. Vol. 170. Graduate Texts in Mathematics. New York, NY: Springer New York, 1997. (Visited on 02/01/2026) (cited on pp. 69, 129).

[ES45] Samuel Eilenberg and Norman E. Steenrod. "Axiomatic Approach to Homology Theory". In: *Proceedings of the National Academy of Sciences* 31.4 (Apr. 1945), pp. 117–120. ISSN: 0027-8424, 1091-6490. (Visited on 04/08/2026) (cited on p. 127).

[GQ22] Jean Gallier and Jocelyn Quaintance. *Homology, Cohomology, and Sheaf Cohomology for Algebraic Topology, Algebraic Geometry, and Differential Geometry*. WORLD SCIENTIFIC, Feb. 2022. (Visited on 12/23/2025) (cited on p. 69).

[God98] Roger Godement. *Topologie algébrique et théorie des faisceaux*. Nouveau tirage. Actualités scientifiques et industrielles 1252. Paris: Hermann, 1998. 283 pp. (cited on p. 69).

[Gro] Alexandre Grothendieck. "Some Aspects of Homological Algebra". In: () (cited on pp. 68, 69).

[GP74] Victor Guillemin and Alan Pollack. *Differential Topology*. Englewood Cliffs, N.J: Prentice-Hall, 1974. 222 pp. (cited on p. 120).

[Har77] Robin Hartshorne. *Algebraic Geometry*. Graduate Texts in Mathematics 52. New York Berlin Paris [etc.]: Springer, 1977 (cited on p. 69).

[Hat02] Allen Hatcher. *Algebraic Topology*. Cambridge ; New York: Cambridge University Press, 2002. 544 pp. (cited on pp. 128, 129).

[KS06] Masaki Kashiwara and Pierre Schapira. *Categories and Sheaves*. Red. by M. Berger et al. Vol. 332. Grundlehren Der Mathematischen Wissenschaften. Berlin, Heidelberg: Springer Berlin Heidelberg, 2006. (Visited on 09/28/2025) (cited on p. 69).

[KS90] Masaki Kashiwara and Pierre Schapira. *Sheaves on Manifolds*. Red. by M. Artin et al. Vol. 292. Grundlehren Der Mathematischen Wissenschaften. Berlin, Heidelberg: Springer Berlin Heidelberg, 1990. (Visited on 11/09/2025) (cited on pp. 69, 108).

[Lee12] John M. Lee. *Introduction to Smooth Manifolds*. Vol. 218. Graduate Texts in Mathematics. New York, NY: Springer New York, 2012. (Visited on 01/26/2026) (cited on pp. 120, 122, 124).

[Mac78] Saunders Mac Lane. *Categories for the Working Mathematician*. Vol. 5. Graduate Texts in Mathematics. New York, NY: Springer New York, 1978. (Visited on 01/19/2026) (cited on p. 49).

[Mil] J Milnor. "On Axiomatic Homology Theory". In: () (cited on p. 128).

[Rie] Emily Riehl. "Category Theory in Context". In: () (cited on pp. 5, 17, 20, 26, 31–33).

[Ros22] Daniel Rosiak. *Sheaf Theory through Examples*. The MIT Press, Oct. 25, 2022. (Visited on 06/01/2026) (cited on p. 69).

[Sch] Pierre Schapira. "An Introduction to Categories and Sheaves". In: ().

[Ser55] Jean-Pierre Serre. "Faisceaux Algebriques Coherents". In: *The Annals of Mathematics* 61.2 (Mar. 1955), p. 197. ISSN: 0003486X. JSTOR: 1969915. (Visited on 06/01/2026) (cited on p. 68).

[Stacks] The Stacks Project Authors. *Stacks Project*. 2018. URL: https://stacks.math.columbia.edu (cited on pp. 72, 106).

[Vak25] Ravi Vakil. *RISING SEA: Foundations of Algebraic Geometry*. S.l.: PRINCETON UNIVERSITY PRES, 2025 (cited on pp. 69, 70, 79).

[Wei94] Charles A. Weibel. *An Introduction to Homological Algebra*. 1st ed. Cambridge University Press, Apr. 29, 1994. (Visited on 12/21/2025) (cited on pp. 49, 61).